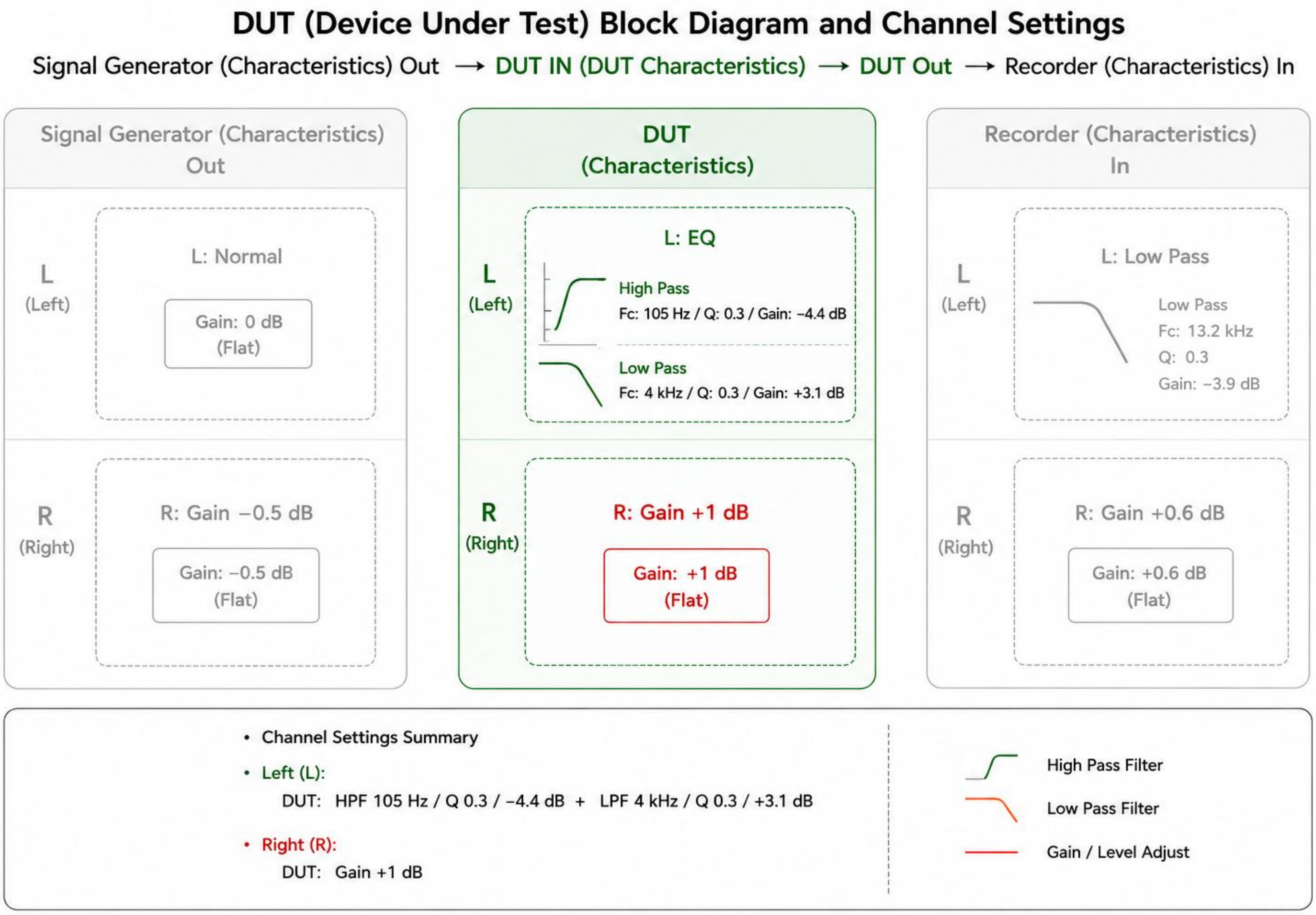
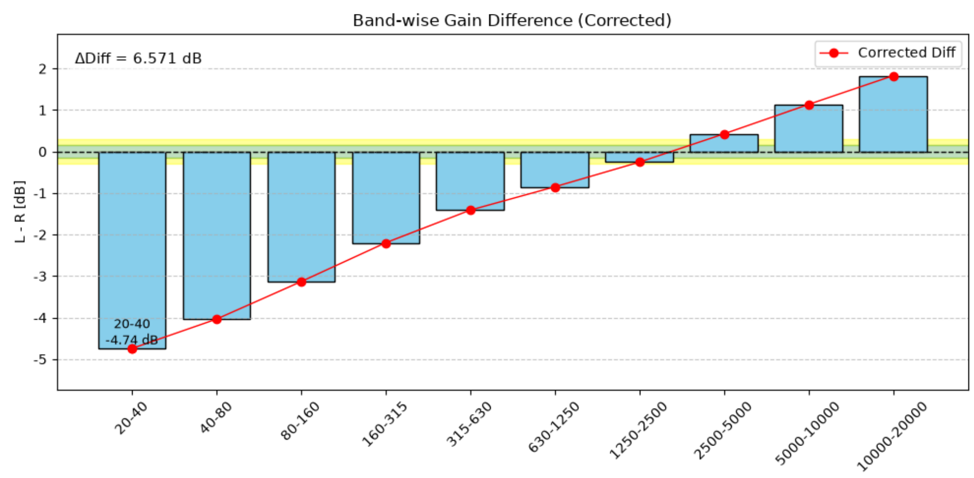
Verification of the calculations used in this analysis script to cancel out factors other than the device under test (DUT). The following sections present the block diagram typically used in measurements and the simulation results for each characteristic.

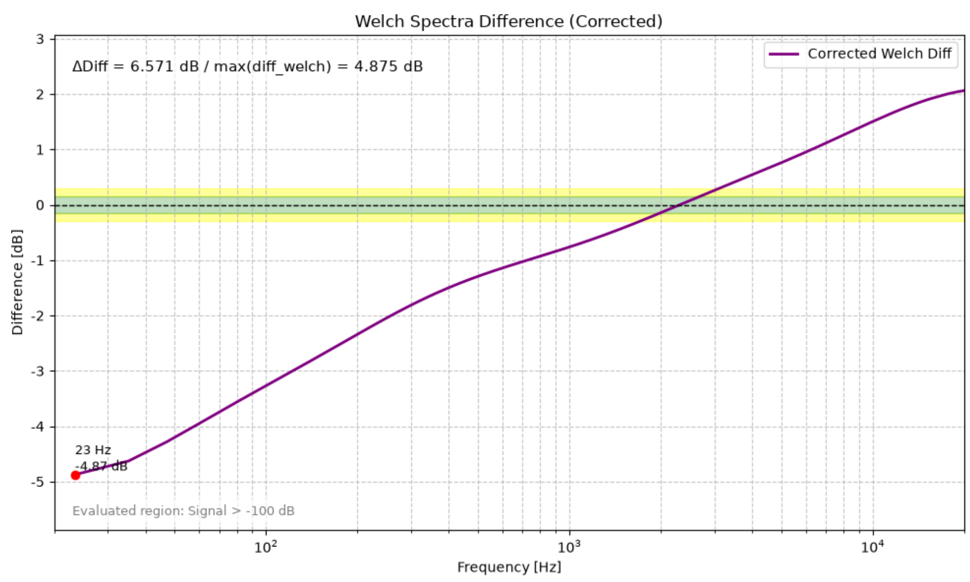


The first dataset shows the characteristics of the DUT alone. These results were obtained by analyzing a file created by applying gain and EQ settings to the base signal in a DAW and mixing it in stereo.

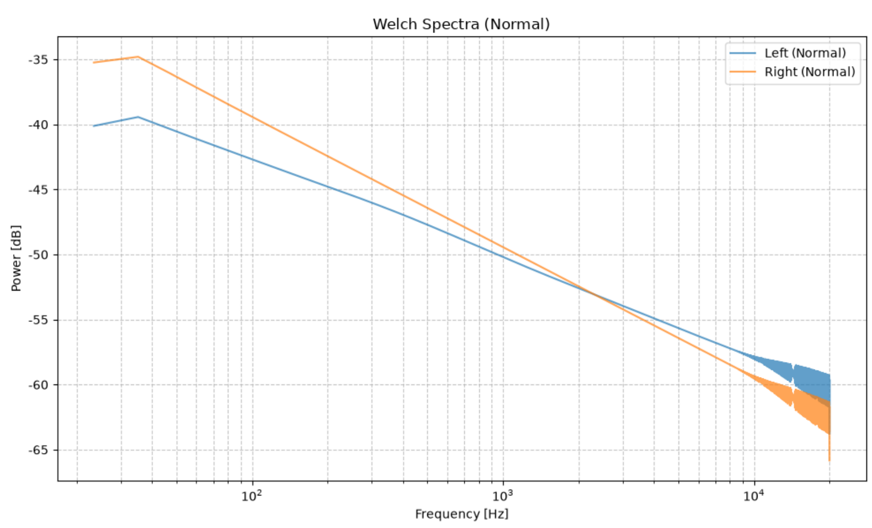
The original L/R signals are at -10 dBFS and have been swept over a 180-second period from 20 Hz to 20 kHz. An EQ setting has been applied to the L signal, and a gain setting has been applied to the R signal.

The following three graphs show the results of an analysis of the Left-Right differences in the DUT characteristics.

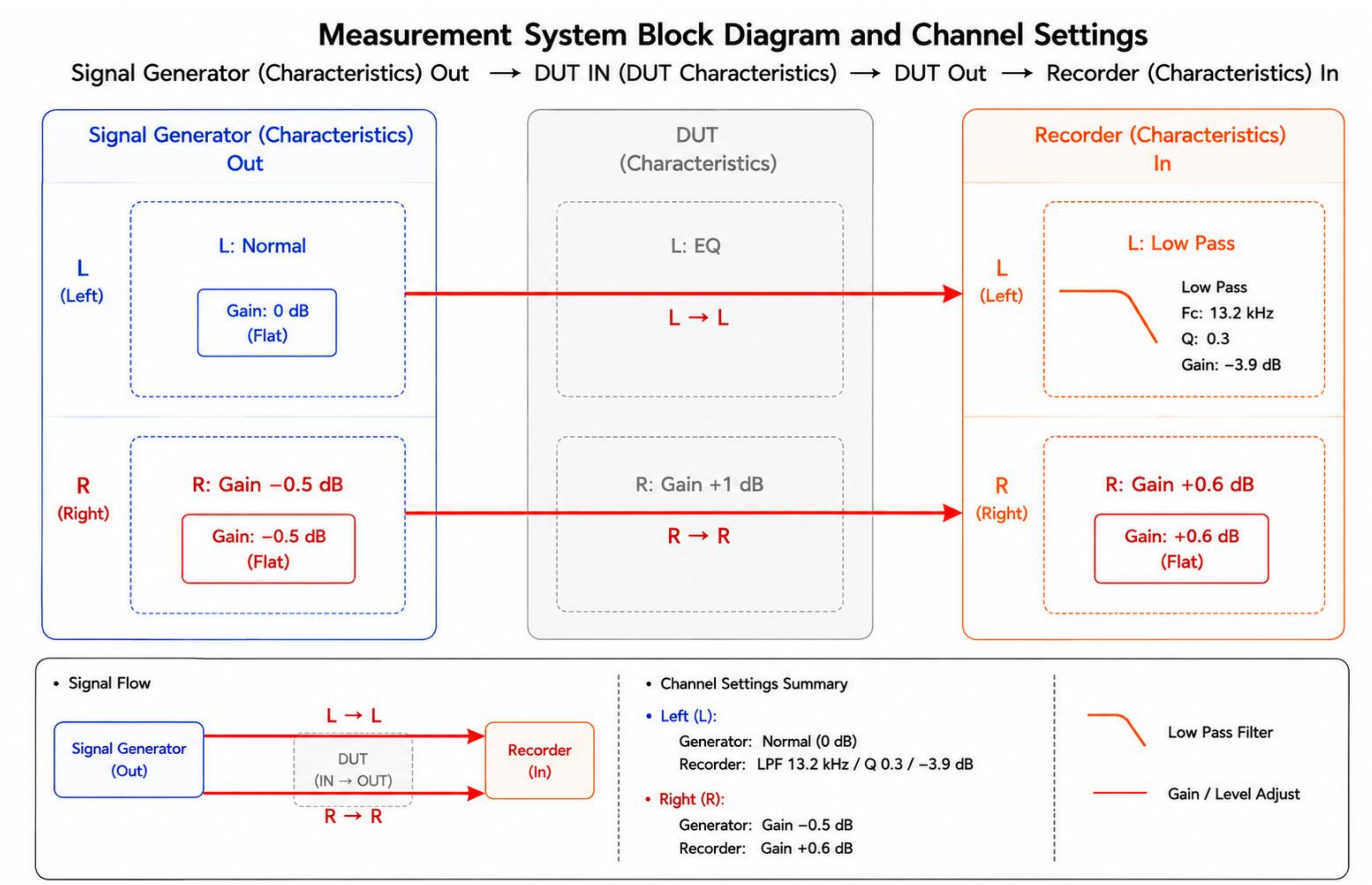




For this isolated simulation, the reverse file was created by swapping the L and R channels of the normal file.



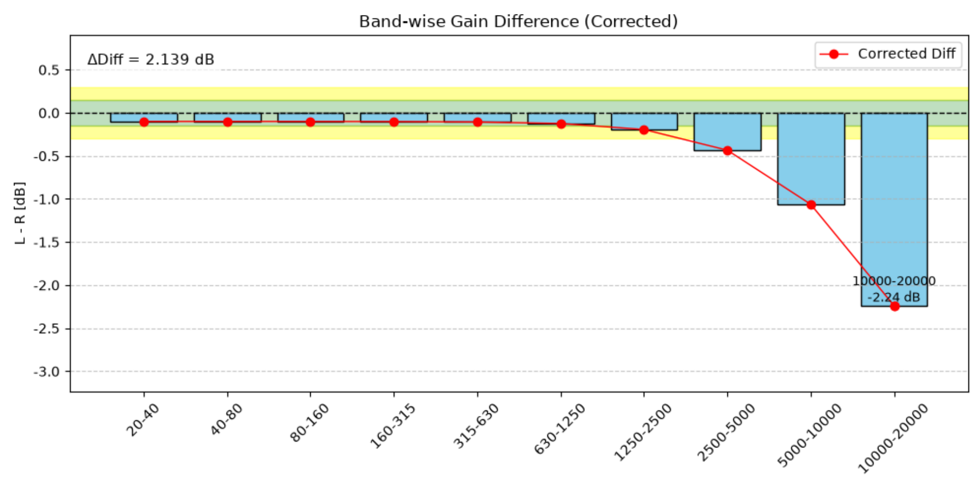
This section simulates the characteristics of the measurement system alone.

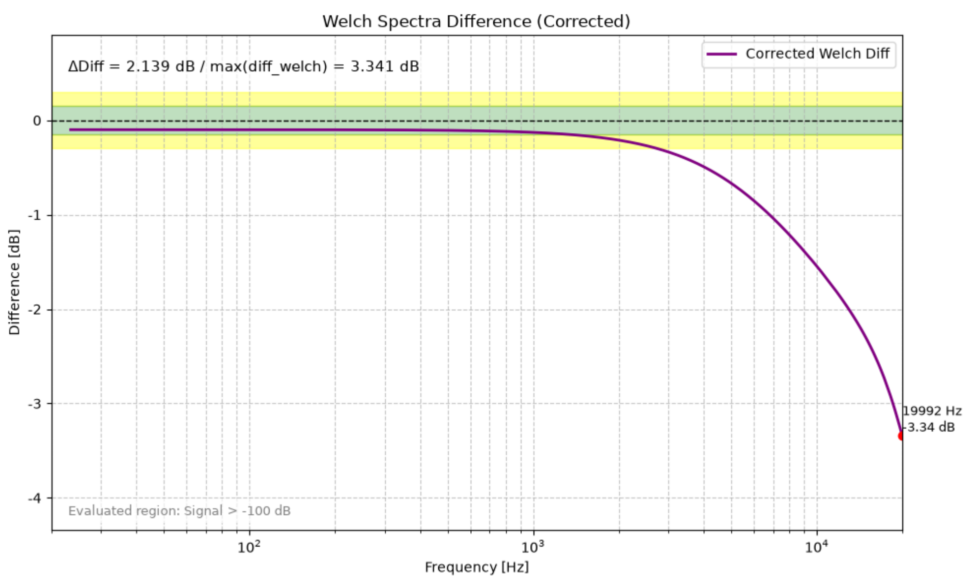


This dataset presents the results of a simulation of the characteristics of a single measurement system. Simulation files were created using generator tracks with their respective characteristics set in DAW and recorder tracks that received the output from those generator tracks. Files created by swapping the L and R channels of these were analyzed as reverse connect signals.

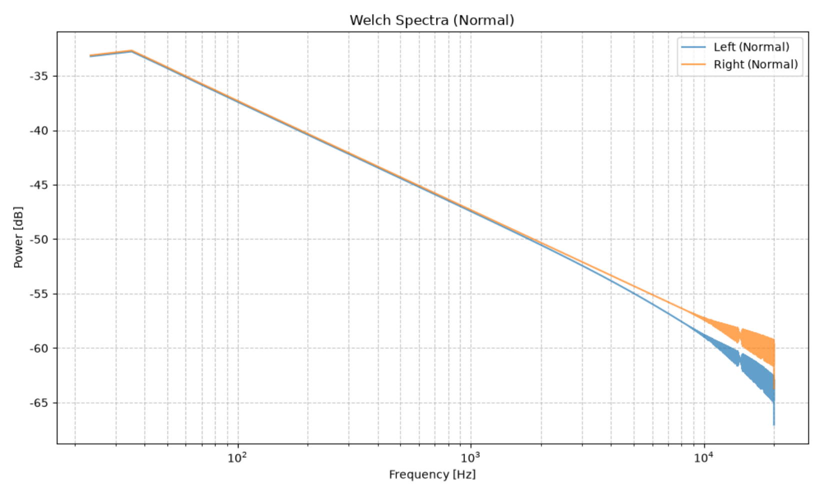
The original L/R signals are at -10 dBFS, and a 180-second sweep is performed from 20 Hz to 20 kHz. Each block has its own gain and EQ settings.

The following three graphs show the results of an analysis of the Left-Right differences in the combined characteristics of the generator and the recorder device.

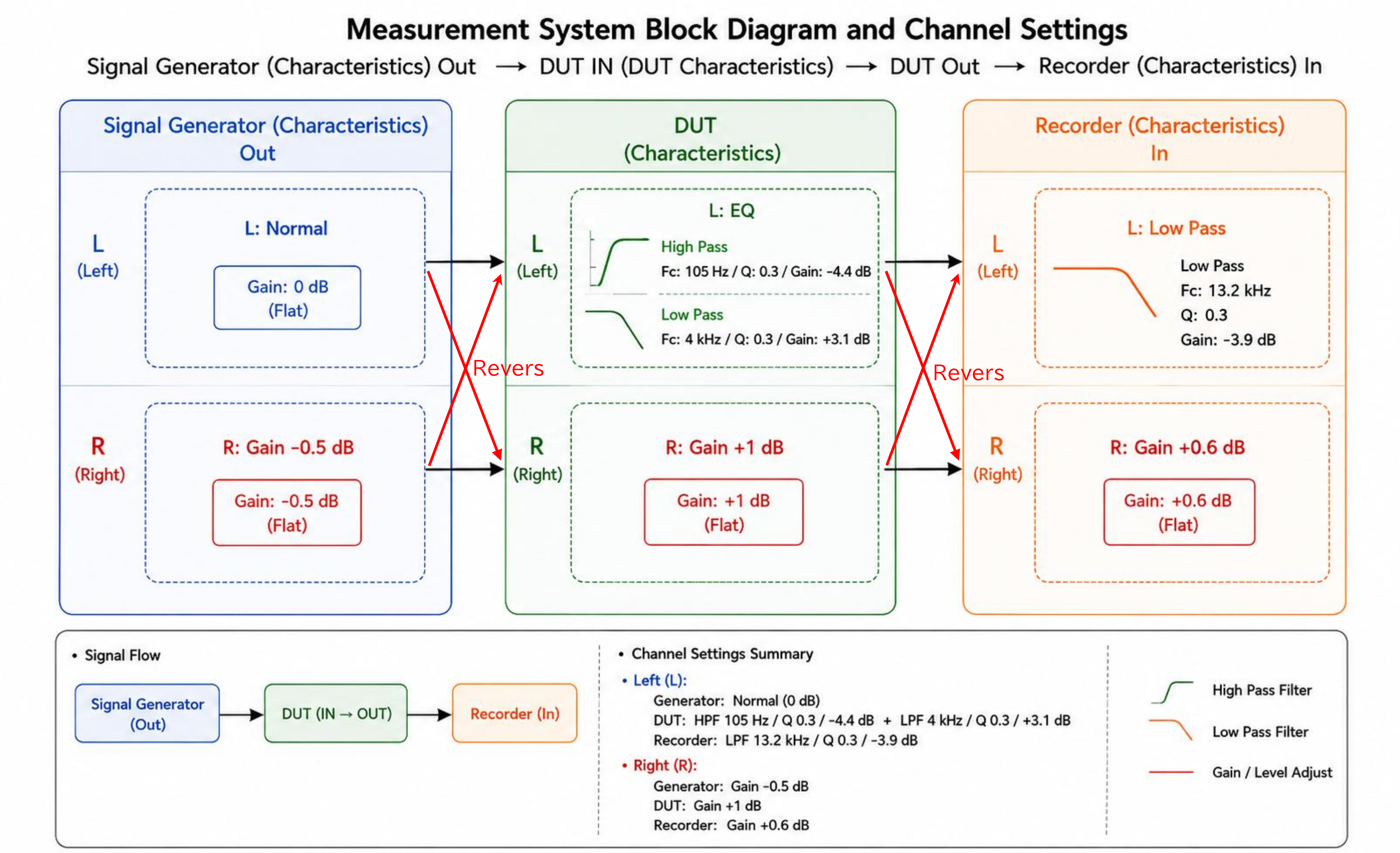




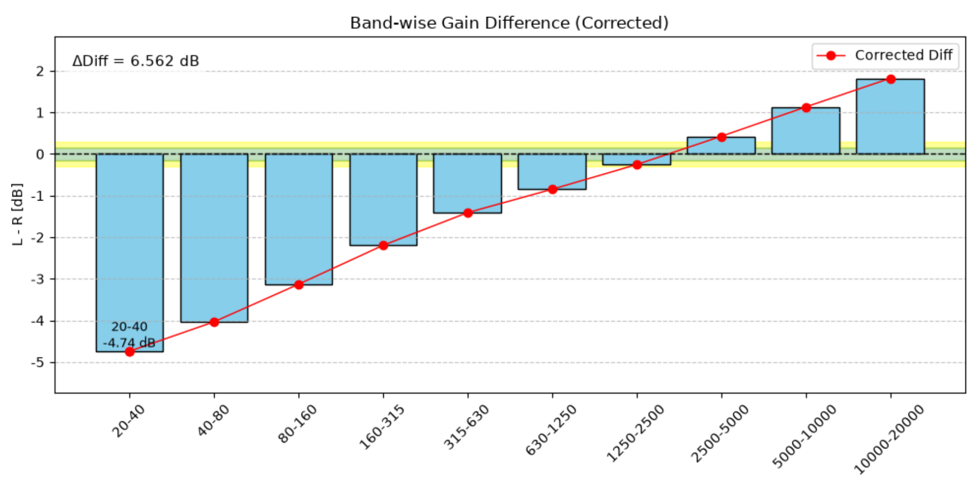
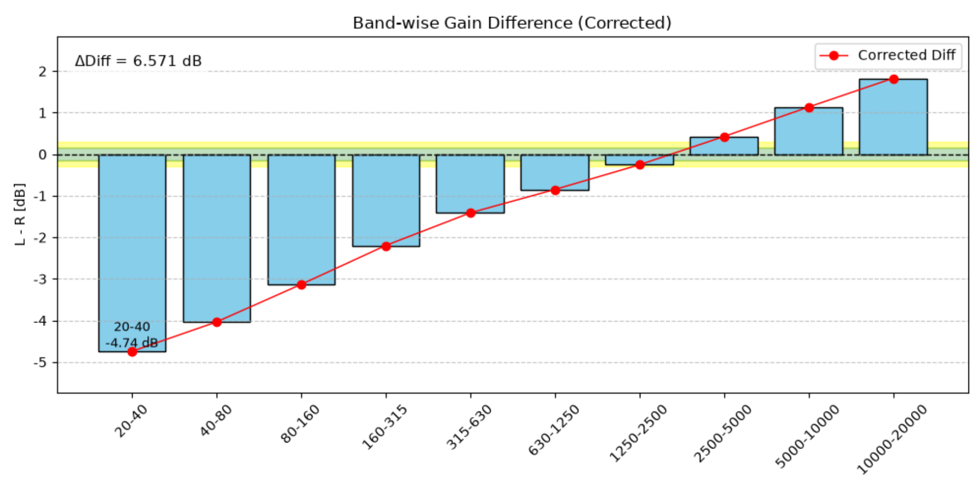
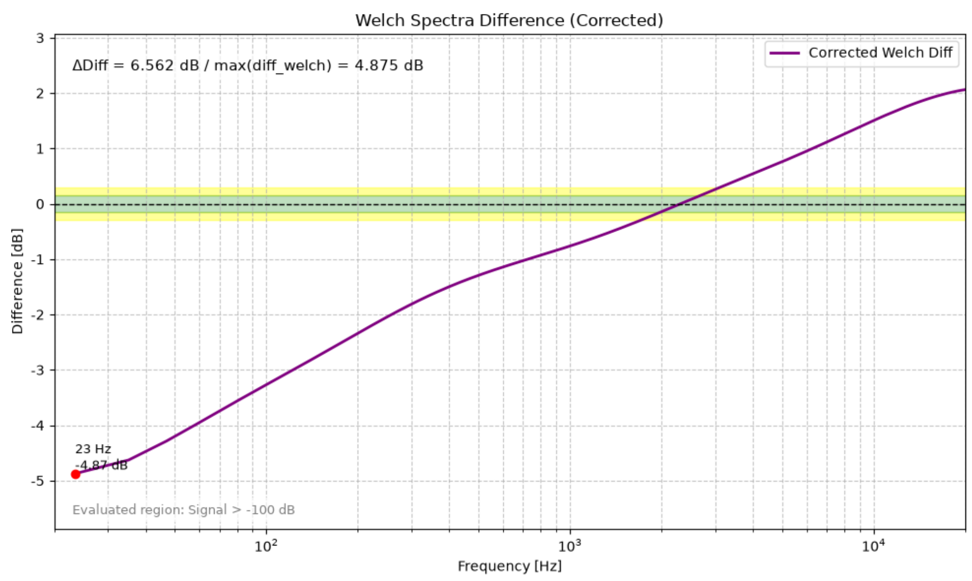
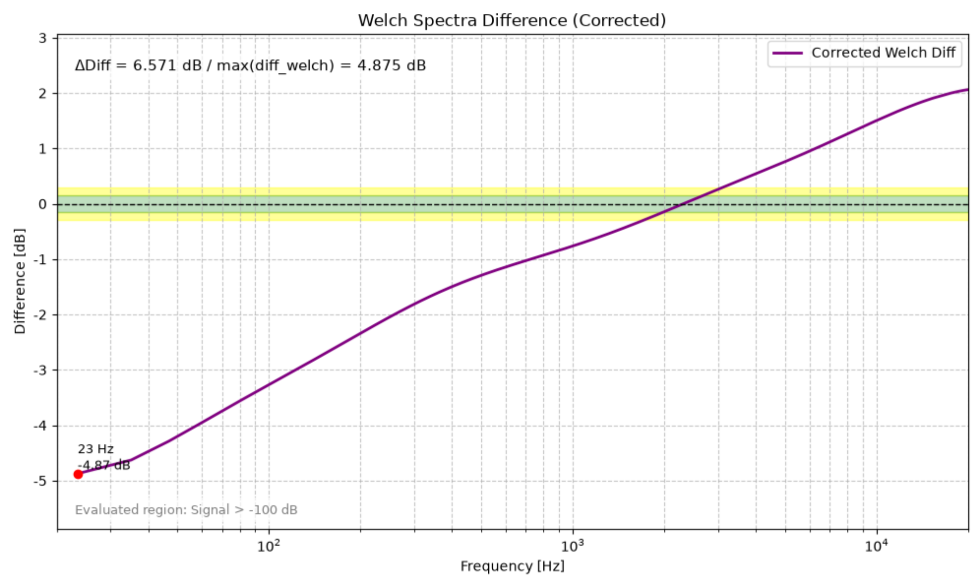
For this isolated simulation, the reverse file was created by swapping the L and R channels of the normal file.

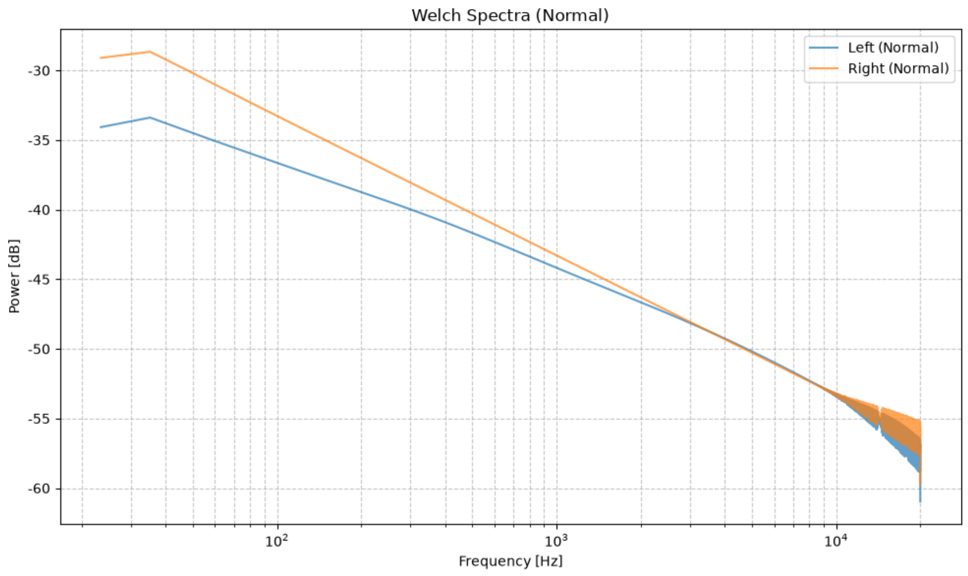
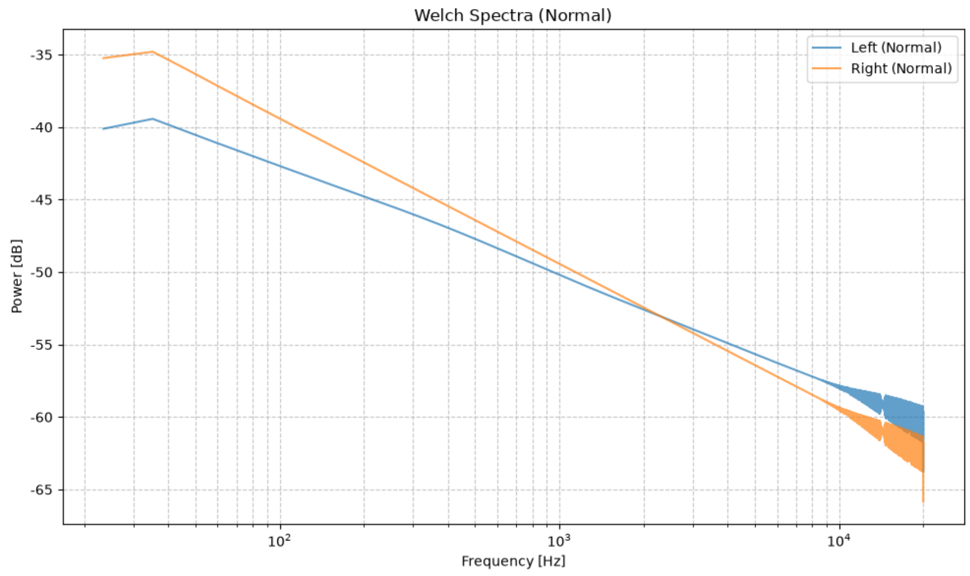
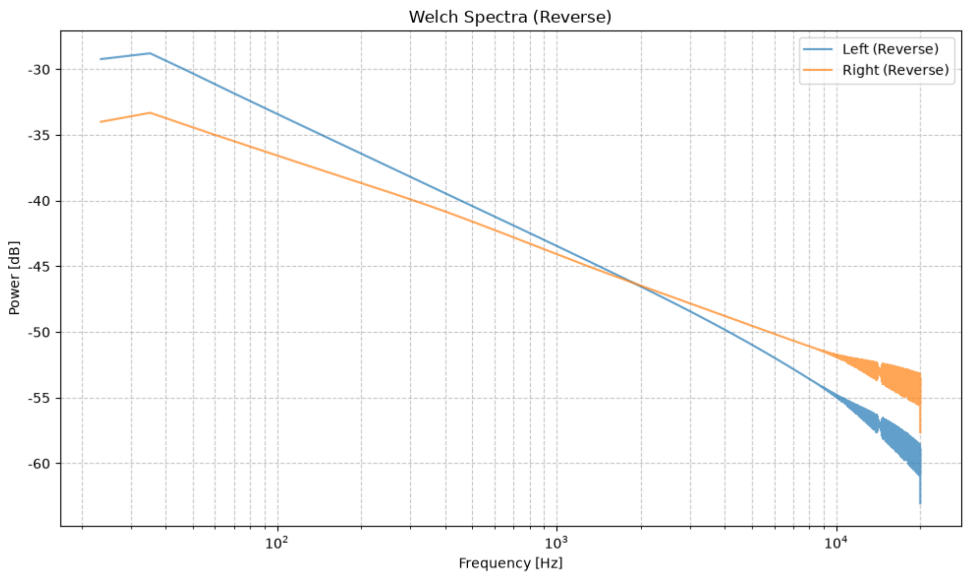
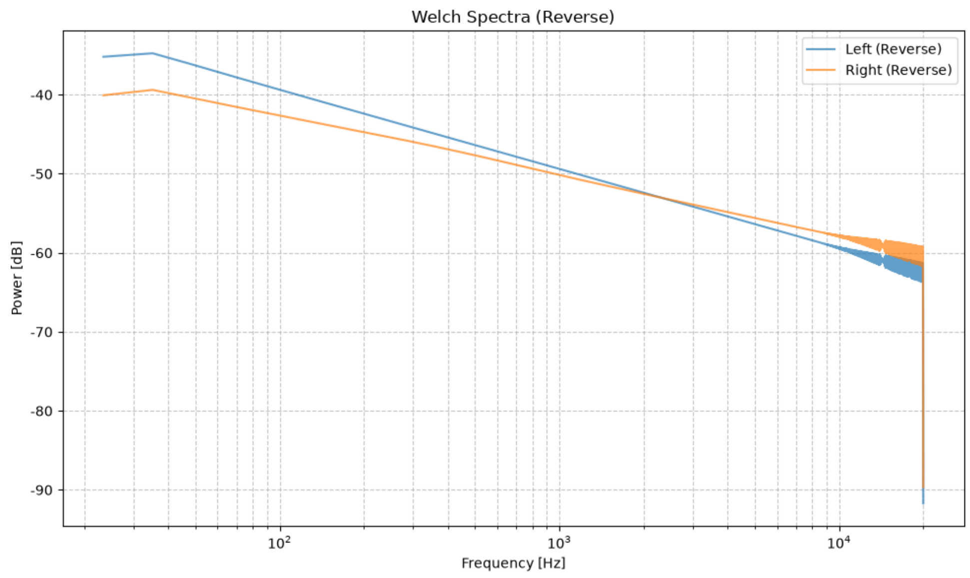


The final section shows the results of calculations that cancel out factors other than the DUT characteristics, starting from a state where the characteristics of the measurement system and those of the DUT overlap.



In the final dataset, we acquired signals for both normal and reverse connections under the combined characteristics of the generator, the DUT, and the recorder. By calculating (normal − reverse)/2 from these two sets of data, we confirmed that characteristics other than those of the DUT were canceled out.

Result of (Normal - Reverse) / 2 for Generator + DUT + Recorder	Result of DUT-Only Characteristics
	
	

<table>
<tr><td>Result of (Normal - Reverse) / 2 for Generator + DUT + Recorder</td><td>Result of DUT-Only Characteristics</td></tr>
<tr><td>Signal spectrum after passing through each block in a Normal Connection

</td><td>Spectral Characteristics of the DUT Under Normal Connection

</td></tr>
<tr><td>Signal spectrum after passing through each block via a reverse connection

</td><td>For this isolated simulation, the reverse file was created by swapping the L and R channels of the normal file.

</td></tr>
<tr><td>Summary.txt
Corrected Gain Difference Analysis

20-40 Hz: Diff=-4.744 dB
40-80 Hz: Diff=-4.036 dB
80-160 Hz: Diff=-3.133 dB
160-315 Hz: Diff=-2.198 dB
315-630 Hz: Diff=-1.412 dB
630-1250 Hz: Diff=-0.850 dB
1250-2500 Hz: Diff=-0.257 dB
2500-5000 Hz: Diff=0.422 dB
5000-10000 Hz: Diff=1.128 dB
10000-20000 Hz: Diff=1.818 dB

--- Summary ---
ΔDiff (Band) = 6.562 dB
ΔDiff (Welch max) = 4.875 dB
Max Diff Freq = 23.4 Hz
Judgement: Check required: Possible equipment or measurement problem.

--- Channel Check ---
Energy Ratio (L/R) = 0.71
Channel balance is within the normal range.</td><td>Summary.txt
Corrected Gain Difference Analysis

20-40 Hz: Diff=-4.744 dB
40-80 Hz: Diff=-4.036 dB
80-160 Hz: Diff=-3.133 dB
160-315 Hz: Diff=-2.198 dB
315-630 Hz: Diff=-1.412 dB
630-1250 Hz: Diff=-0.849 dB
1250-2500 Hz: Diff=-0.256 dB
2500-5000 Hz: Diff=0.425 dB
5000-10000 Hz: Diff=1.134 dB
10000-20000 Hz: Diff=1.826 dB

--- Summary ---
ΔDiff (Band) = 6.571 dB
ΔDiff (Welch max) = 4.875 dB
Max Diff Freq = 23.4 Hz
Judgement: Check required: Possible equipment or measurement problem.

--- Channel Check ---
Energy Ratio (L/R) = 0.82
Channel balance is within the normal range.</td></tr>
</table>

The corrected result closely matches the DUT-only result, confirming that the channel differences introduced by the simulated measurement system are effectively cancelled.

The following results are from a simulation of how the script makes decisions when the Left and Right signals are identical or when some discrepancy occurs.

Responses to the following simulation signals have been verified using the stereo bandwidth gain difference analyzer.

******* Test signals generated using Python.**
-10 dBFS 1 kHz Sine Wave, Symmetrical
-10 dBFS 1 kHz Sine Wave, 1% THD (Left)
-10 dBFS 1 kHz Sine Wave, 3% THD (Left)
-10 dBFS 1 kHz Sine Wave, 5% THD (Left)
-10 dBFS 1 kHz Sine Wave, Gain -0.3 dB (Left)
-10 dBFS 1 kHz Sine Wave, 1% Noise (Left)
-10 dBFS 1 kHz Sine Wave, 3% Noise (Left)
-10 dBFS 1 kHz Sine Wave, 5% Noise (Left)
-10 dBFS 1 kHz Sine Wave, PEQ +1 dB (Left)
-10 dBFS 1 kHz Sine Wave, HSF 5 kHz -1 dB (Left)
-10 dBFS 1 kHz Sine Wave, Gain -0.3 dB (Right)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Symmetrical
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% Noise (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% Noise (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% Noise (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, PEQ 125 Hz Q10 +1 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, HSF 5 kHz -1 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Right)

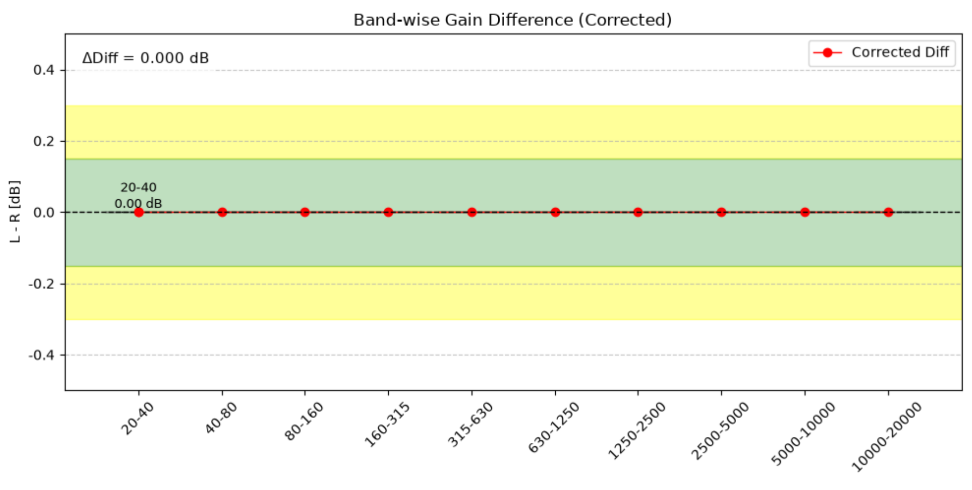
******** Test signals generated and processed using WaveGene and Cakewalk SONAR + GClip**
-10 dBFS 1 kHz Sine Wave, 1% THD (Left)
-10 dBFS 1 kHz Sine Wave, 3% THD (Left)
-10 dBFS 1 kHz Sine Wave, 5% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% THD (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, HSF 5 kHz -1 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, PEQ 125 Hz Q10 +1 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, WhiteNoise Mix -26 dB (Left)
-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Symmetrical

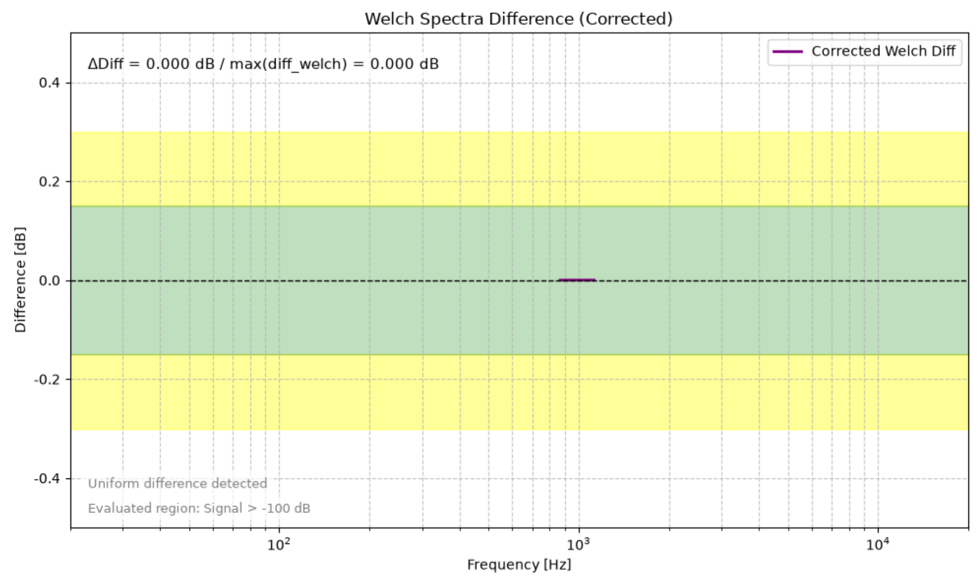
Graphs and summaries of the respective analysis results are presented on the following pages.

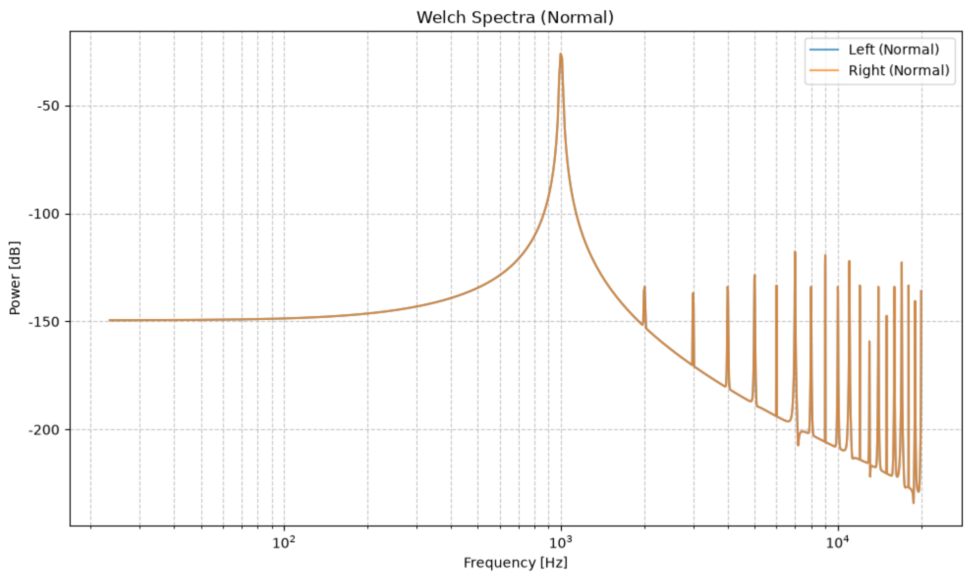
Test signals generated using Python.

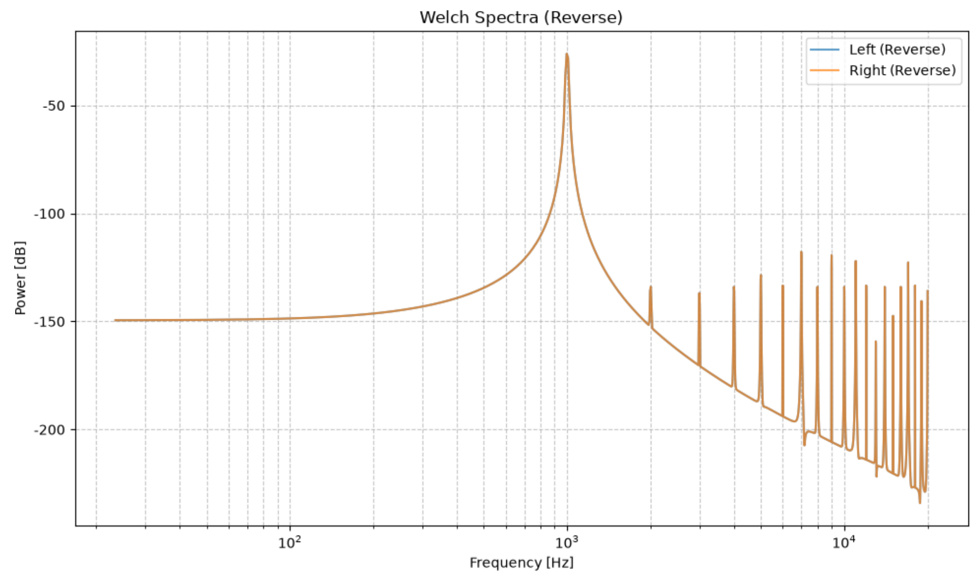
-10 dBFS 1 kHz Sine Wave, Symmetrical

For these single-tone simulations, Band-analysis values outside the frequency regions containing meaningful signal energy should not be interpreted as broadband DUT characteristics. They are included here to show the script's behavior with narrowband input signals.









Summary.txt

Corrected Gain Difference Analysis

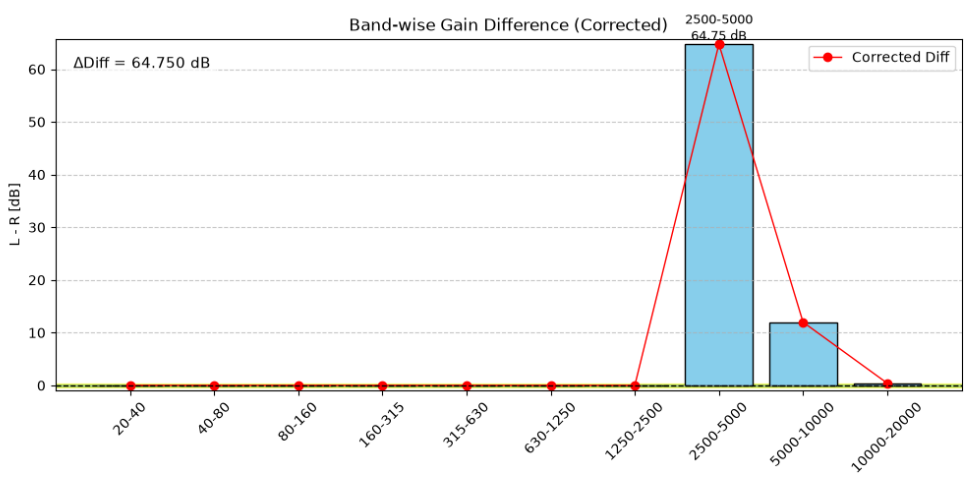
20-40 Hz: Diff=0.000 dB
40-80 Hz: Diff=0.000 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.000 dB
5000-10000 Hz: Diff=0.000 dB
10000-20000 Hz: Diff=0.000 dB

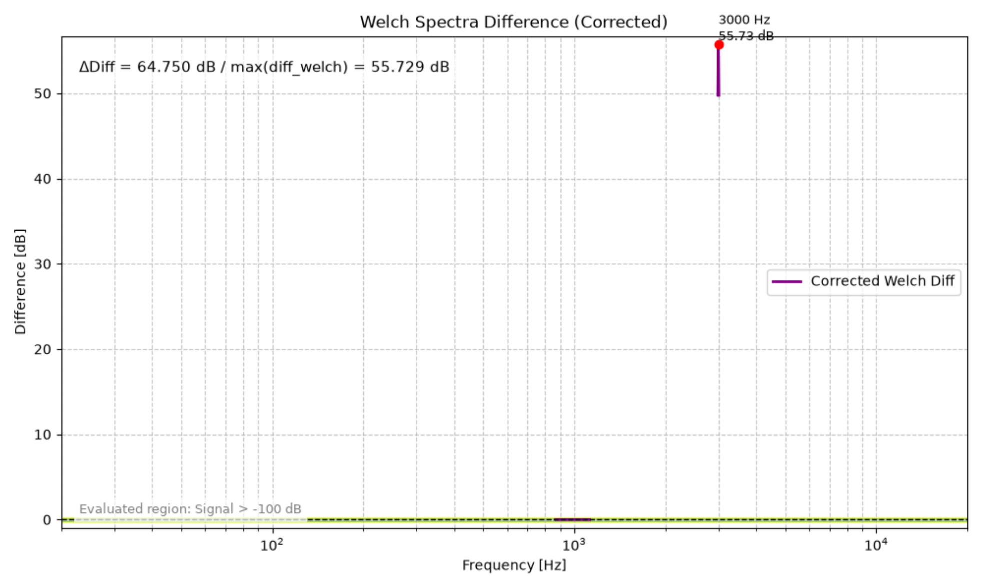
--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.000 dB
Judgement: OK: Channel difference is within measurement error.

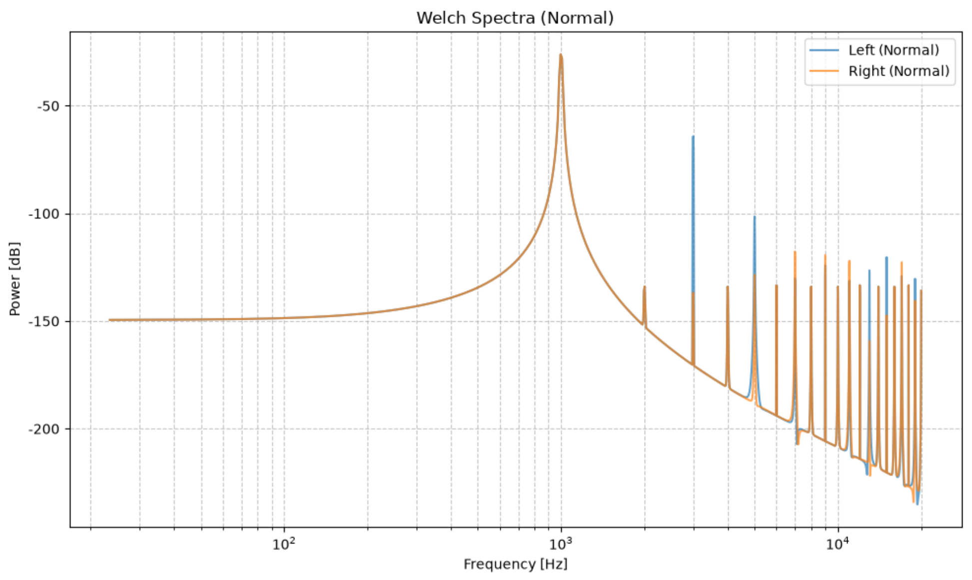
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

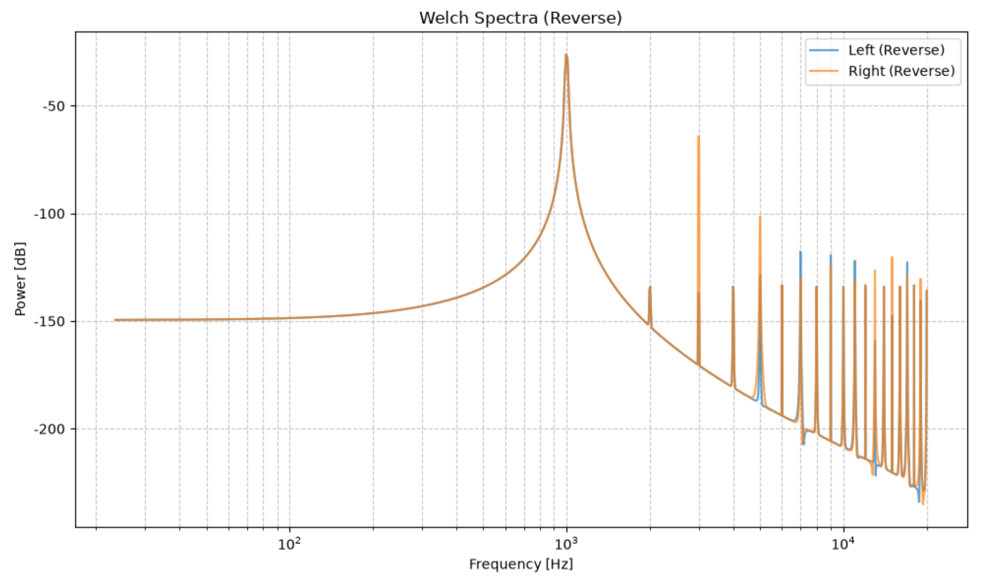
-10 dBFS 1 kHz Sine Wave, 1% THD (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device’s bandwidth characteristics. It was used to evaluate the script’s decision-making logic for narrowband signals.









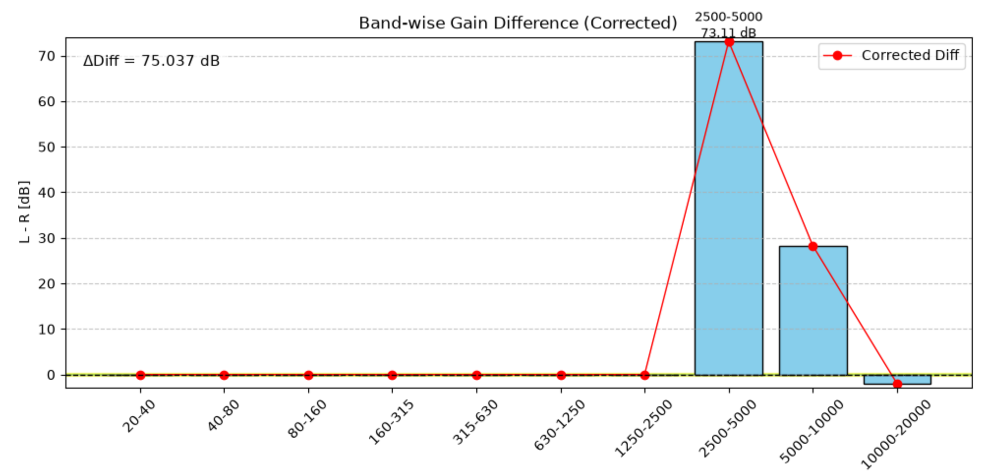
Summary.txt

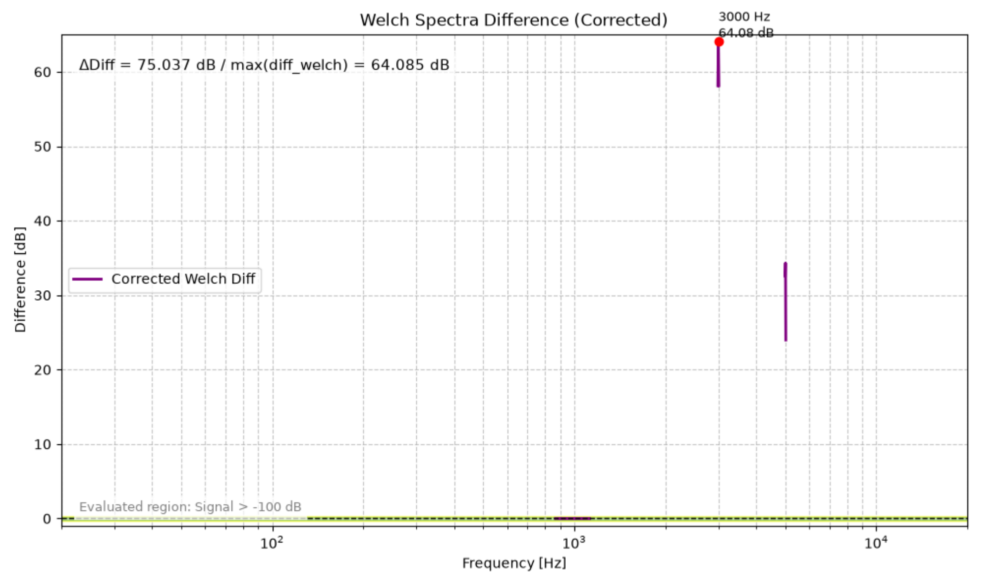
Corrected Gain Difference Analysis

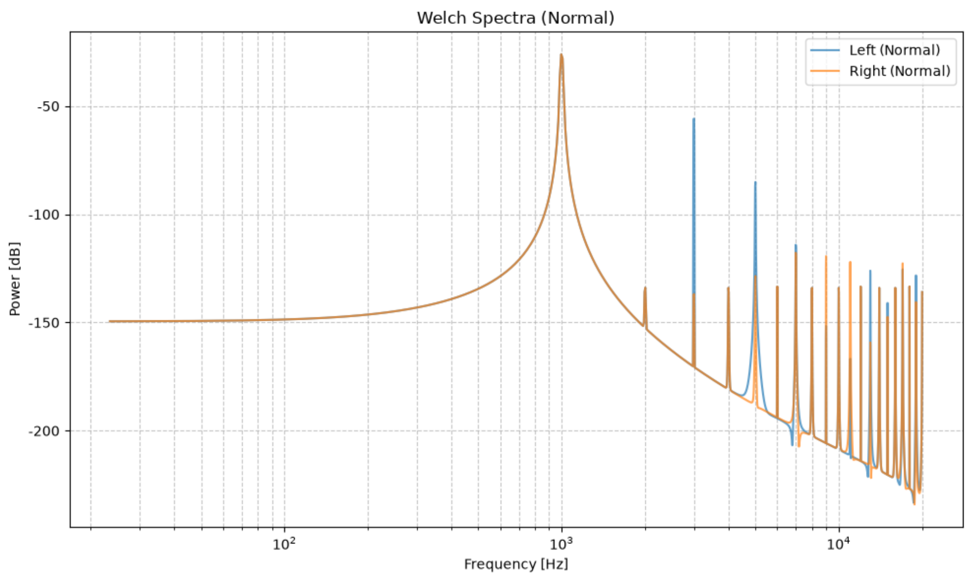
20-40 Hz: Diff=-0.001 dB
40-80 Hz: Diff=-0.001 dB
80-160 Hz: Diff=-0.001 dB
160-315 Hz: Diff=-0.001 dB
315-630 Hz: Diff=-0.001 dB
630-1250 Hz: Diff=-0.001 dB
1250-2500 Hz: Diff=-0.001 dB
2500-5000 Hz: Diff=64.749 dB
5000-10000 Hz: Diff=12.001 dB
10000-20000 Hz: Diff=0.429 dB

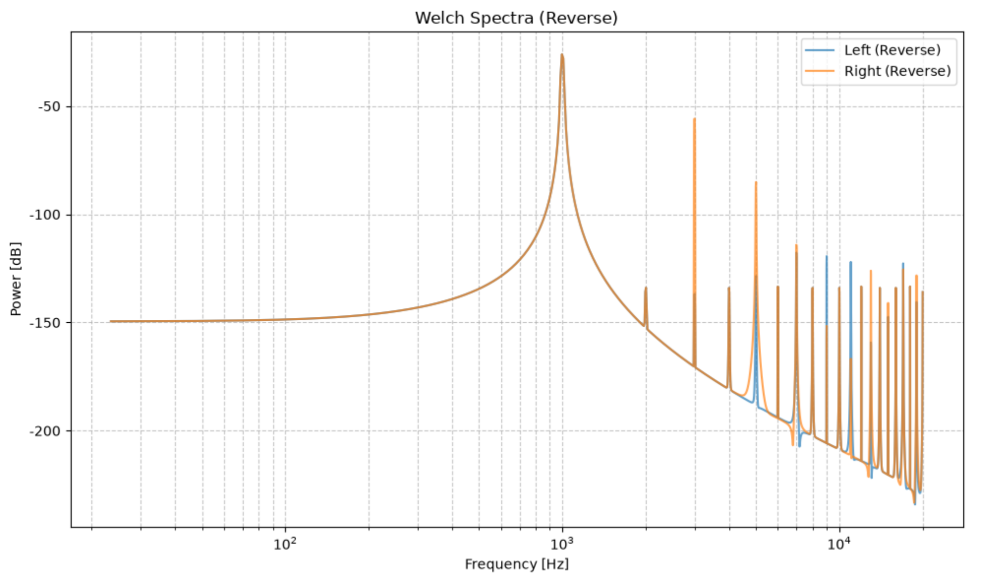
--- Summary ---
ΔDiff (Band) = 64.750 dB
ΔDiff (Welch max) = 55.729 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.
The automatic Judgement is a simple screening aid based on numerical thresholds. It does not identify the physical cause of a detected difference, and its interpretation depends on the test signal and measurement conditions.
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, 3% THD (Left)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

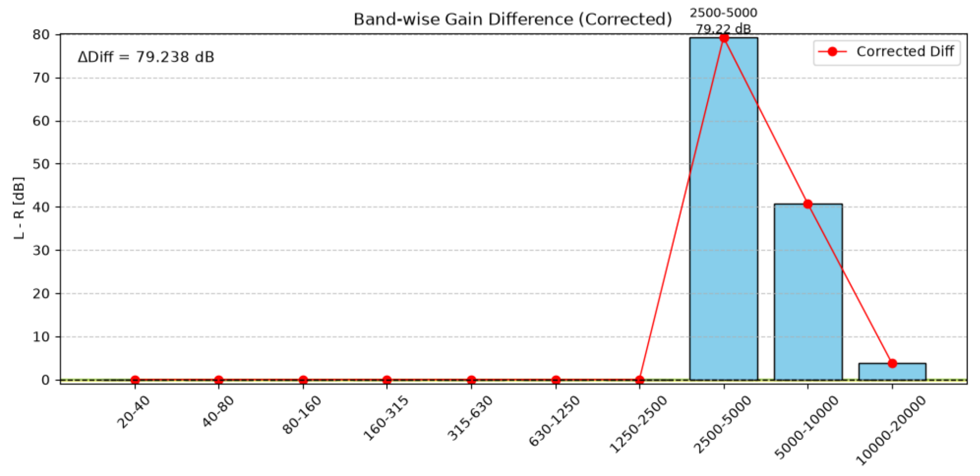
Corrected Gain Difference Analysis

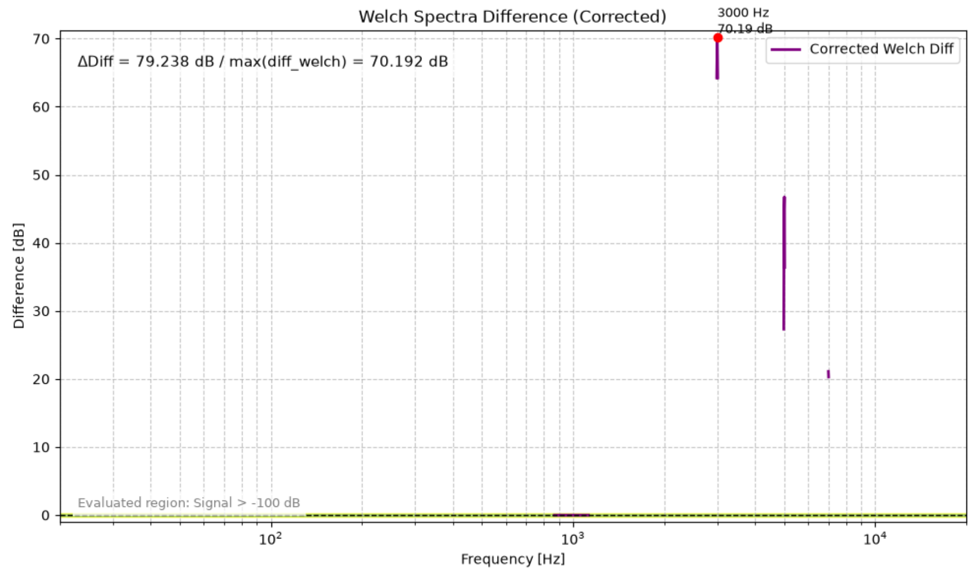
20-40 Hz: Diff=-0.004 dB
40-80 Hz: Diff=-0.004 dB
80-160 Hz: Diff=-0.004 dB
160-315 Hz: Diff=-0.004 dB
315-630 Hz: Diff=-0.004 dB
630-1250 Hz: Diff=-0.004 dB
1250-2500 Hz: Diff=-0.004 dB
2500-5000 Hz: Diff=73.107 dB
5000-10000 Hz: Diff=28.291 dB
10000-20000 Hz: Diff=-1.930 dB

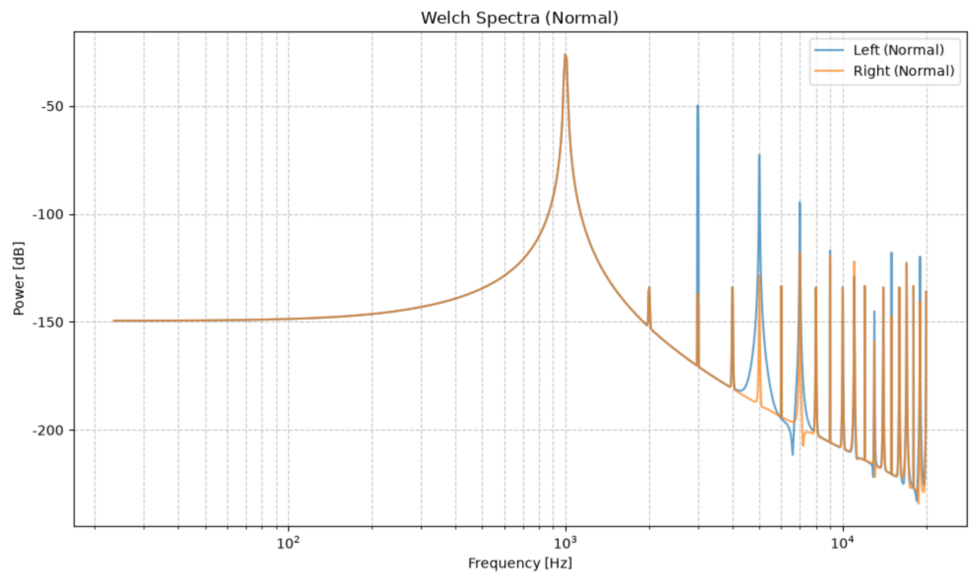
--- Summary ---
ΔDiff (Band) = 75.037 dB
ΔDiff (Welch max) = 64.085 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.

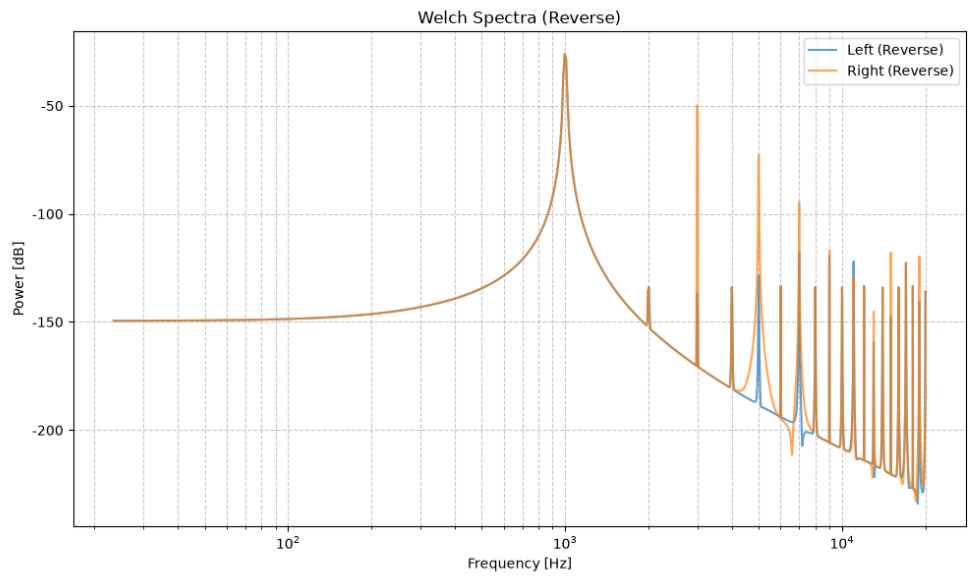
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, 5% THD (Left)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

Corrected Gain Difference Analysis

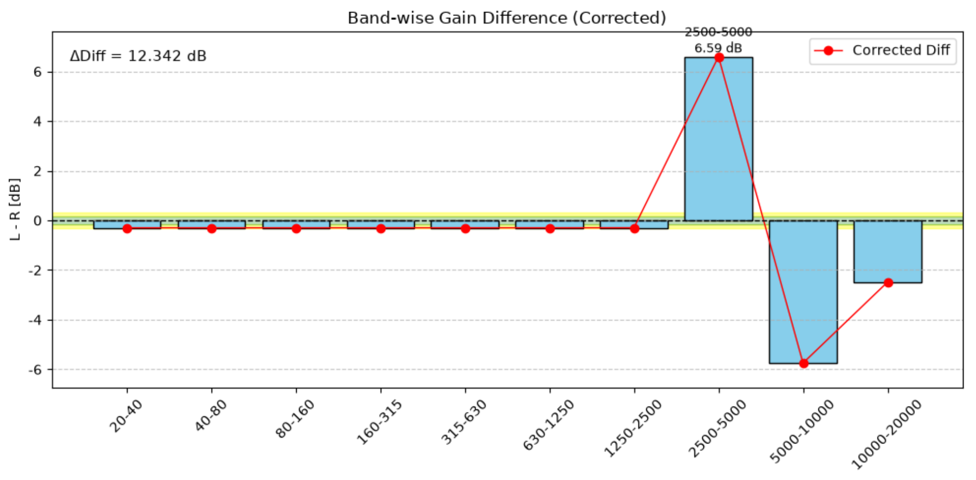
20-40 Hz: Diff=-0.016 dB
40-80 Hz: Diff=-0.016 dB
80-160 Hz: Diff=-0.016 dB
160-315 Hz: Diff=-0.016 dB
315-630 Hz: Diff=-0.016 dB
630-1250 Hz: Diff=-0.016 dB
1250-2500 Hz: Diff=-0.016 dB
2500-5000 Hz: Diff=79.222 dB
5000-10000 Hz: Diff=40.753 dB
10000-20000 Hz: Diff=3.819 dB

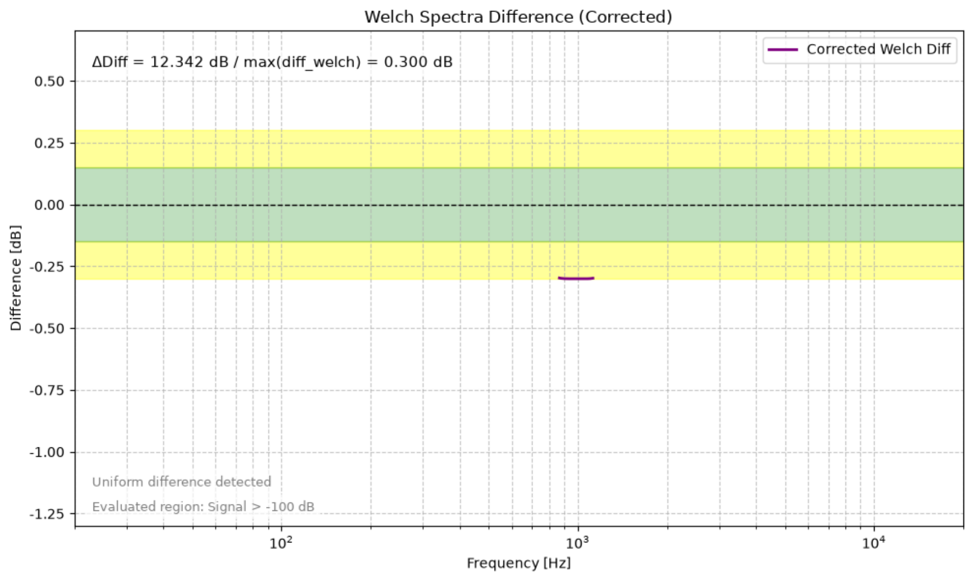
--- Summary ---
ΔDiff (Band) = 79.238 dB
ΔDiff (Welch max) = 70.192 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.

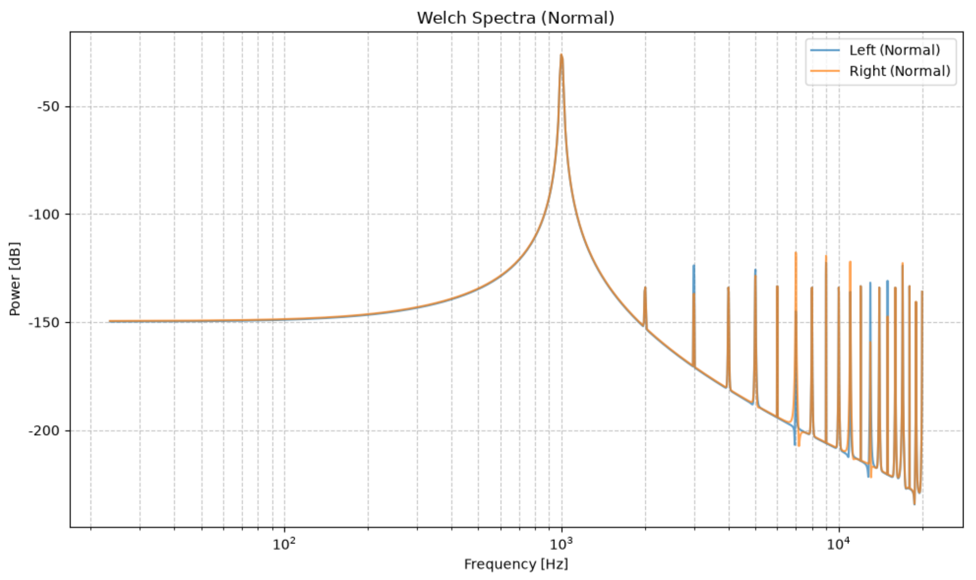
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

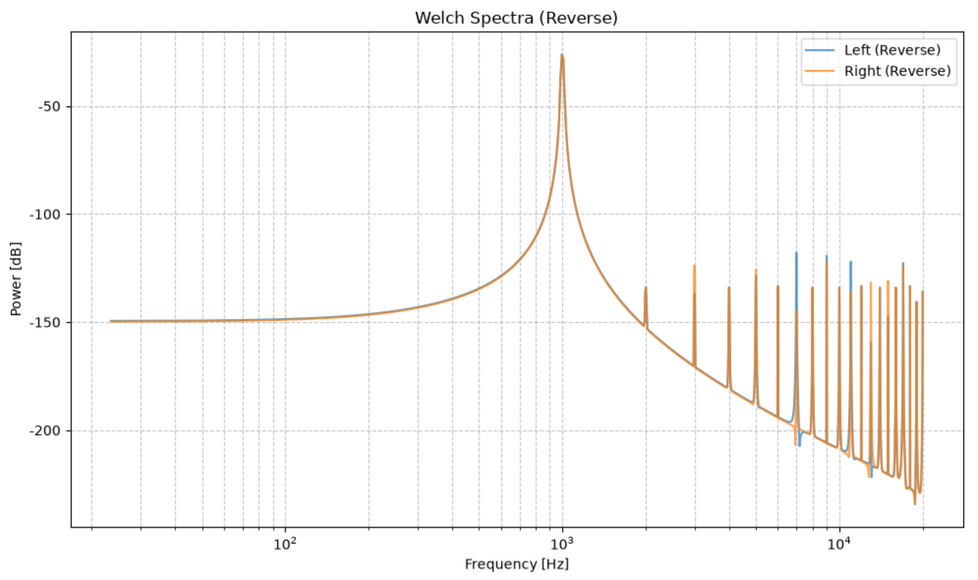
-10 dBFS 1 kHz Sine Wave, Gain -0.3 dB (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device’s bandwidth characteristics. It was used to evaluate the script’s decision-making logic for narrowband signals.









Summary.txt

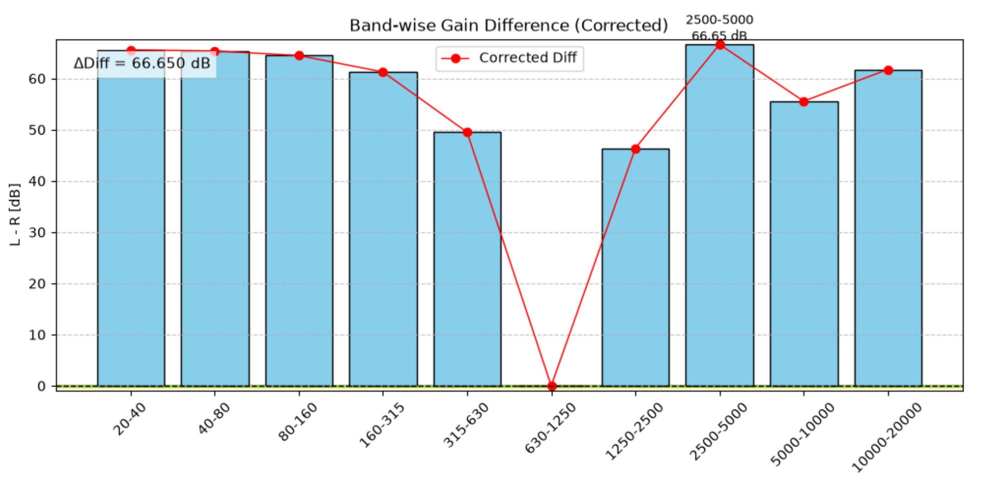
Corrected Gain Difference Analysis

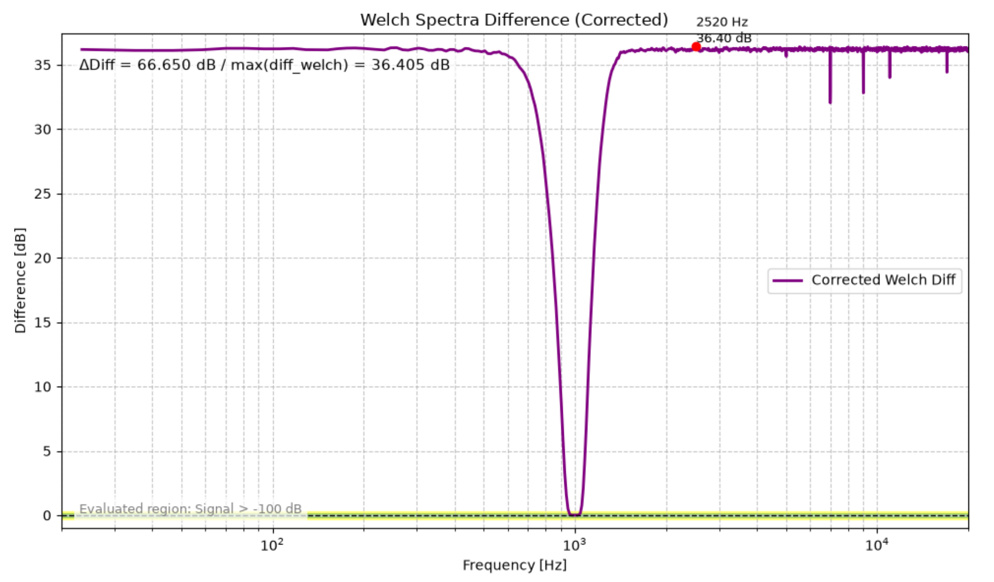
20-40 Hz: Diff=-0.300 dB
40-80 Hz: Diff=-0.300 dB
80-160 Hz: Diff=-0.300 dB
160-315 Hz: Diff=-0.300 dB
315-630 Hz: Diff=-0.300 dB
630-1250 Hz: Diff=-0.300 dB
1250-2500 Hz: Diff=-0.298 dB
2500-5000 Hz: Diff=6.595 dB
5000-10000 Hz: Diff=-5.748 dB
10000-20000 Hz: Diff=-2.487 dB

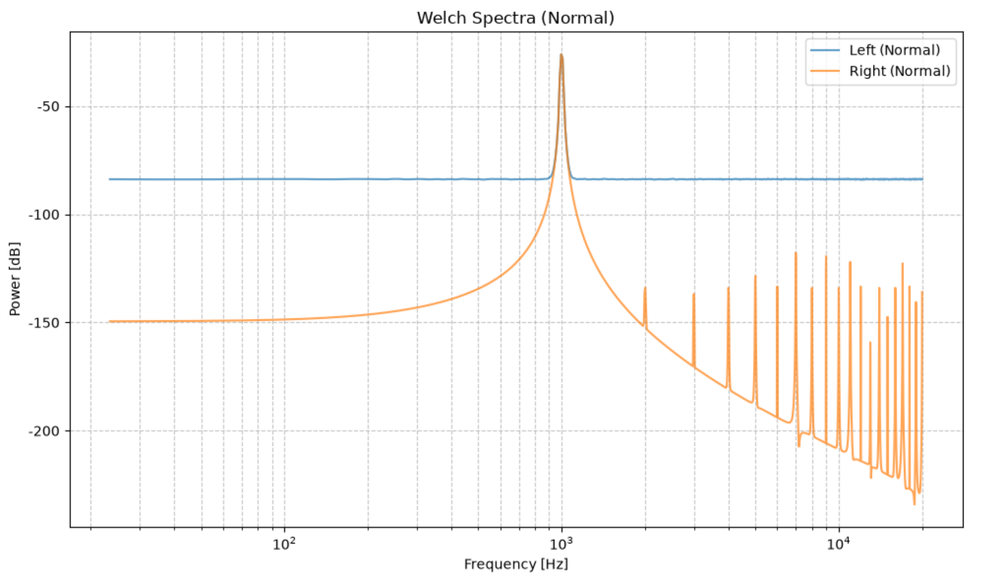
--- Summary ---
ΔDiff (Band) = 12.342 dB
ΔDiff (Welch max) = 0.300 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

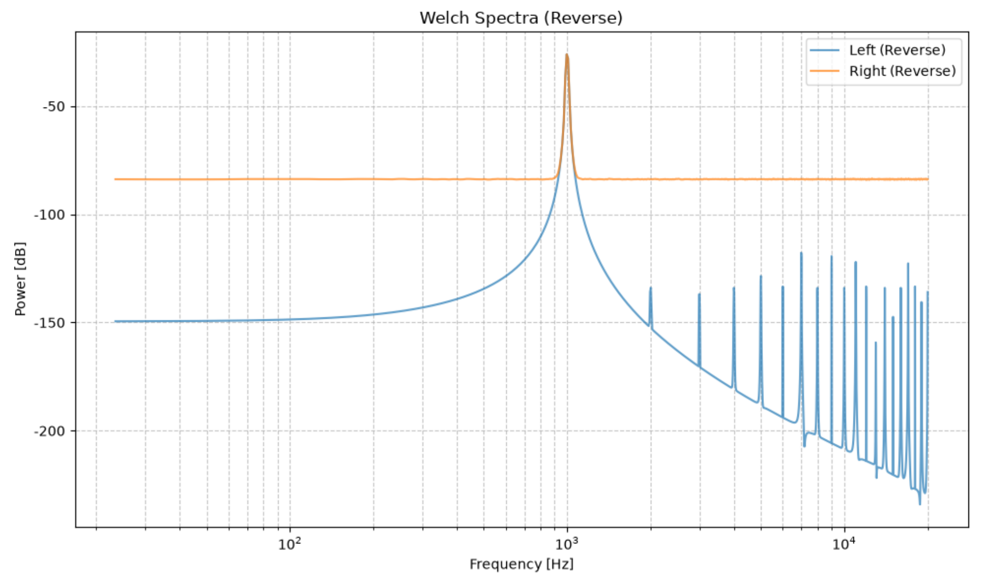
--- Channel Check ---
Energy Ratio (L/R) = 0.93
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, 1% Noise (Left)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device’s bandwidth characteristics. It was used to evaluate the script’s decision-making logic for narrowband signals.









Summary.txt

Corrected Gain Difference Analysis

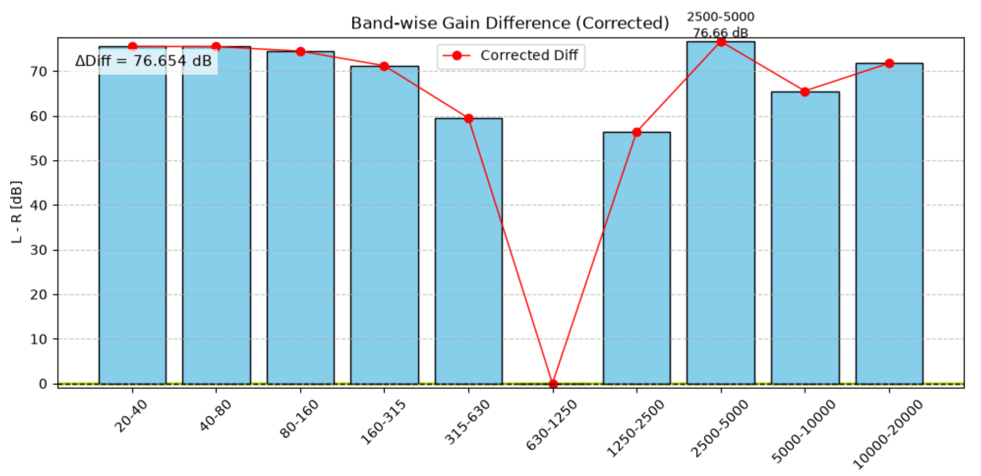
20-40 Hz: Diff=65.574 dB
40-80 Hz: Diff=65.374 dB
80-160 Hz: Diff=64.479 dB
160-315 Hz: Diff=61.275 dB
315-630 Hz: Diff=49.507 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=46.372 dB
2500-5000 Hz: Diff=66.651 dB
5000-10000 Hz: Diff=55.554 dB
10000-20000 Hz: Diff=61.761 dB

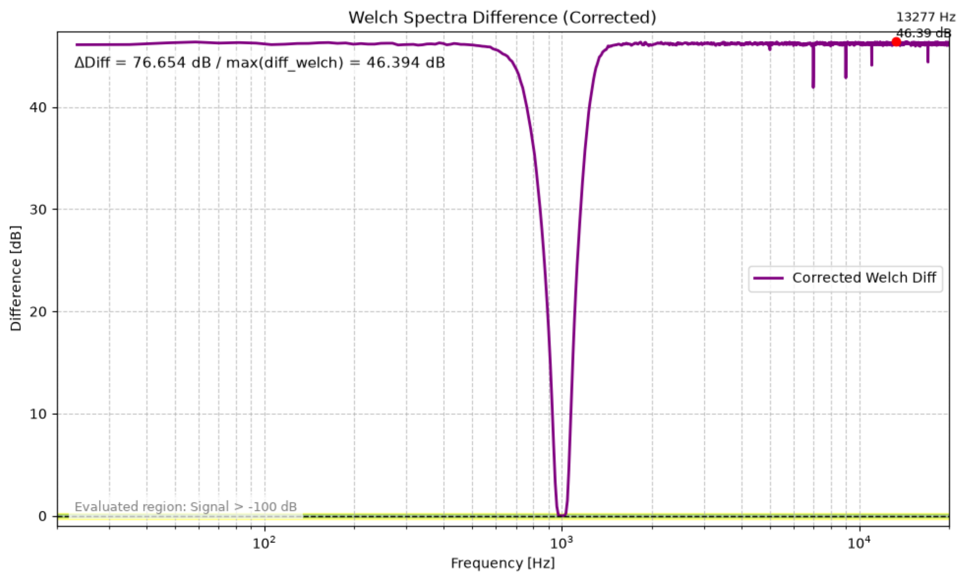
--- Summary ---
ΔDiff (Band) = 66.650 dB
ΔDiff (Welch max) = 36.405 dB
Max Diff Freq = 2519.5 Hz
Judgement: Check required: Possible equipment or measurement problem.

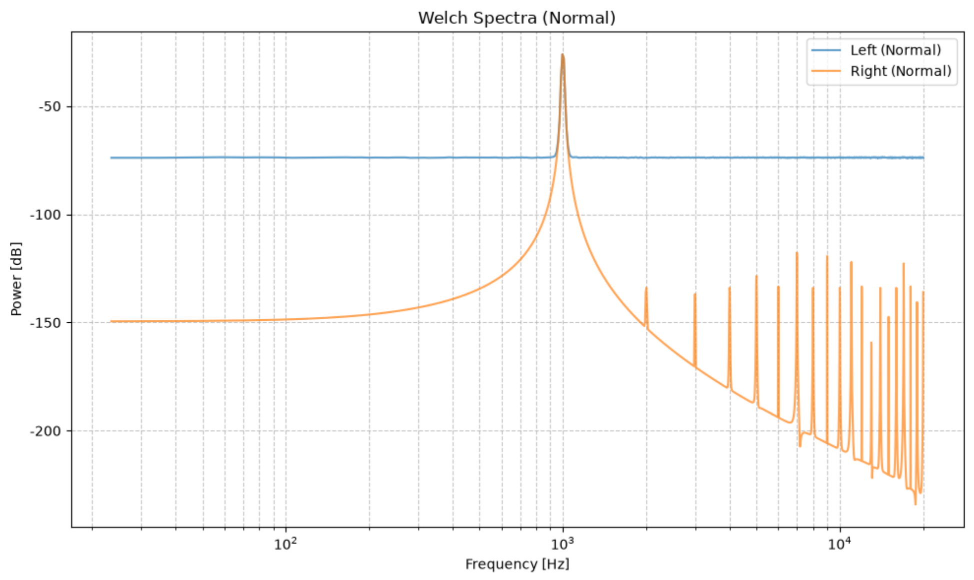
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

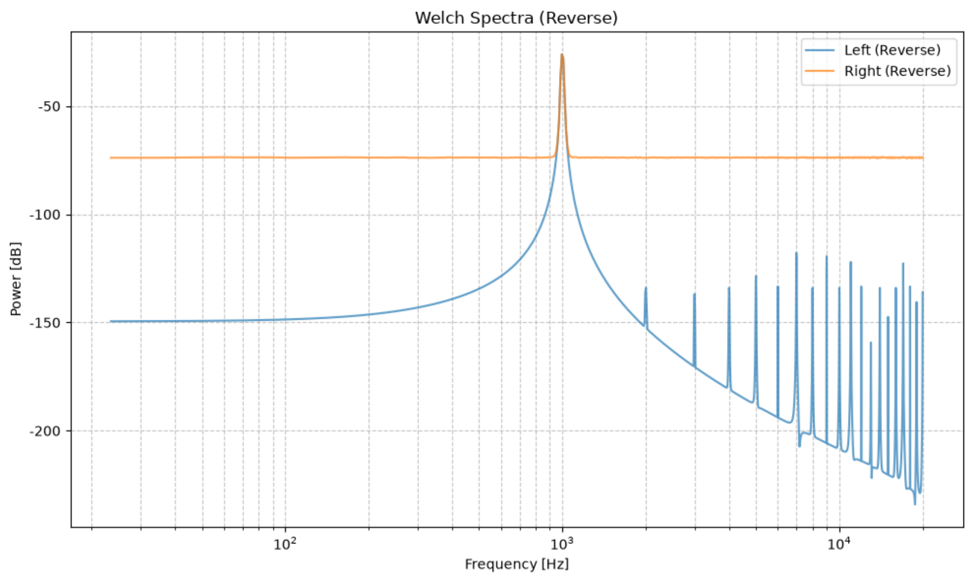
-10 dBFS 1 kHz Sine Wave, 3% Noise (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

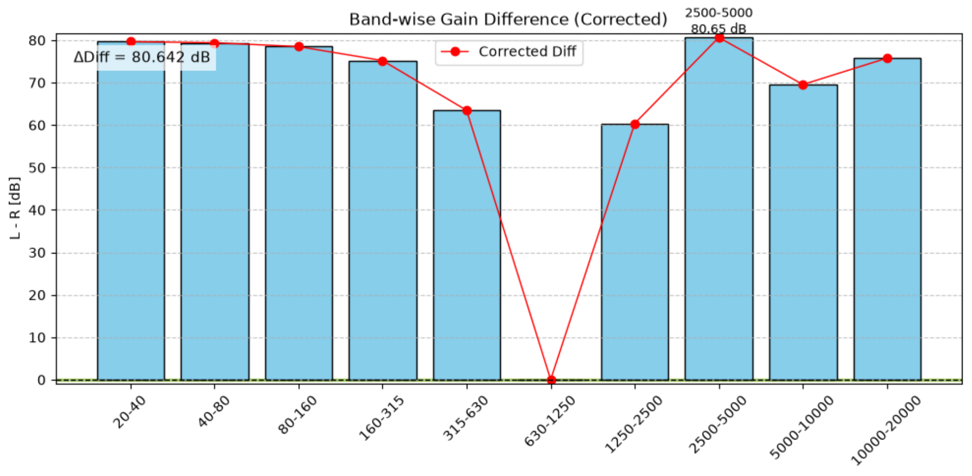
Corrected Gain Difference Analysis

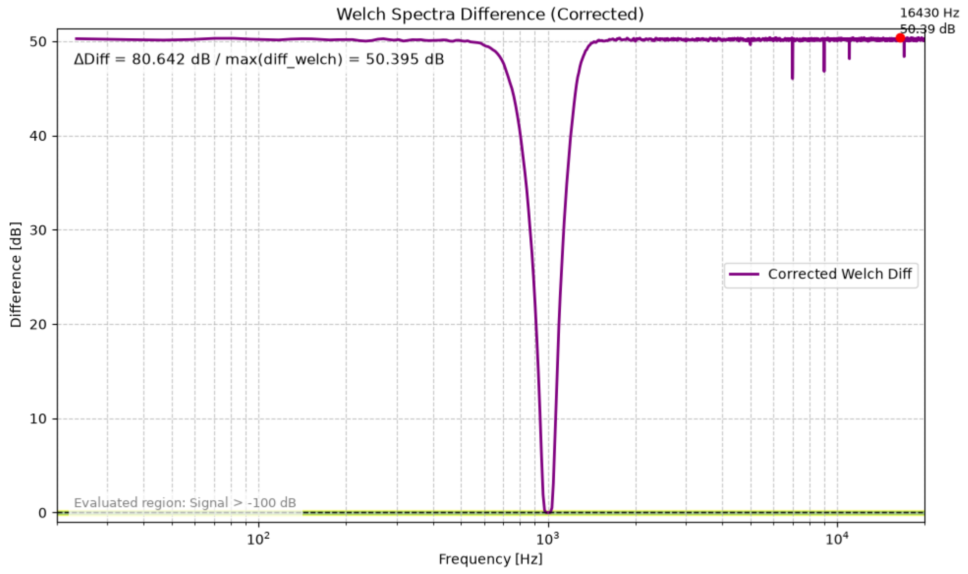
20-40 Hz: Diff=75.532 dB
40-80 Hz: Diff=75.488 dB
80-160 Hz: Diff=74.485 dB
160-315 Hz: Diff=71.227 dB
315-630 Hz: Diff=59.499 dB
630-1250 Hz: Diff=0.002 dB
1250-2500 Hz: Diff=56.378 dB
2500-5000 Hz: Diff=76.656 dB
5000-10000 Hz: Diff=65.551 dB
10000-20000 Hz: Diff=71.764 dB

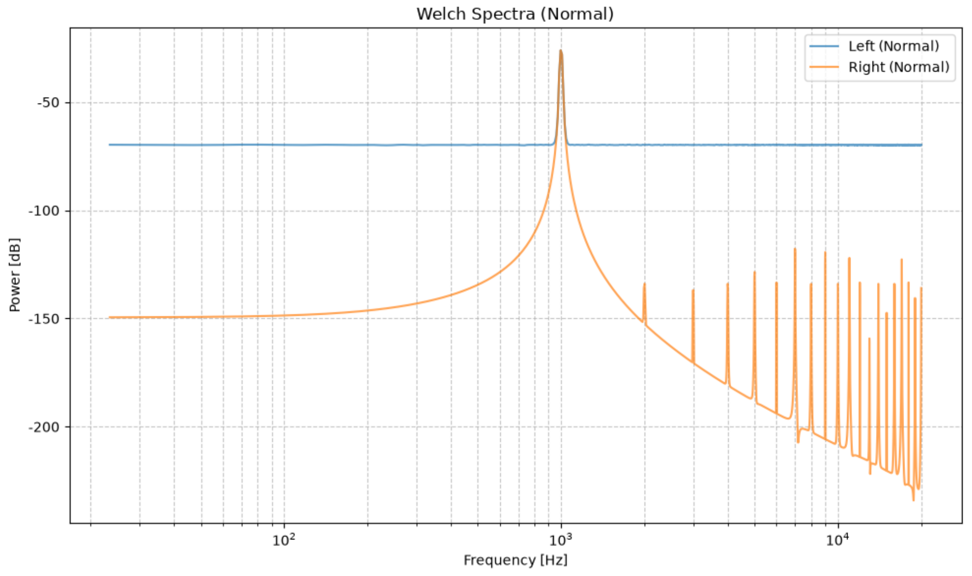
--- Summary ---
ΔDiff (Band) = 76.654 dB
ΔDiff (Welch max) = 46.394 dB
Max Diff Freq = 13277.3 Hz
Judgement: Check required: Possible equipment or measurement problem.

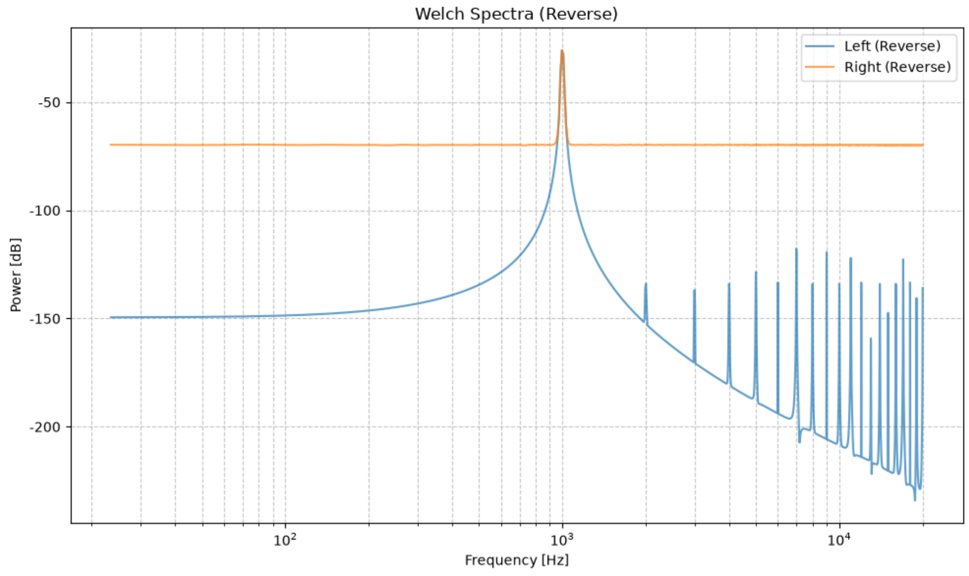
--- Channel Check ---
Energy Ratio (L/R) = 1.02
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, 5% Noise (Left)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device’s bandwidth characteristics. It was used to evaluate the script’s decision-making logic for narrowband signals.









Summary.txt

Corrected Gain Difference Analysis

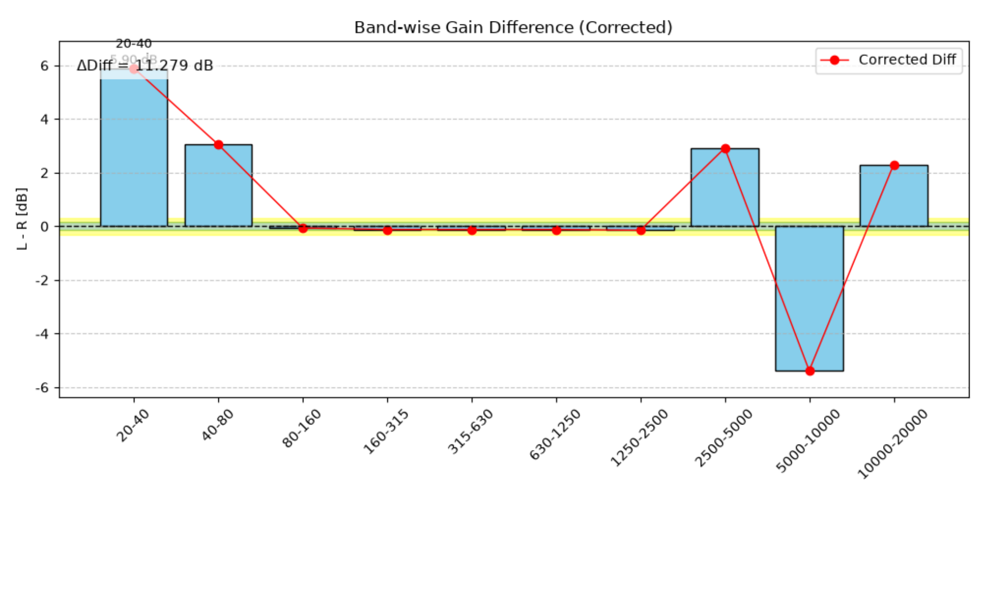
20-40 Hz: Diff=79.655 dB
40-80 Hz: Diff=79.402 dB
80-160 Hz: Diff=78.497 dB
160-315 Hz: Diff=75.195 dB
315-630 Hz: Diff=63.477 dB
630-1250 Hz: Diff=0.007 dB
1250-2500 Hz: Diff=60.391 dB
2500-5000 Hz: Diff=80.648 dB
5000-10000 Hz: Diff=69.549 dB
10000-20000 Hz: Diff=75.766 dB

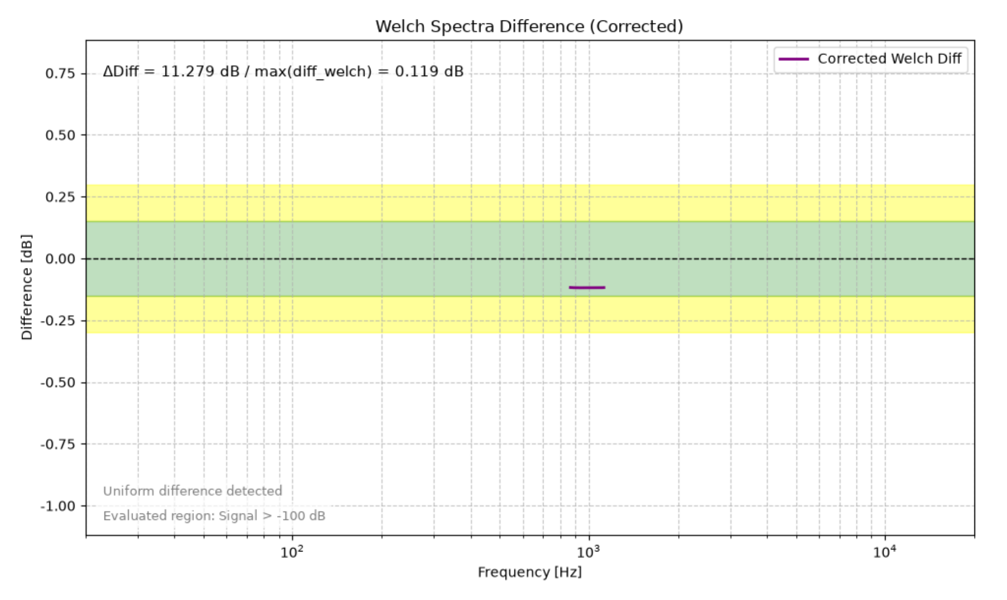
--- Summary ---
ΔDiff (Band) = 80.642 dB
ΔDiff (Welch max) = 50.395 dB
Max Diff Freq = 16429.7 Hz
Judgement: Check required: Possible equipment or measurement problem.

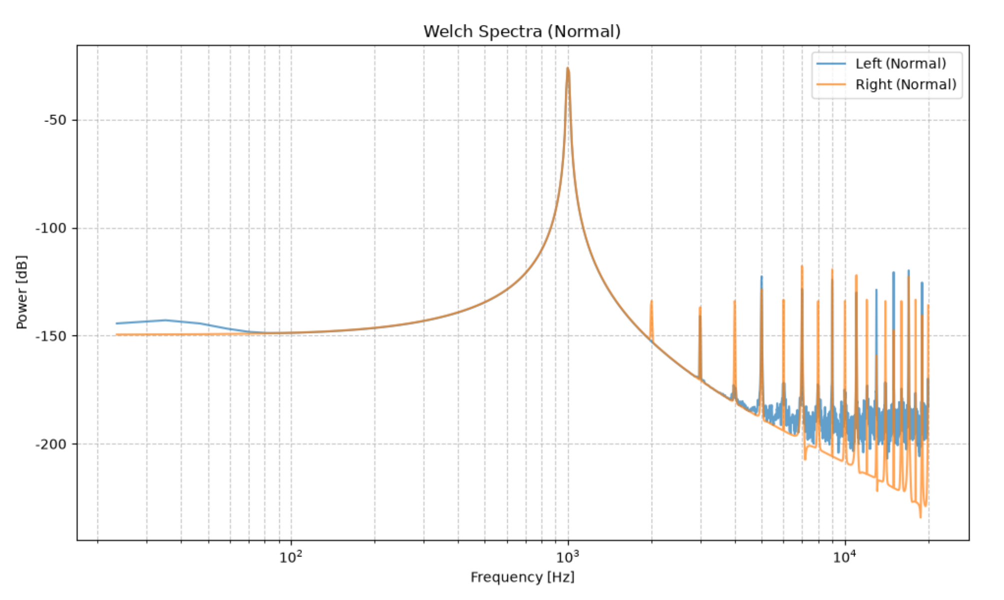
--- Channel Check ---
Energy Ratio (L/R) = 1.05
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

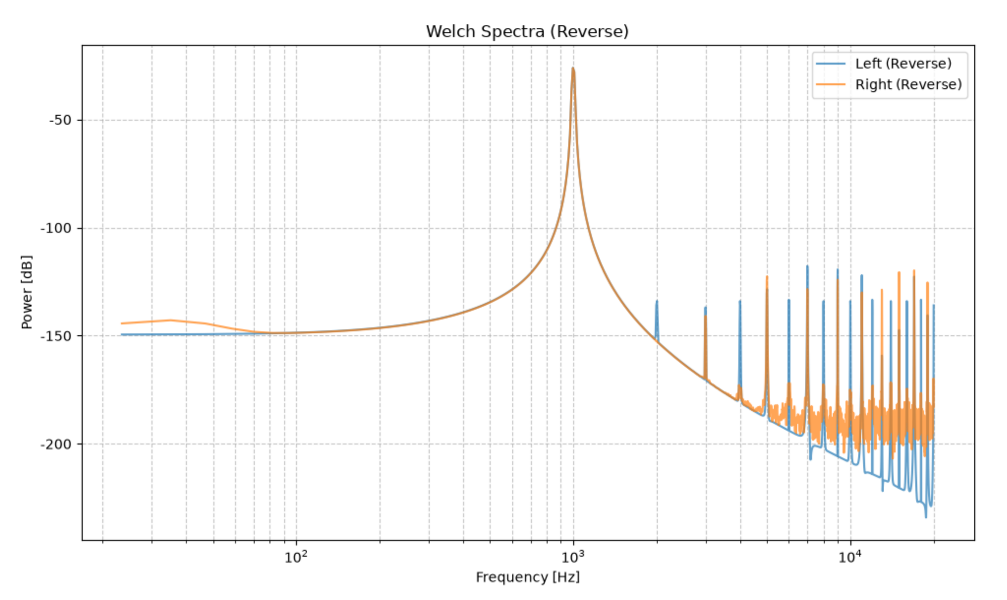
-10 dBFS 1 kHz Sine Wave, PEQ +1 dB (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device’s bandwidth characteristics. It was used to evaluate the script’s decision-making logic for narrowband signals.









Summary.txt

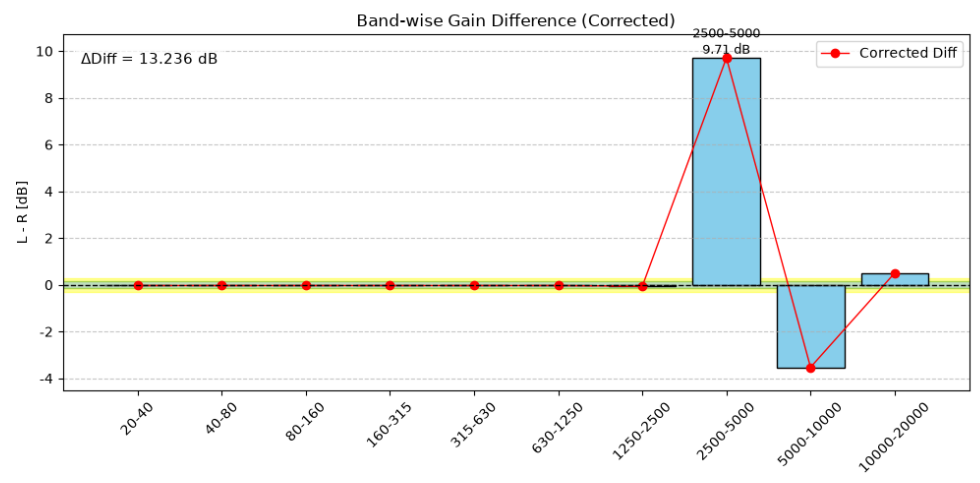
Corrected Gain Difference Analysis

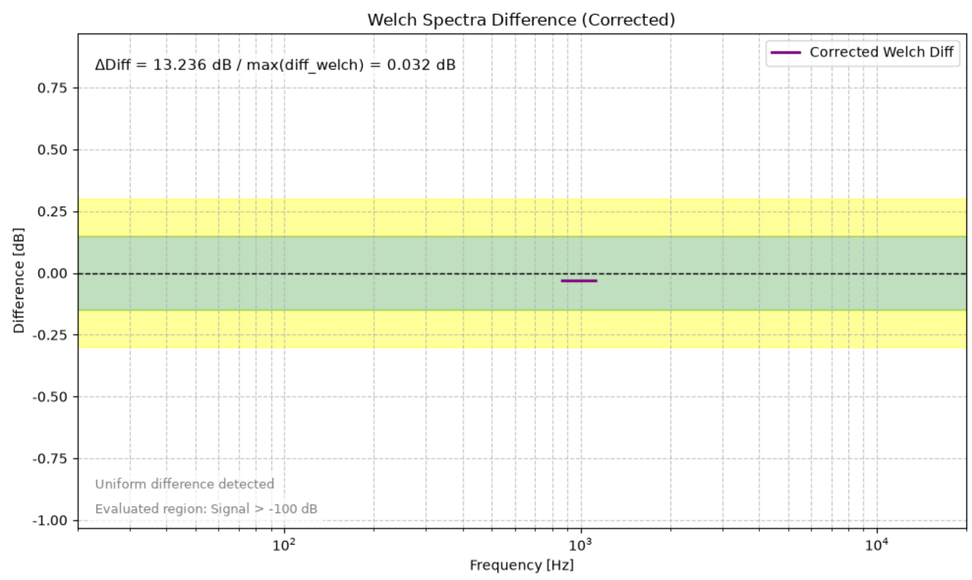
20-40 Hz: Diff=5.899 dB
40-80 Hz: Diff=3.051 dB
80-160 Hz: Diff=-0.062 dB
160-315 Hz: Diff=-0.119 dB
315-630 Hz: Diff=-0.118 dB
630-1250 Hz: Diff=-0.119 dB
1250-2500 Hz: Diff=-0.148 dB
2500-5000 Hz: Diff=2.916 dB
5000-10000 Hz: Diff=-5.380 dB
10000-20000 Hz: Diff=2.301 dB

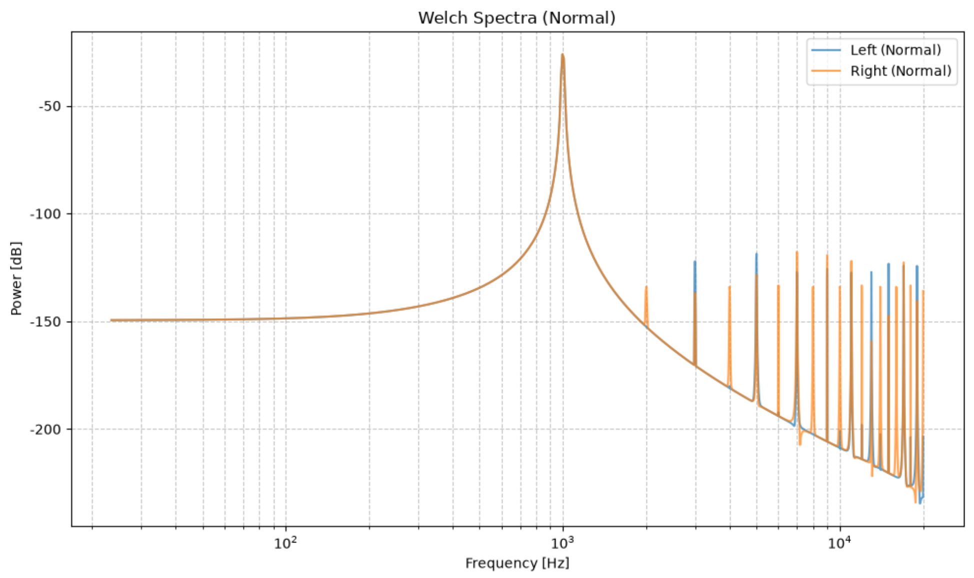
--- Summary ---
ΔDiff (Band) = 11.279 dB
ΔDiff (Welch max) = 0.119 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

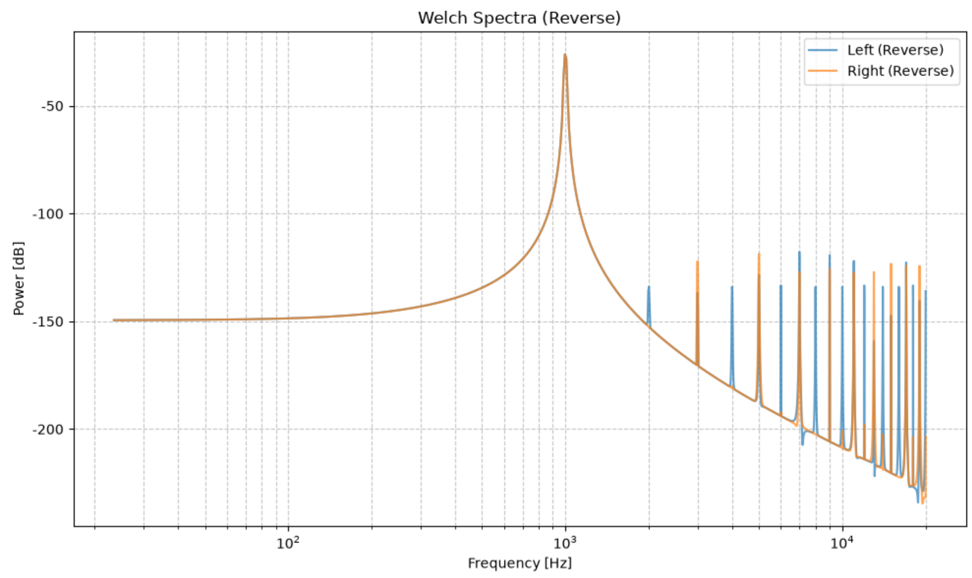
--- Channel Check ---
Energy Ratio (L/R) = 0.97
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, HSF 5 kHz -1 dB (Left)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

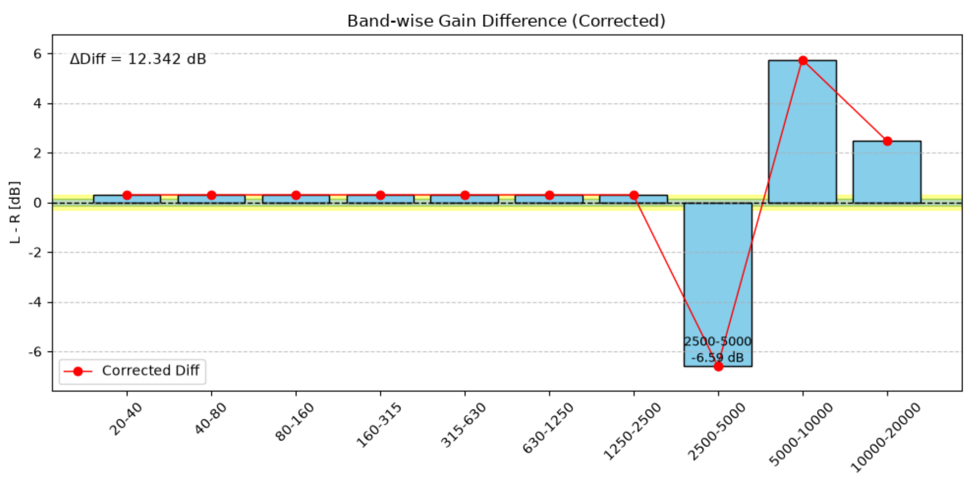
Corrected Gain Difference Analysis

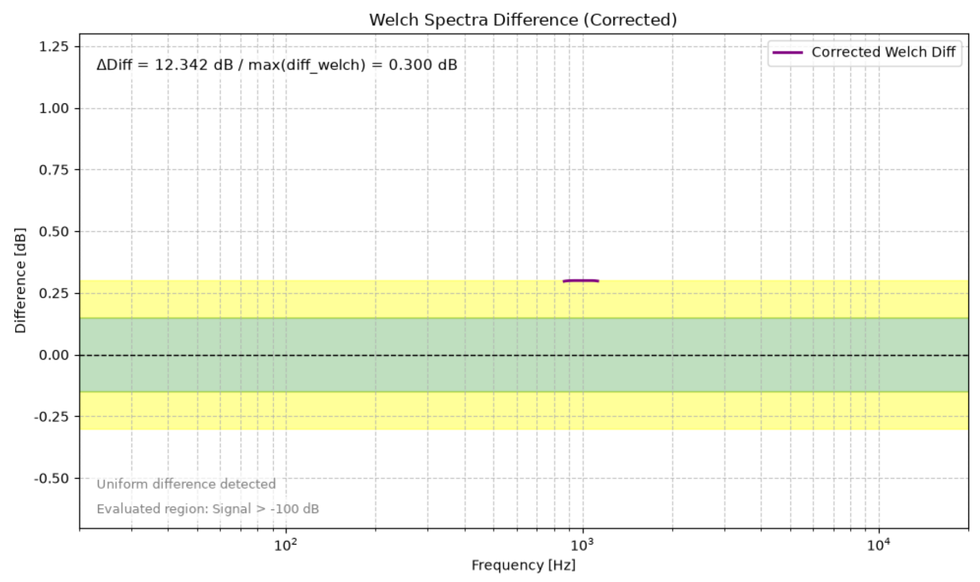
20-40 Hz: Diff=-0.032 dB
40-80 Hz: Diff=-0.032 dB
80-160 Hz: Diff=-0.032 dB
160-315 Hz: Diff=-0.032 dB
315-630 Hz: Diff=-0.032 dB
630-1250 Hz: Diff=-0.032 dB
1250-2500 Hz: Diff=-0.062 dB
2500-5000 Hz: Diff=9.708 dB
5000-10000 Hz: Diff=-3.529 dB
10000-20000 Hz: Diff=0.503 dB

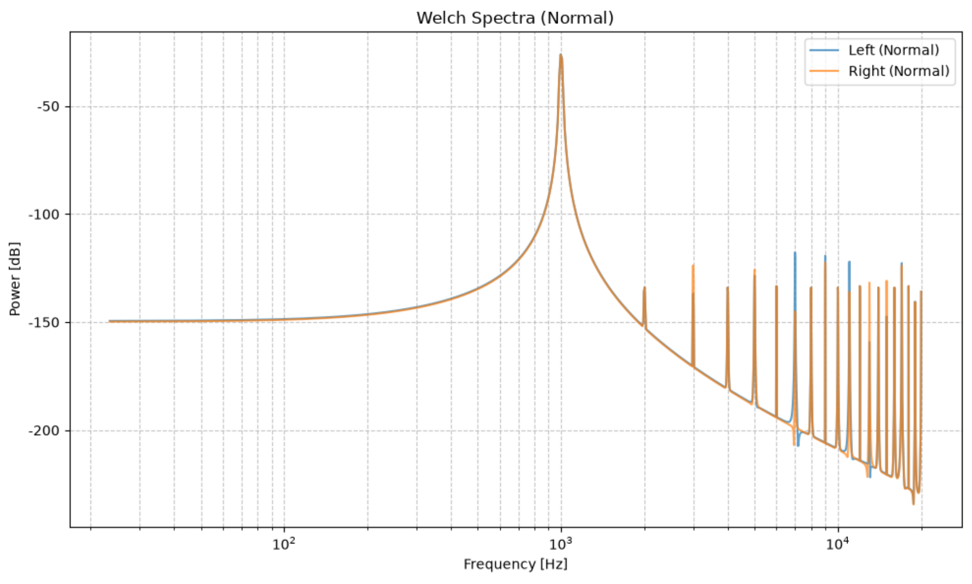
--- Summary ---
ΔDiff (Band) = 13.236 dB
ΔDiff (Welch max) = 0.032 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

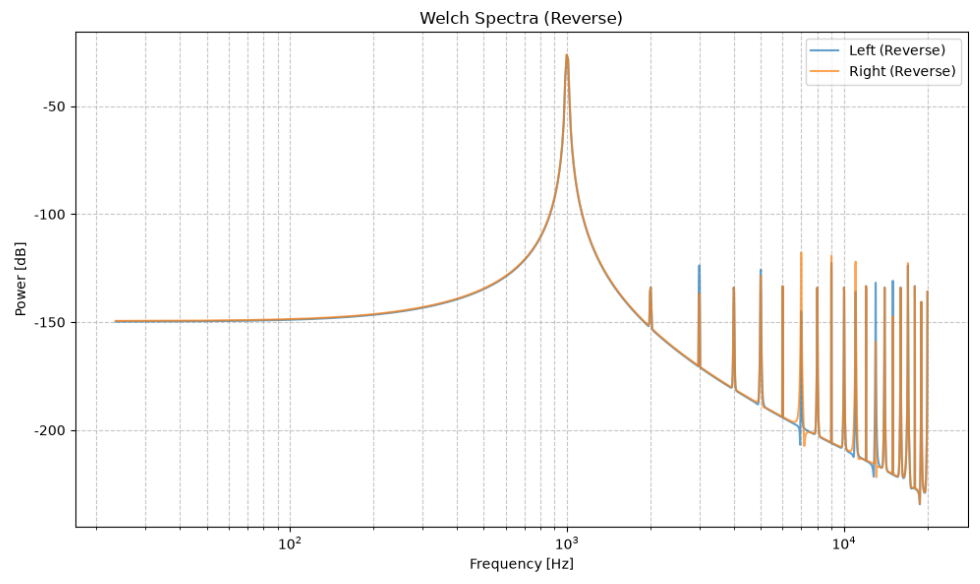
--- Channel Check ---
Energy Ratio (L/R) = 0.99
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 1 kHz Sine Wave, Gain -0.3 dB (Right)
The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

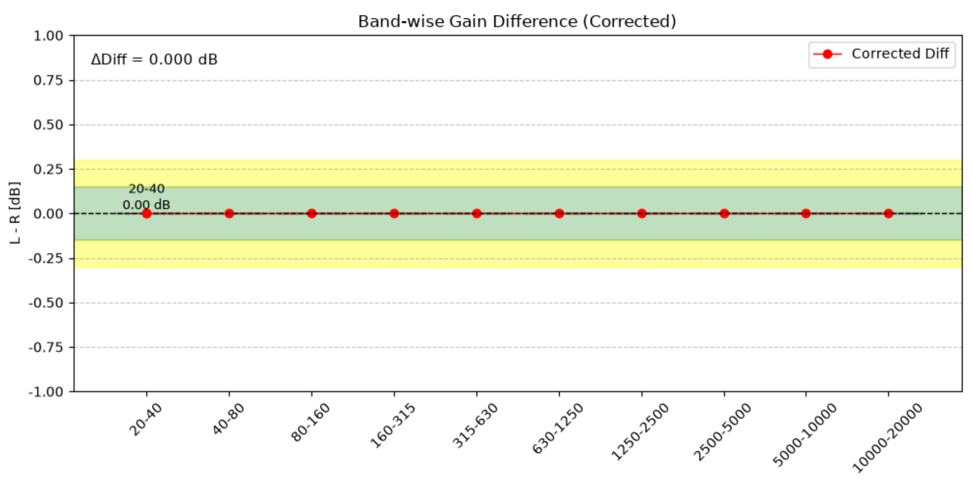
Corrected Gain Difference Analysis

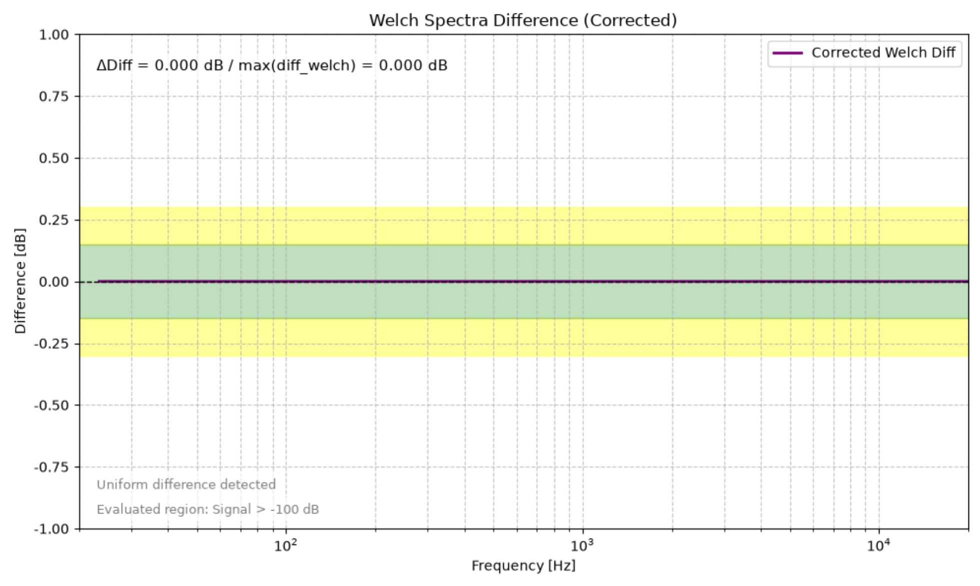
20-40 Hz: Diff=0.300 dB
40-80 Hz: Diff=0.300 dB
80-160 Hz: Diff=0.300 dB
160-315 Hz: Diff=0.300 dB
315-630 Hz: Diff=0.300 dB
630-1250 Hz: Diff=0.300 dB
1250-2500 Hz: Diff=0.298 dB
2500-5000 Hz: Diff=-6.595 dB
5000-10000 Hz: Diff=5.748 dB
10000-20000 Hz: Diff=2.487 dB

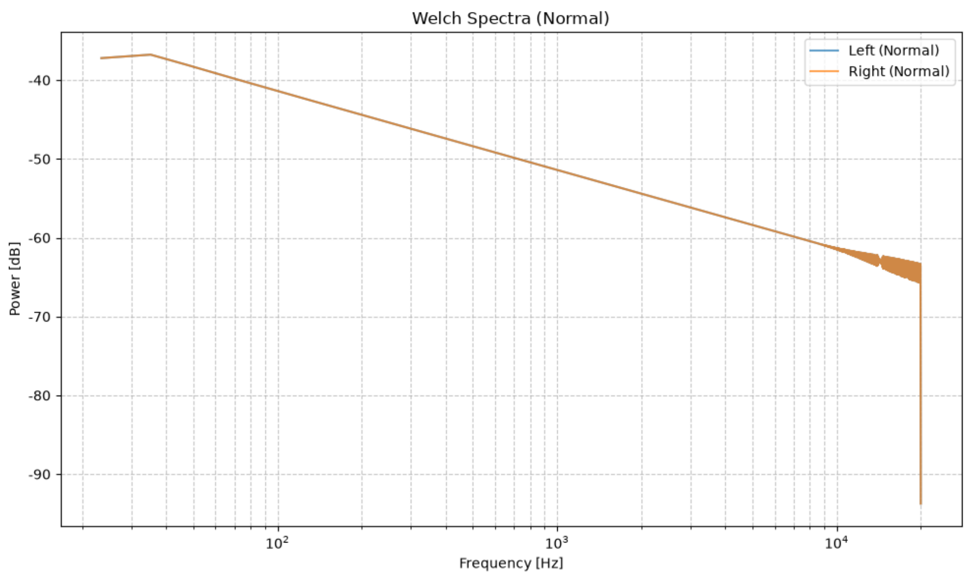
--- Summary ---
ΔDiff (Band) = 12.342 dB
ΔDiff (Welch max) = 0.300 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

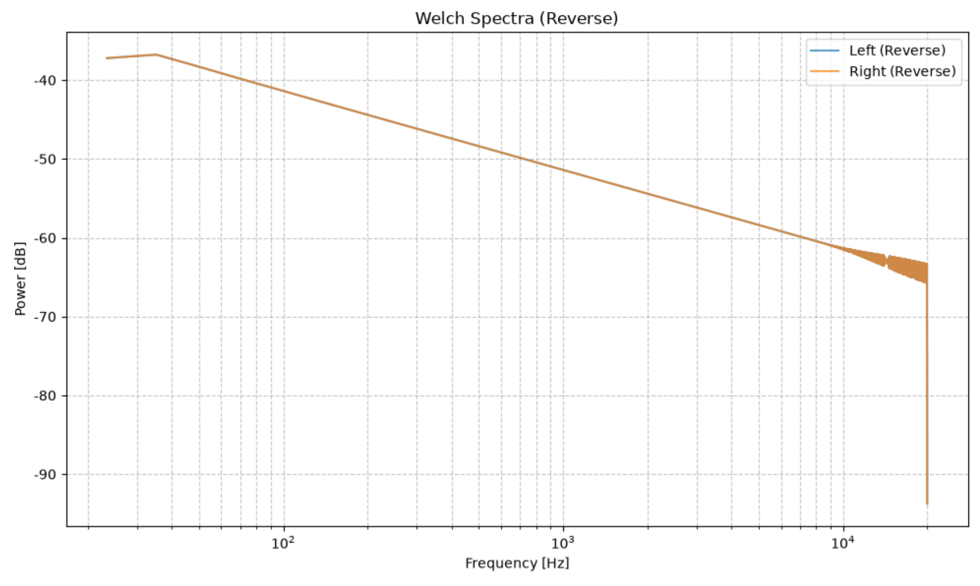
--- Channel Check ---
Energy Ratio (L/R) = 1.07
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Symmetrical









Summary.txt

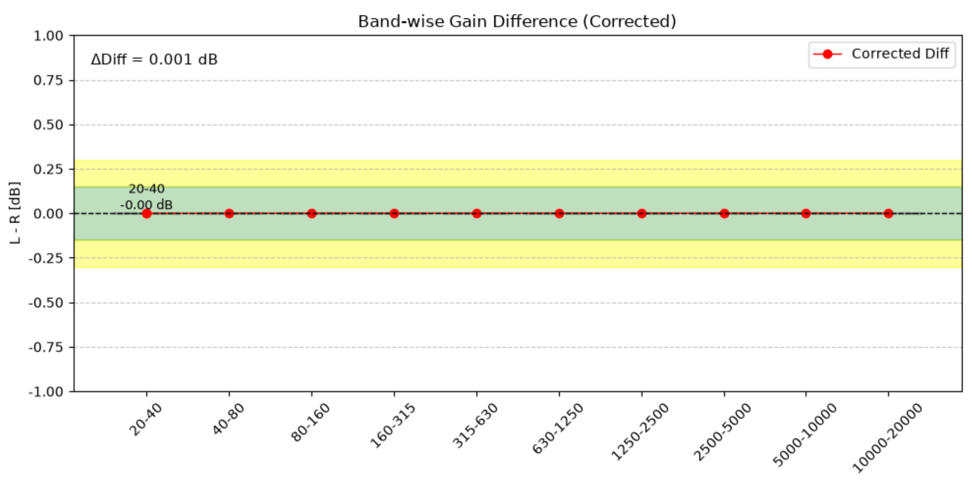
Corrected Gain Difference Analysis

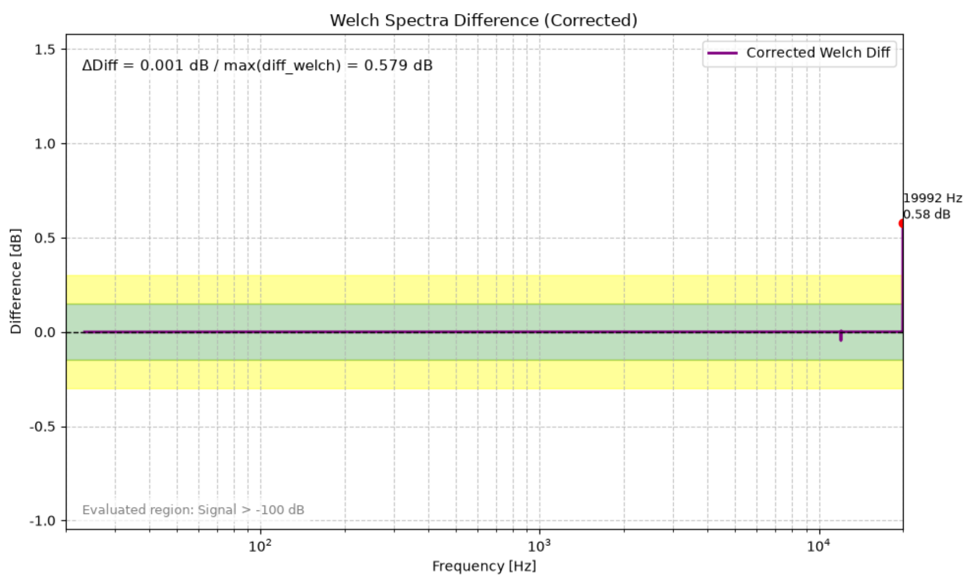
20-40 Hz: Diff=0.000 dB
40-80 Hz: Diff=0.000 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.000 dB
5000-10000 Hz: Diff=0.000 dB
10000-20000 Hz: Diff=0.000 dB

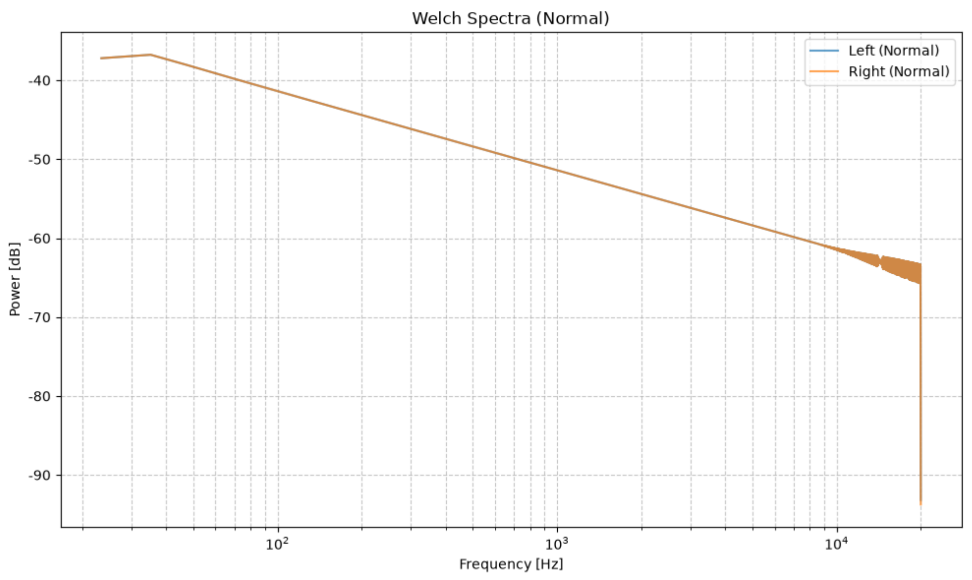
--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.000 dB
Judgement: OK: Channel difference is within measurement error.

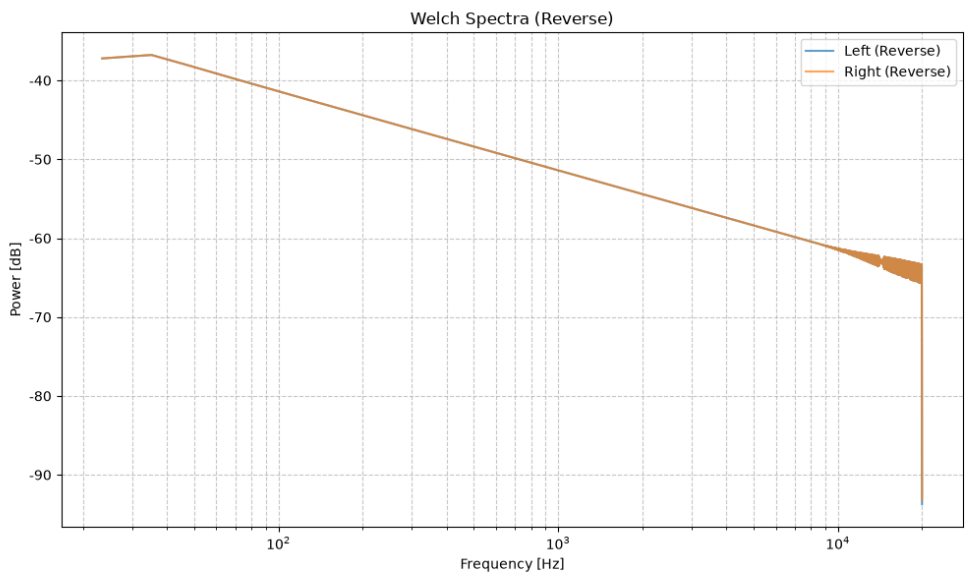
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% THD (Left)









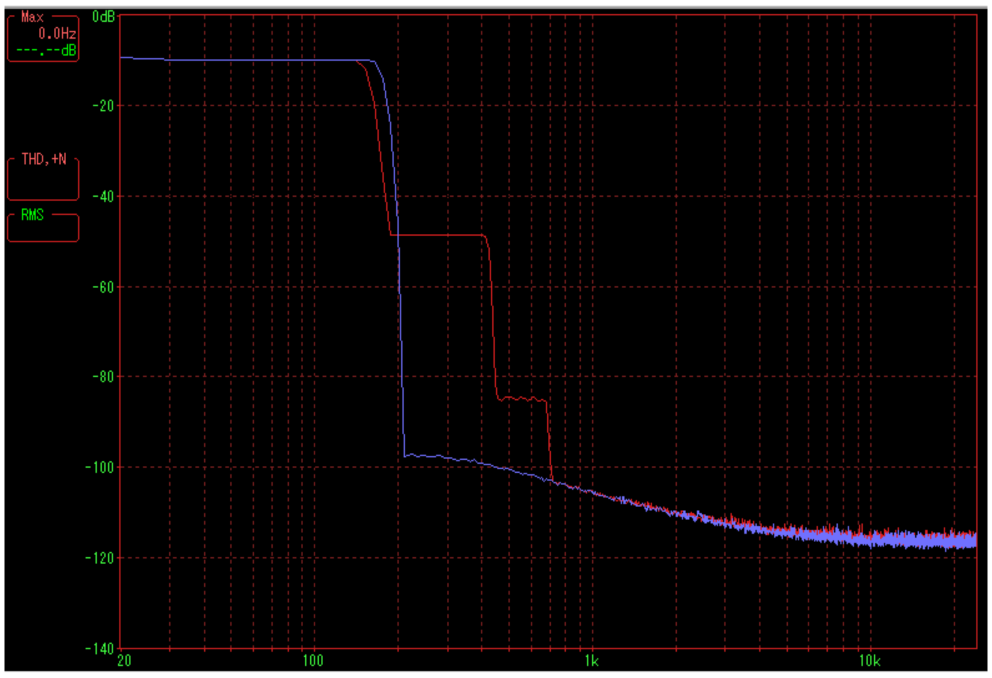
Summary.txt
Corrected Gain Difference Analysis

20-40 Hz: Diff=-0.001 dB
40-80 Hz: Diff=-0.000 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.000 dB
5000-10000 Hz: Diff=0.000 dB
10000-20000 Hz: Diff=0.000 dB

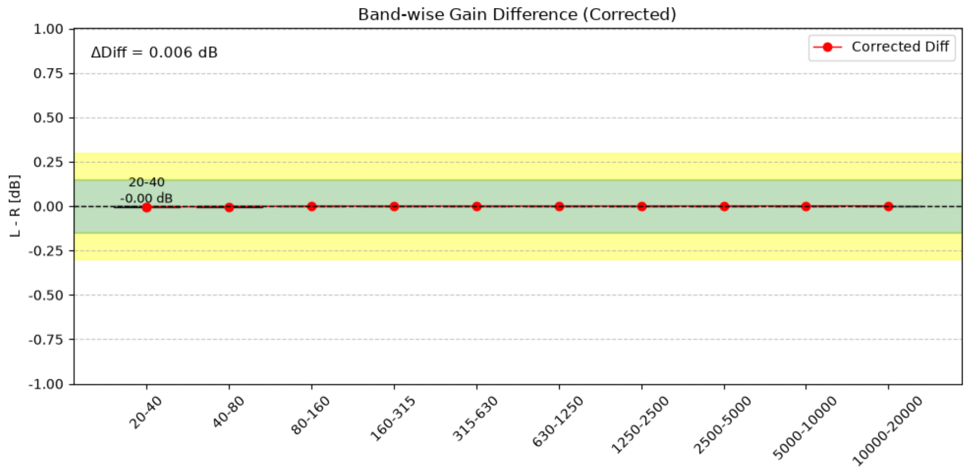
--- Summary ---
ΔDiff (Band) = 0.001 dB
ΔDiff (Welch max) = 0.579 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

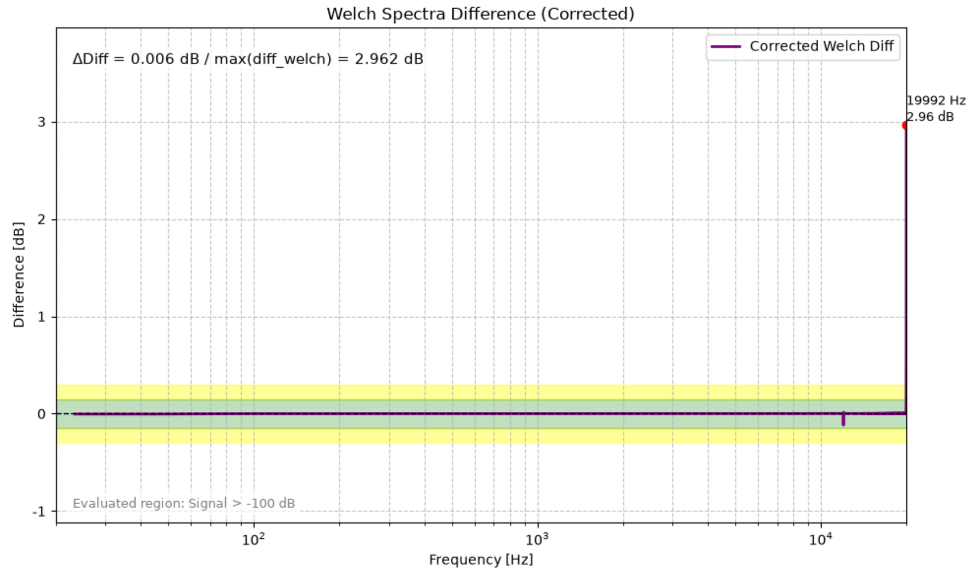
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

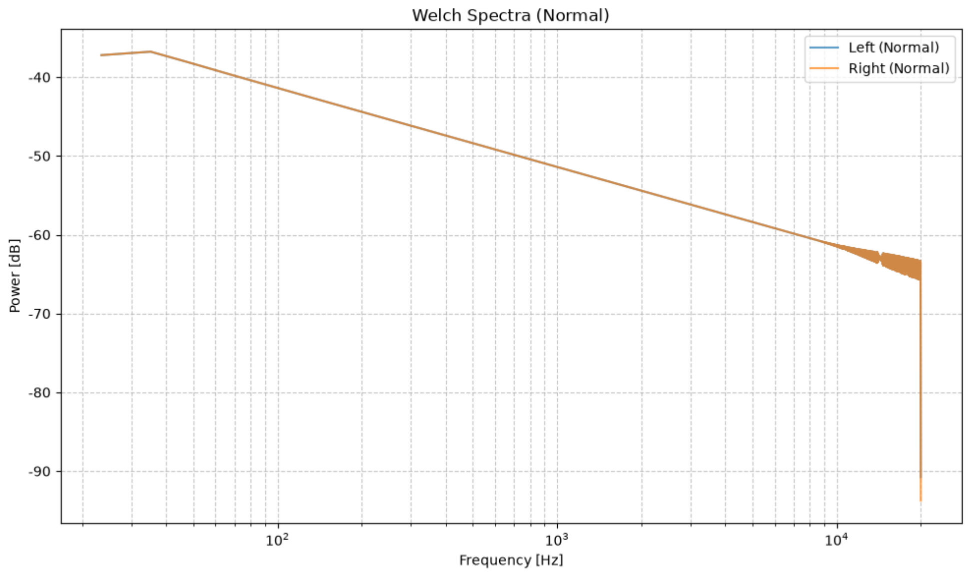
Shows the level of harmonics superimposed on the Left-channel signal in an image taken during a sweep

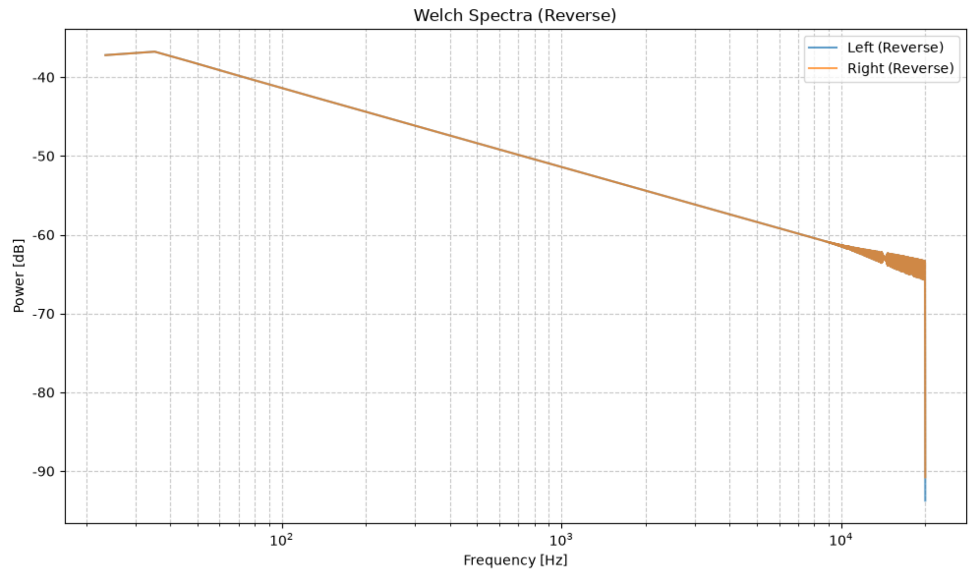


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% THD (Left)

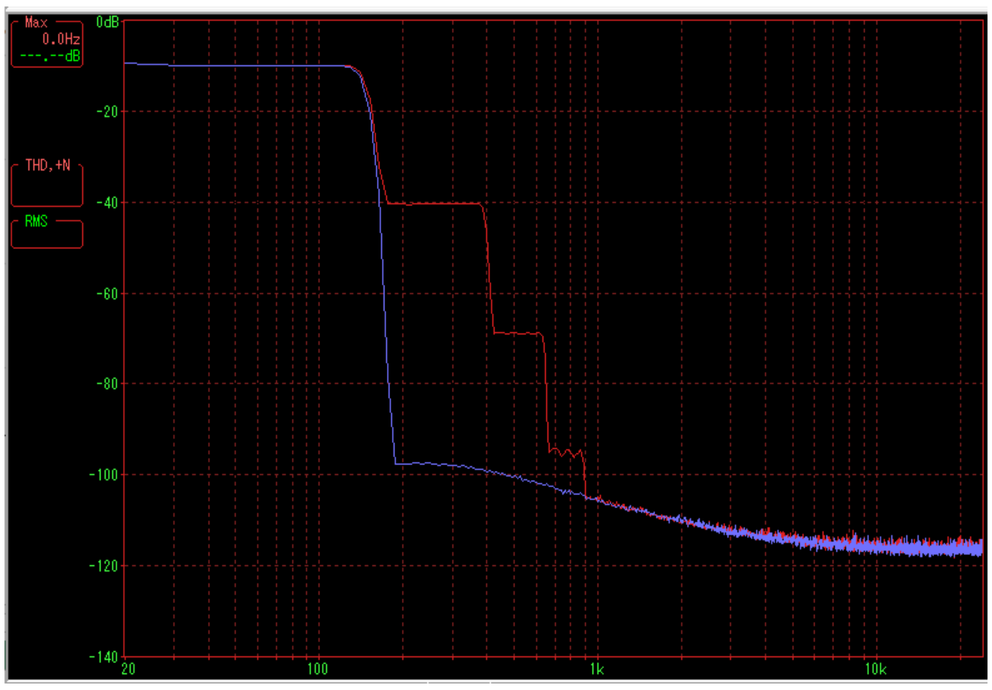








Shows the level of harmonics superimposed on the Left-channel signal in an image taken during a sweep



Summary.txt

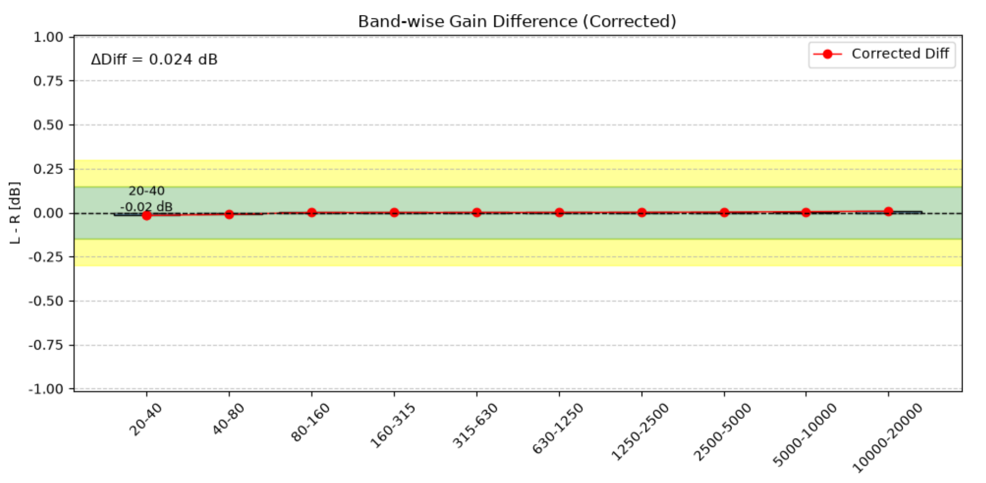
Corrected Gain Difference Analysis

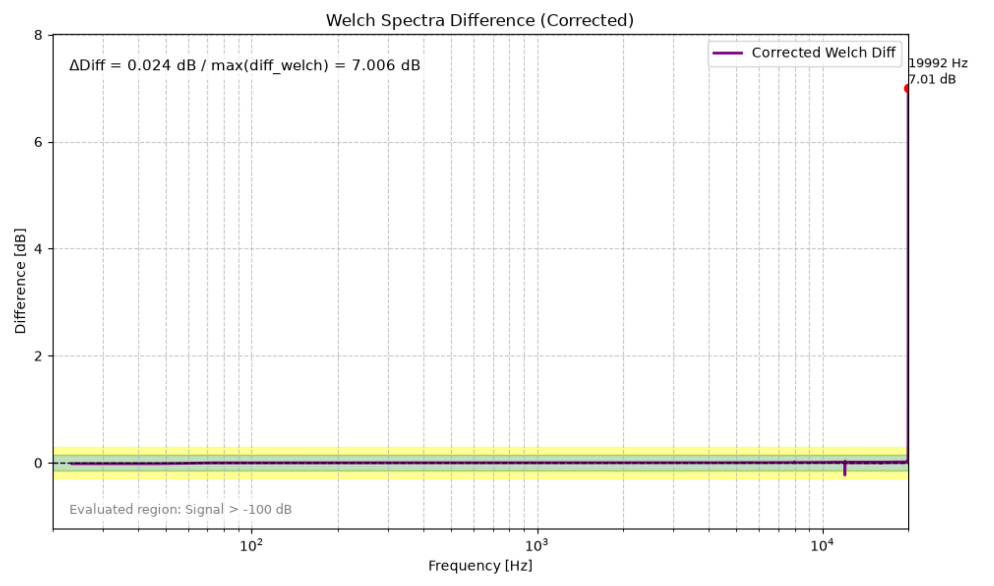
20-40 Hz: Diff=-0.004 dB
40-80 Hz: Diff=-0.002 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.001 dB
5000-10000 Hz: Diff=0.001 dB
10000-20000 Hz: Diff=0.002 dB

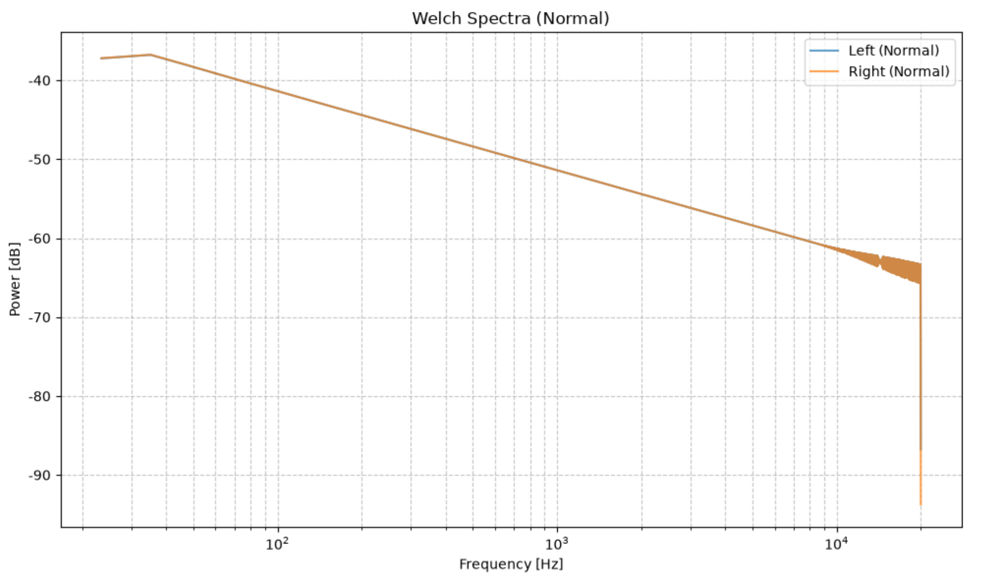
--- Summary ---
ΔDiff (Band) = 0.006 dB
ΔDiff (Welch max) = 2.962 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

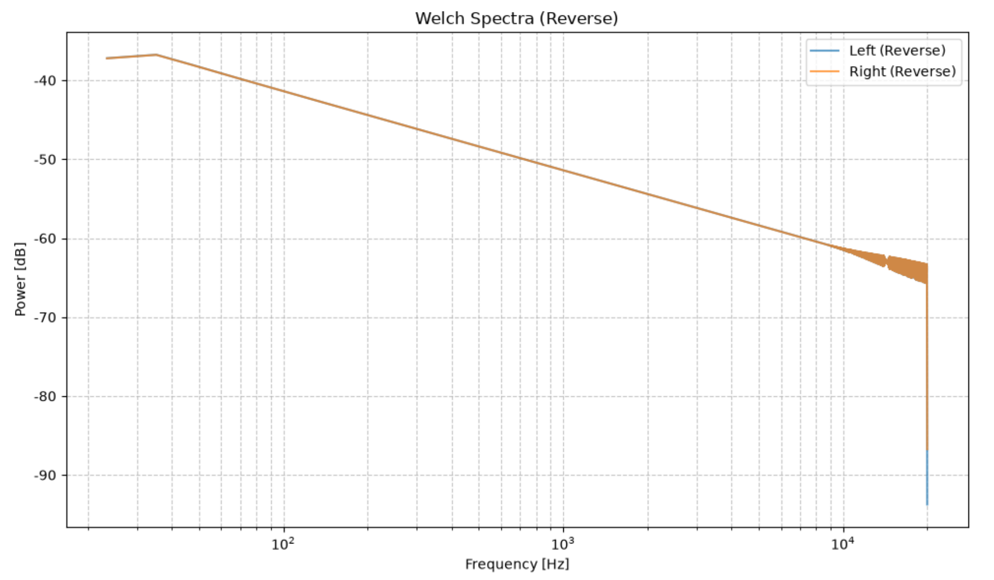
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% THD (Left)

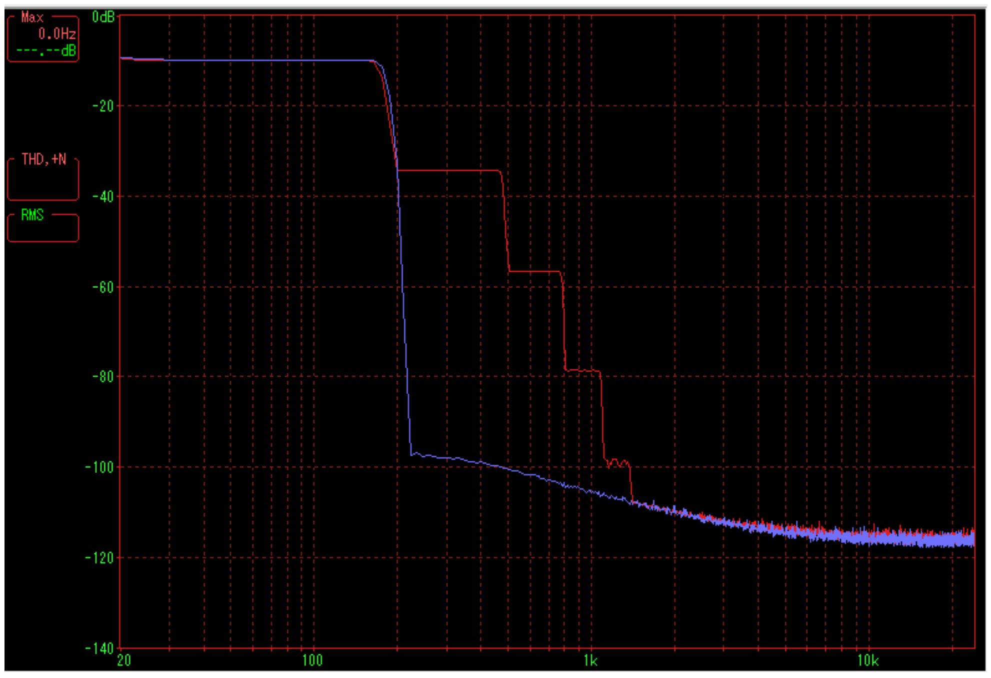








Shows the level of harmonics superimposed on the Left-channel signal in an image taken during a sweep



Summary.txt

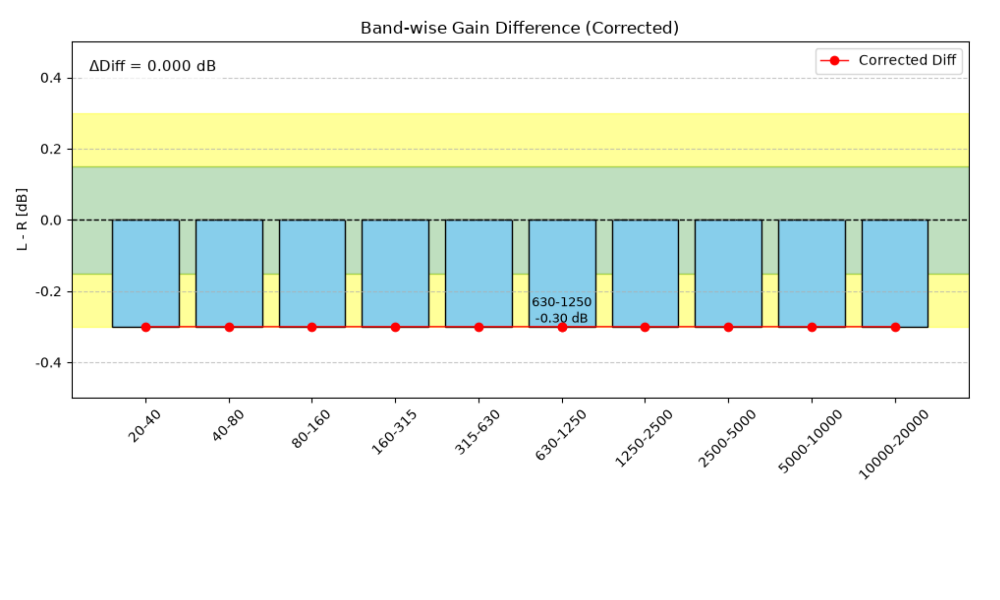
Corrected Gain Difference Analysis

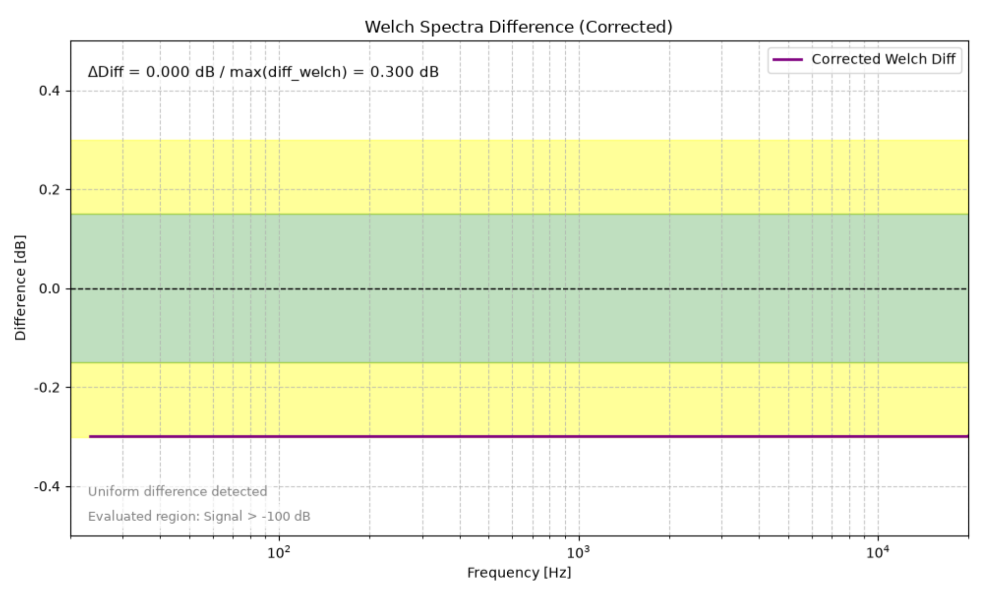
20-40 Hz: Diff=-0.016 dB
40-80 Hz: Diff=-0.010 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.001 dB
1250-2500 Hz: Diff=0.001 dB
2500-5000 Hz: Diff=0.003 dB
5000-10000 Hz: Diff=0.005 dB
10000-20000 Hz: Diff=0.007 dB

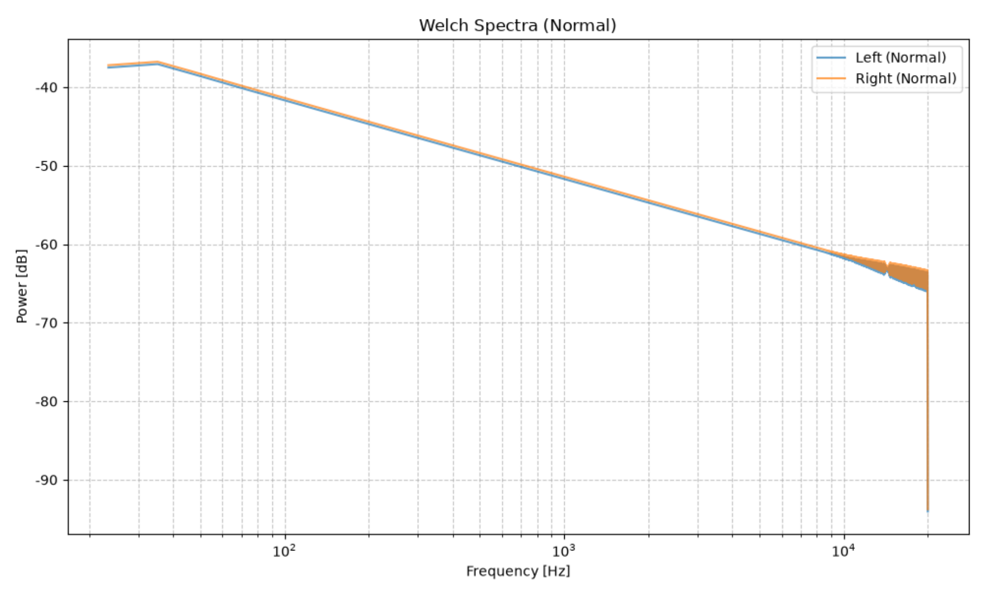
--- Summary ---
ΔDiff (Band) = 0.024 dB
ΔDiff (Welch max) = 7.006 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

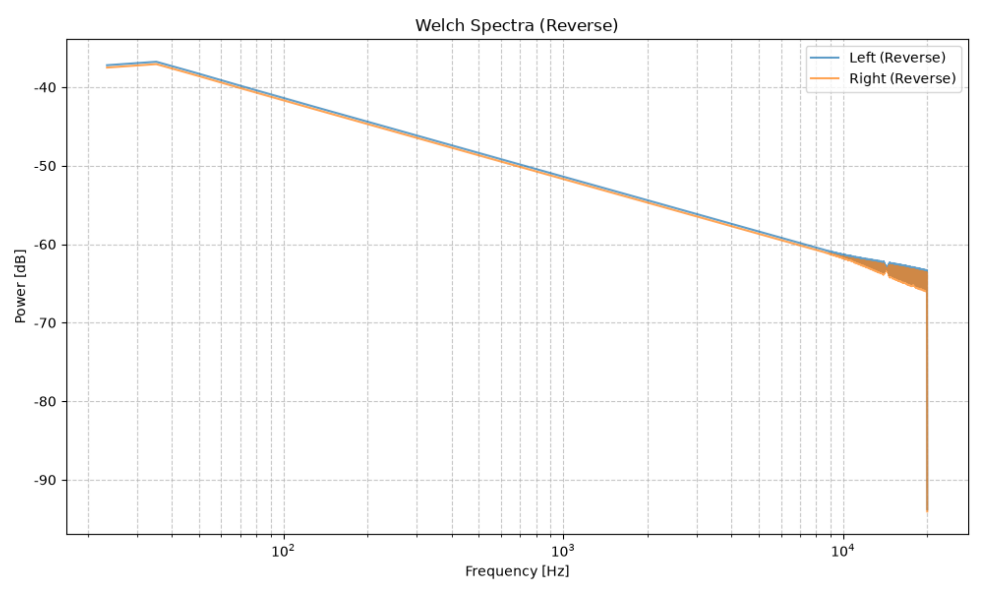
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Left)









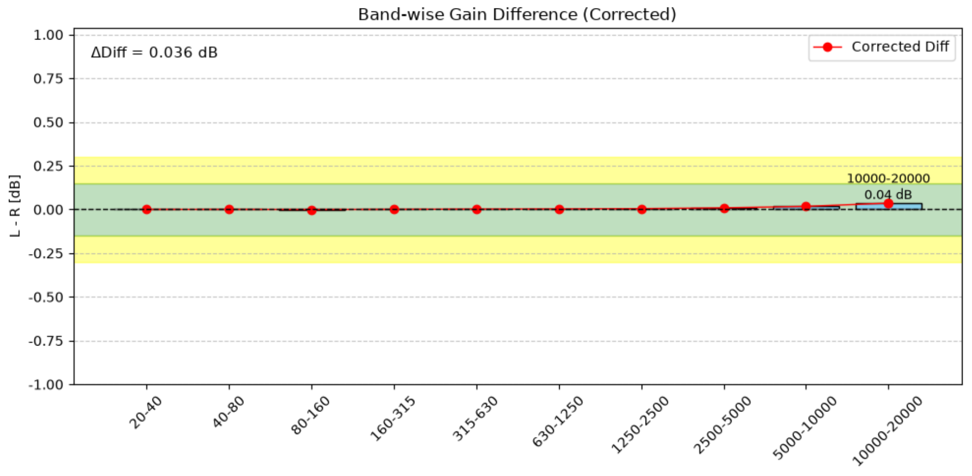
Summary.txt
Corrected Gain Difference Analysis

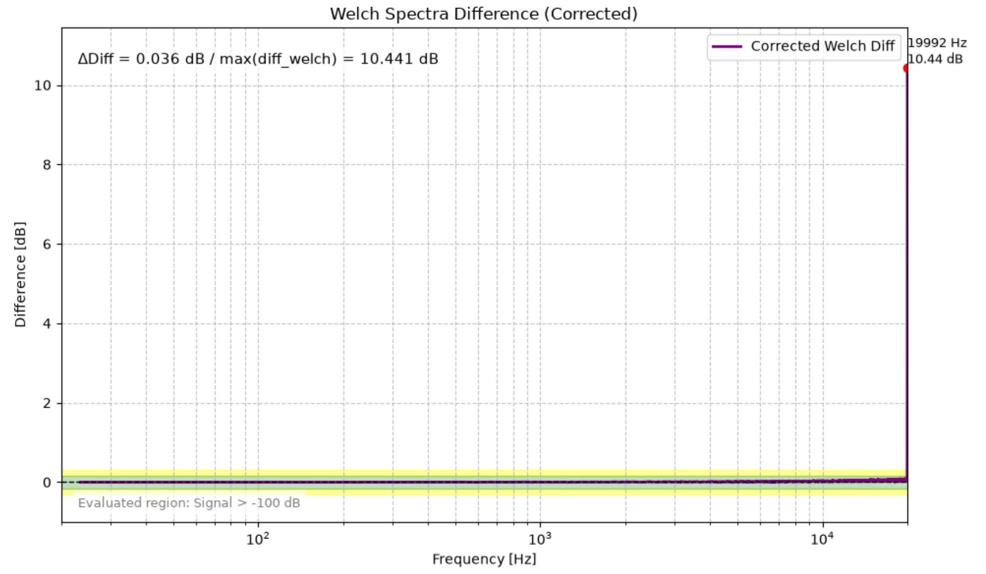
20-40 Hz: Diff=-0.300 dB
40-80 Hz: Diff=-0.300 dB
80-160 Hz: Diff=-0.300 dB
160-315 Hz: Diff=-0.300 dB
315-630 Hz: Diff=-0.300 dB
630-1250 Hz: Diff=-0.300 dB
1250-2500 Hz: Diff=-0.300 dB
2500-5000 Hz: Diff=-0.300 dB
5000-10000 Hz: Diff=-0.300 dB
10000-20000 Hz: Diff=-0.300 dB

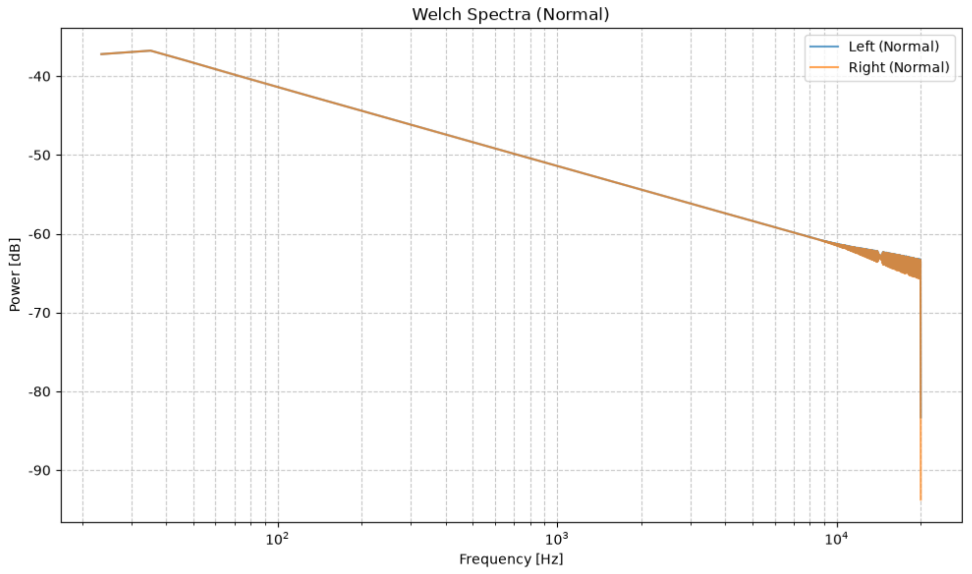
--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.300 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

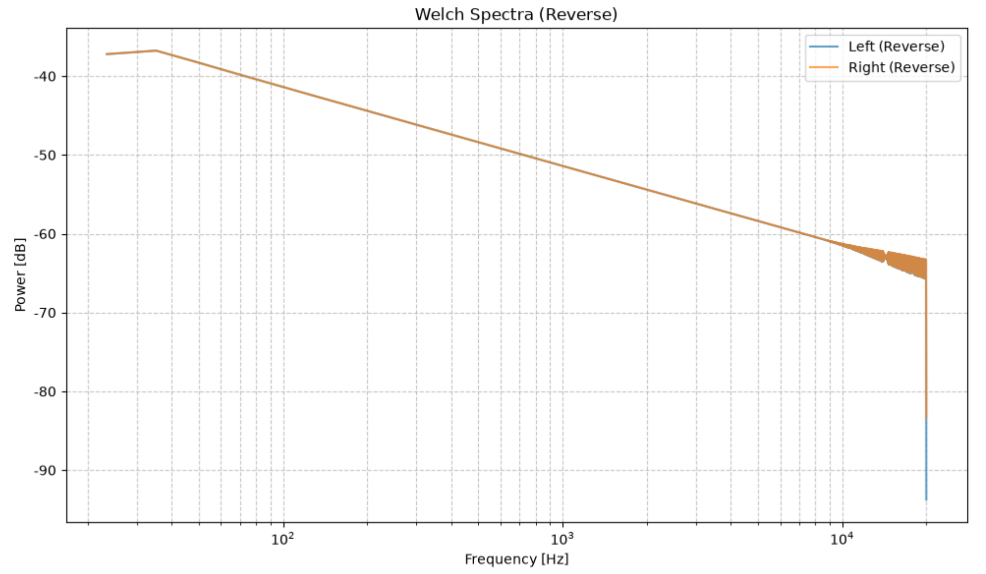
--- Channel Check ---
Energy Ratio (L/R) = 0.93
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% Noise (Left)









Summary.txt

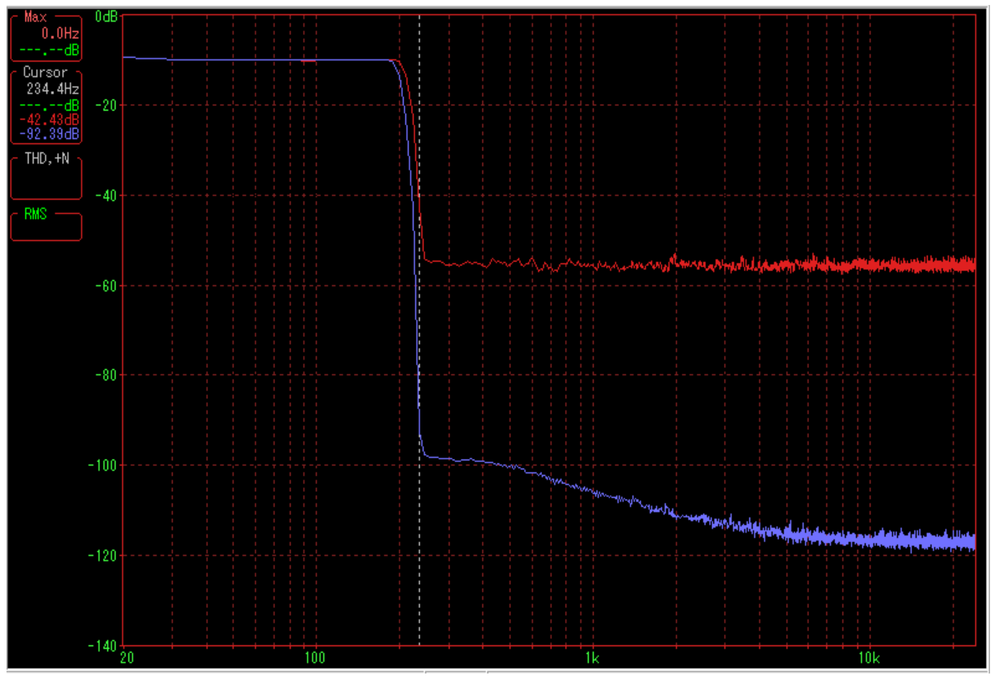
Corrected Gain Difference Analysis

20-40 Hz: Diff=-0.000 dB
40-80 Hz: Diff=0.000 dB
80-160 Hz: Diff=-0.001 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.002 dB
630-1250 Hz: Diff=0.003 dB
1250-2500 Hz: Diff=0.004 dB
2500-5000 Hz: Diff=0.009 dB
5000-10000 Hz: Diff=0.017 dB
10000-20000 Hz: Diff=0.036 dB

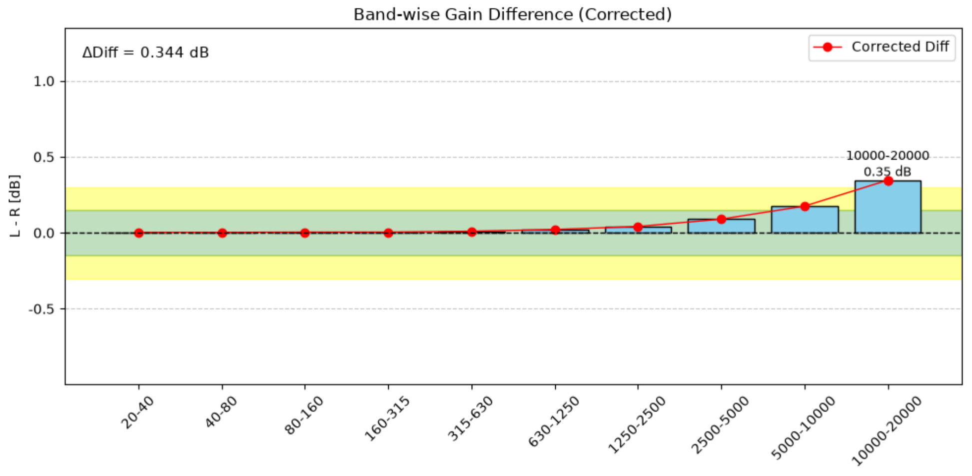
--- Summary ---
ΔDiff (Band) = 0.036 dB
ΔDiff (Welch max) = 10.441 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

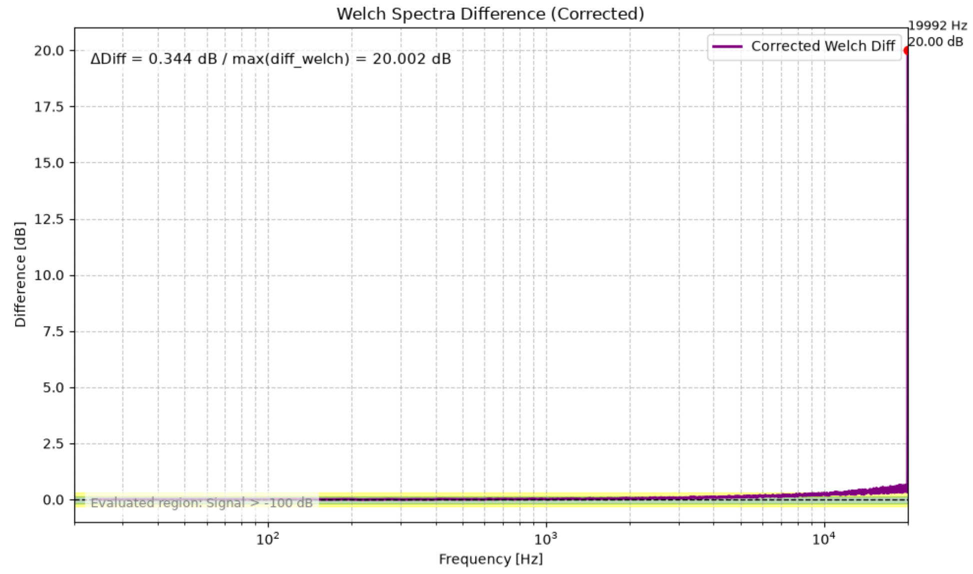
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

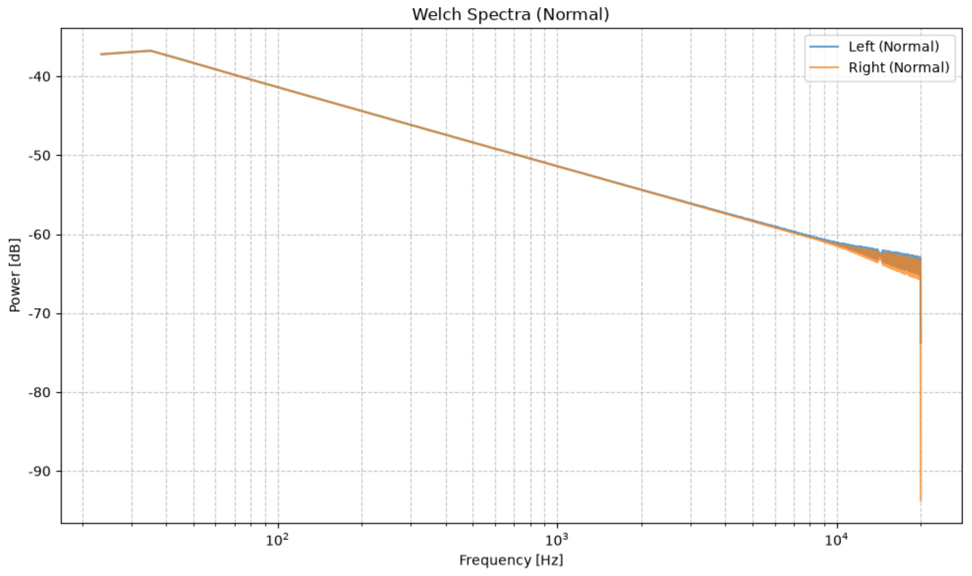
Shows the level of noise superimposed on the Left-channel signal in an image taken during a sweep

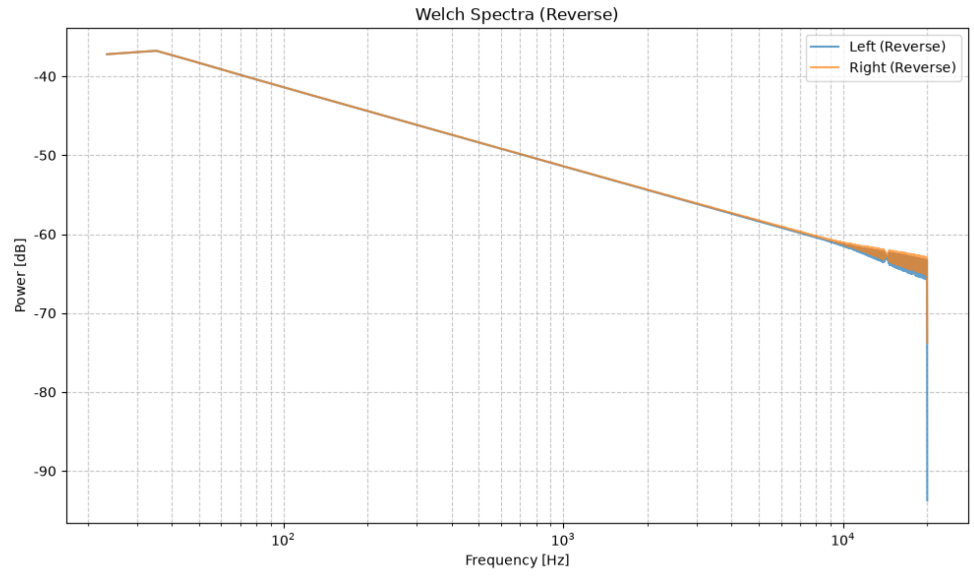


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% Noise (Left)









Summary.txt

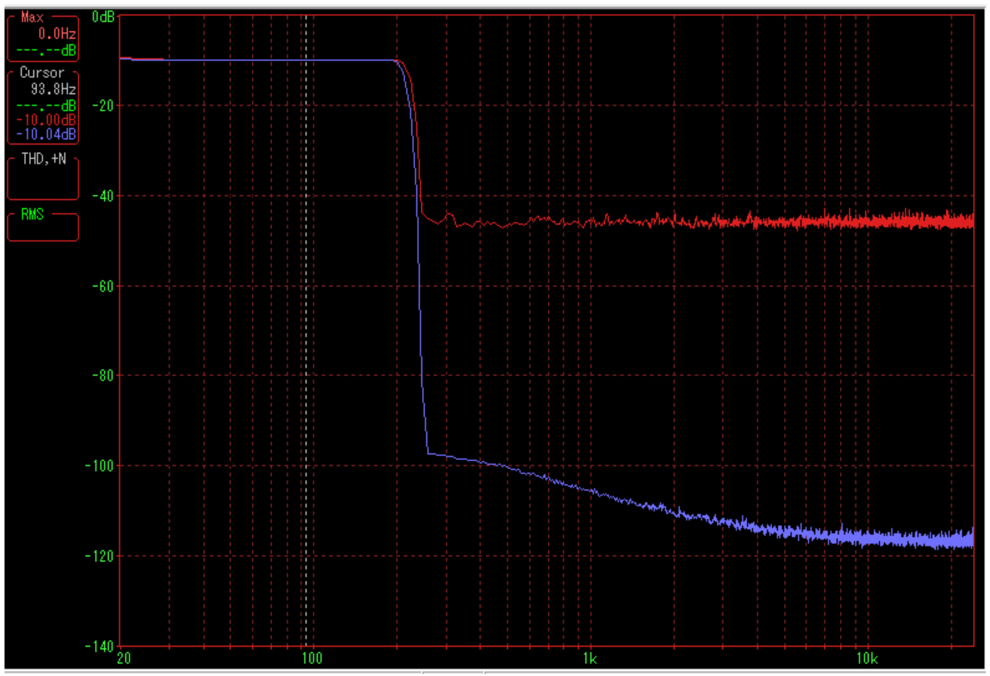
Corrected Gain Difference Analysis

20-40 Hz: Diff=0.002 dB
40-80 Hz: Diff=0.003 dB
80-160 Hz: Diff=0.004 dB
160-315 Hz: Diff=0.004 dB
315-630 Hz: Diff=0.010 dB
630-1250 Hz: Diff=0.022 dB
1250-2500 Hz: Diff=0.042 dB
2500-5000 Hz: Diff=0.090 dB
5000-10000 Hz: Diff=0.175 dB
10000-20000 Hz: Diff=0.346 dB

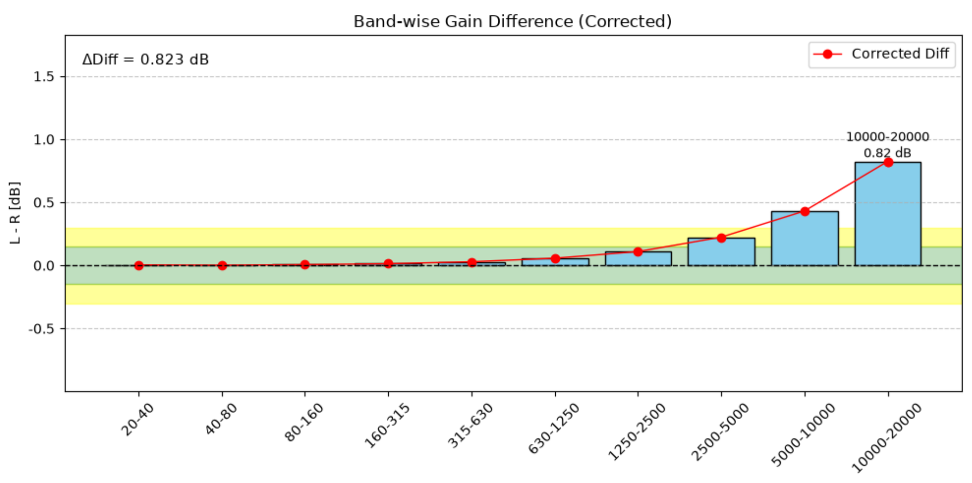
--- Summary ---
ΔDiff (Band) = 0.344 dB
ΔDiff (Welch max) = 20.002 dB
Max Diff Freq = 19992.2 Hz
Judgement: Check required: Possible equipment or measurement problem.

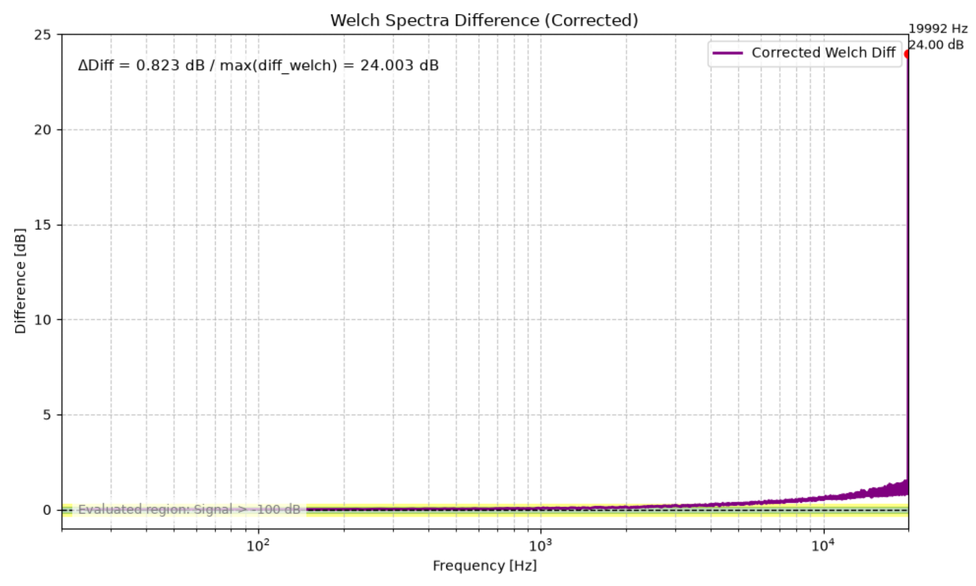
--- Channel Check ---
Energy Ratio (L/R) = 1.02
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

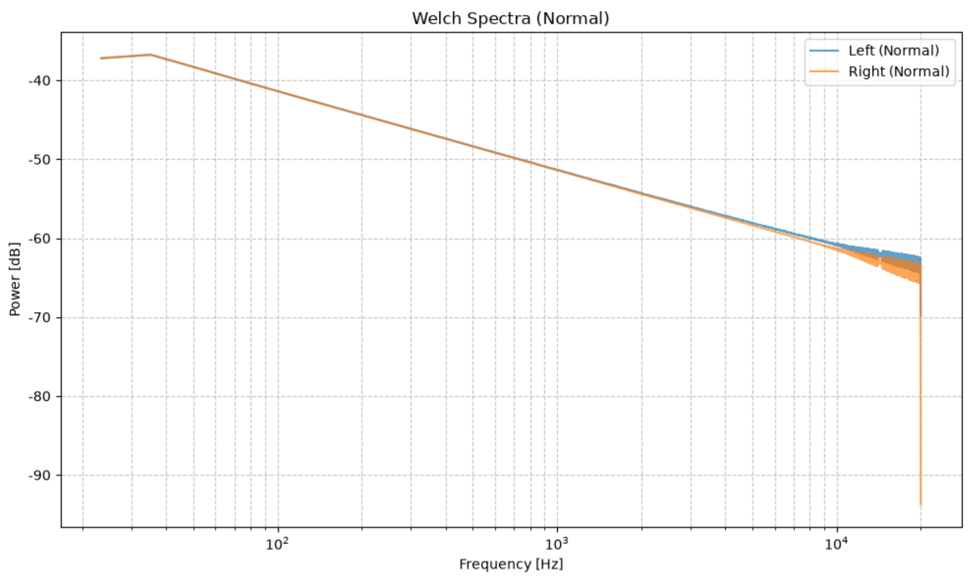
Shows the level of noise superimposed on the Left-channel signal in an image taken during a sweep

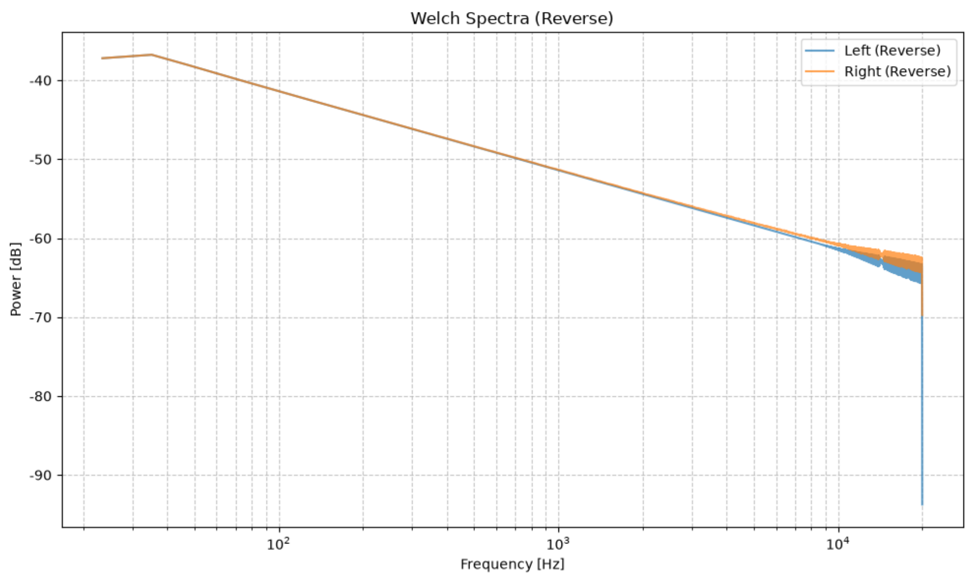


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% Noise (Left)









Summary.txt

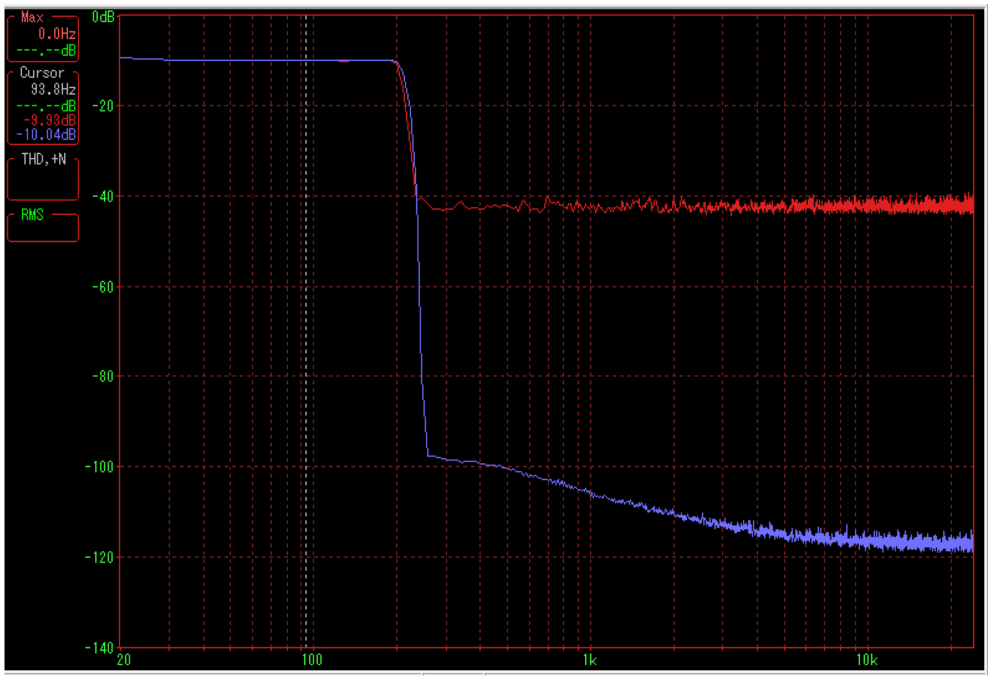
Corrected Gain Difference Analysis

20-40 Hz: Diff=0.004 dB
40-80 Hz: Diff=0.001 dB
80-160 Hz: Diff=0.006 dB
160-315 Hz: Diff=0.013 dB
315-630 Hz: Diff=0.029 dB
630-1250 Hz: Diff=0.057 dB
1250-2500 Hz: Diff=0.109 dB
2500-5000 Hz: Diff=0.224 dB
5000-10000 Hz: Diff=0.431 dB
10000-20000 Hz: Diff=0.824 dB

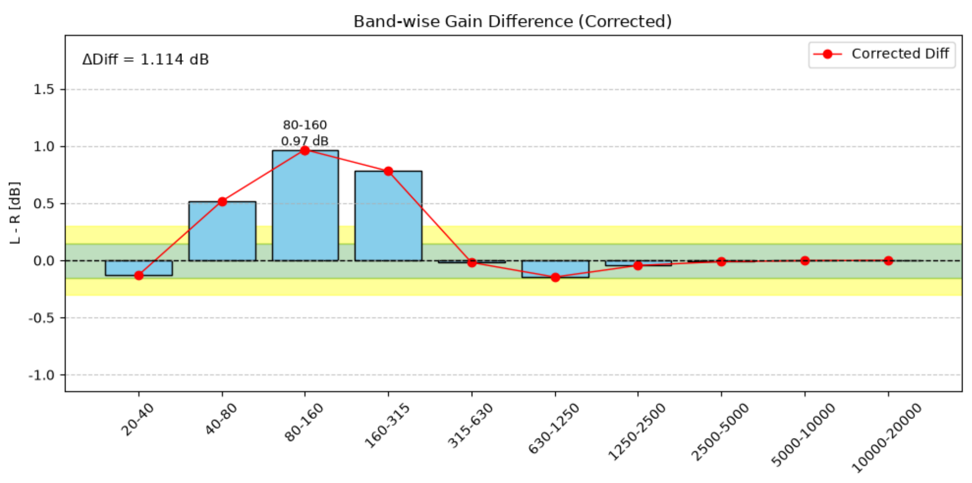
--- Summary ---
ΔDiff (Band) = 0.823 dB
ΔDiff (Welch max) = 24.003 dB
Max Diff Freq = 19992.2 Hz
Judgement: Check required: Possible equipment or measurement problem.

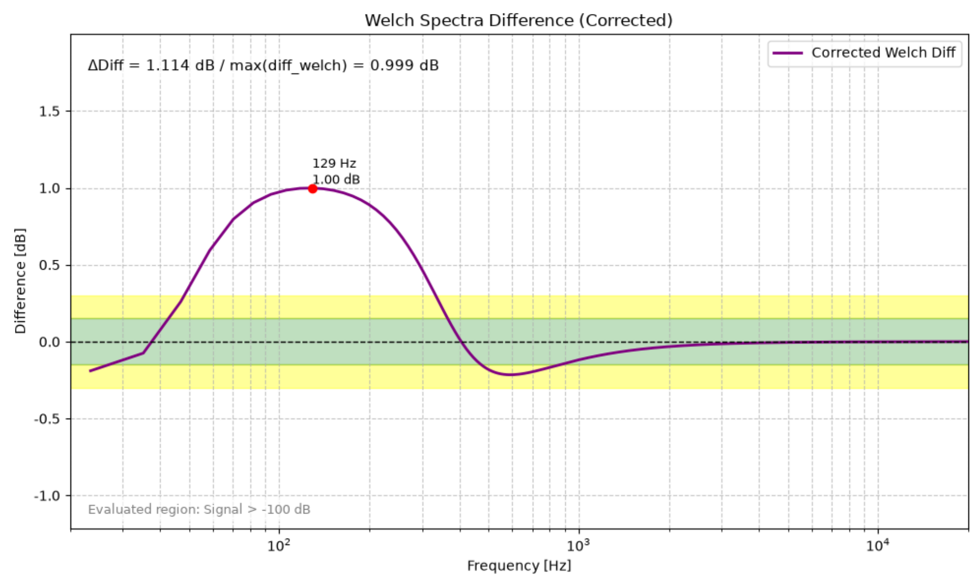
--- Channel Check ---
Energy Ratio (L/R) = 1.05
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

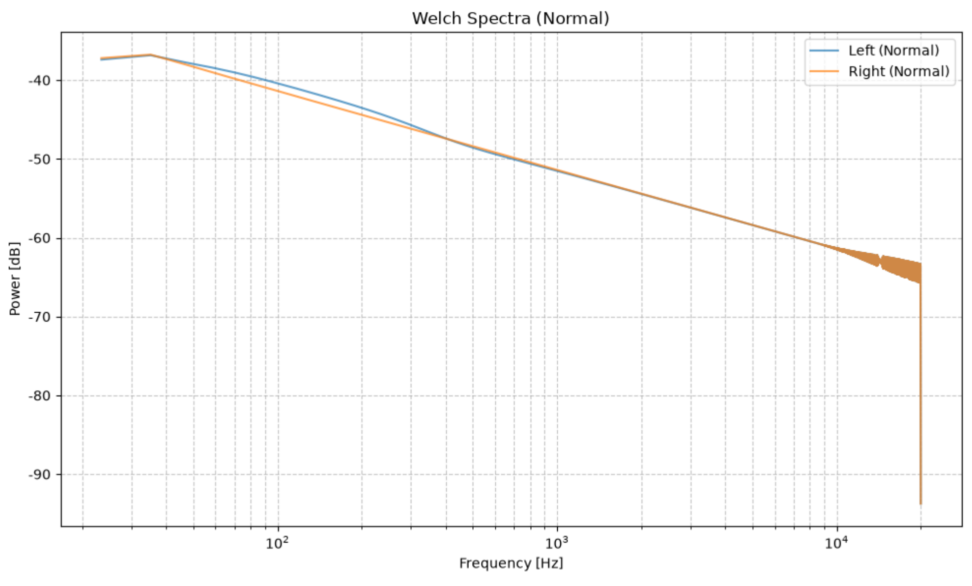
Shows the level of noise superimposed on the Left-channel signal in an image taken during a sweep

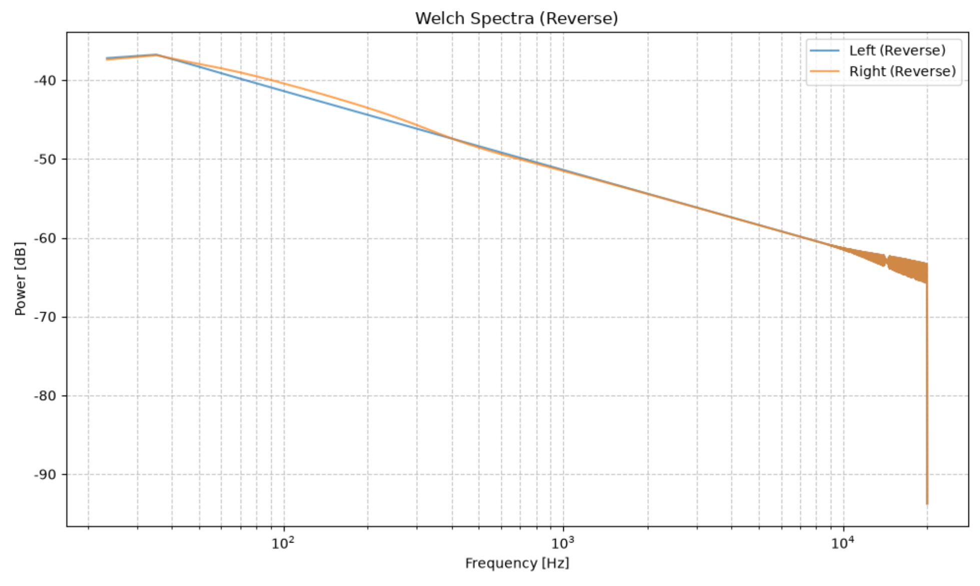


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, PEQ 125 Hz Q10 +1 dB (Left)









Summary.txt

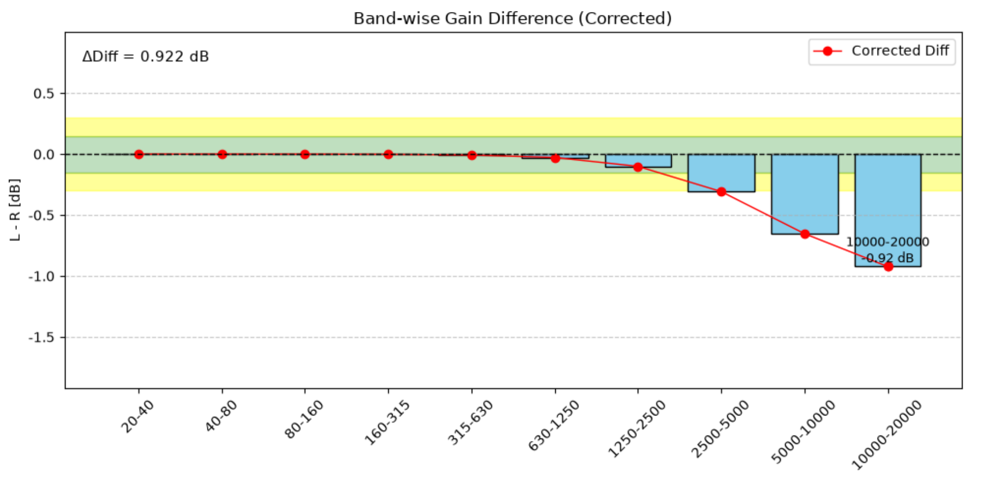
Corrected Gain Difference Analysis

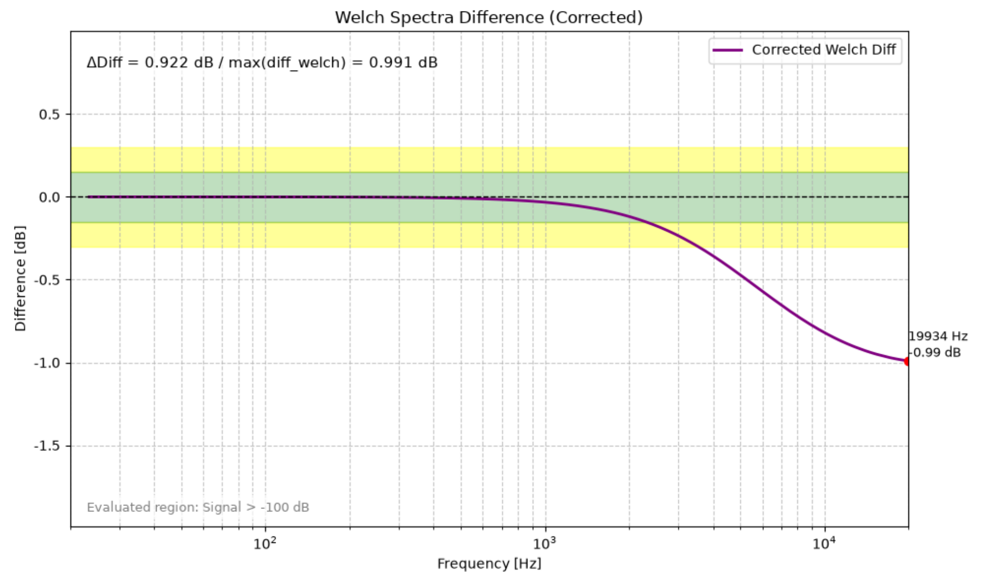
20-40 Hz: Diff=-0.129 dB
40-80 Hz: Diff=0.517 dB
80-160 Hz: Diff=0.968 dB
160-315 Hz: Diff=0.781 dB
315-630 Hz: Diff=-0.018 dB
630-1250 Hz: Diff=-0.146 dB
1250-2500 Hz: Diff=-0.044 dB
2500-5000 Hz: Diff=-0.011 dB
5000-10000 Hz: Diff=-0.003 dB
10000-20000 Hz: Diff=-0.000 dB

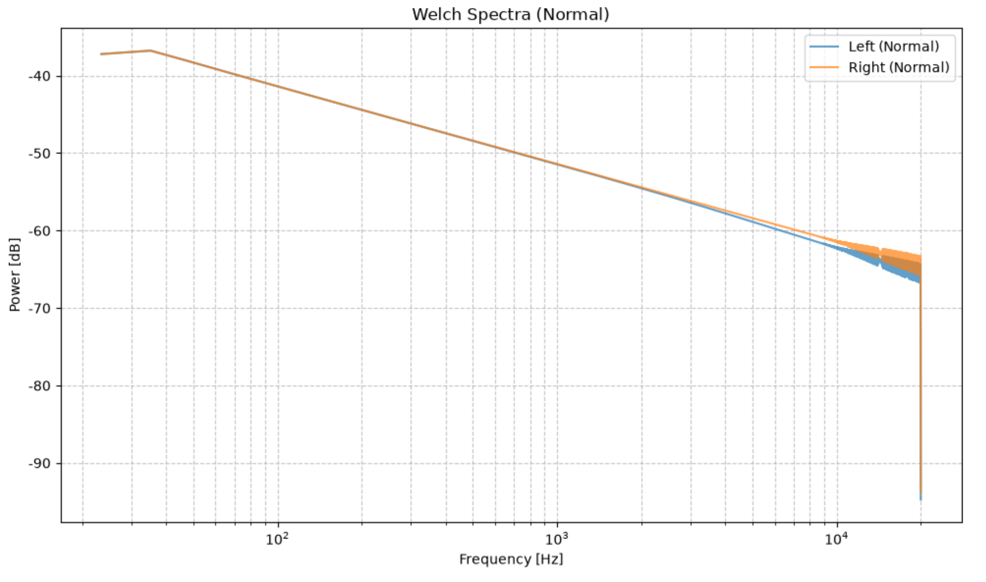
--- Summary ---
ΔDiff (Band) = 1.114 dB
ΔDiff (Welch max) = 0.999 dB
Max Diff Freq = 128.9 Hz
Judgement: Check required: Possible equipment or measurement problem.

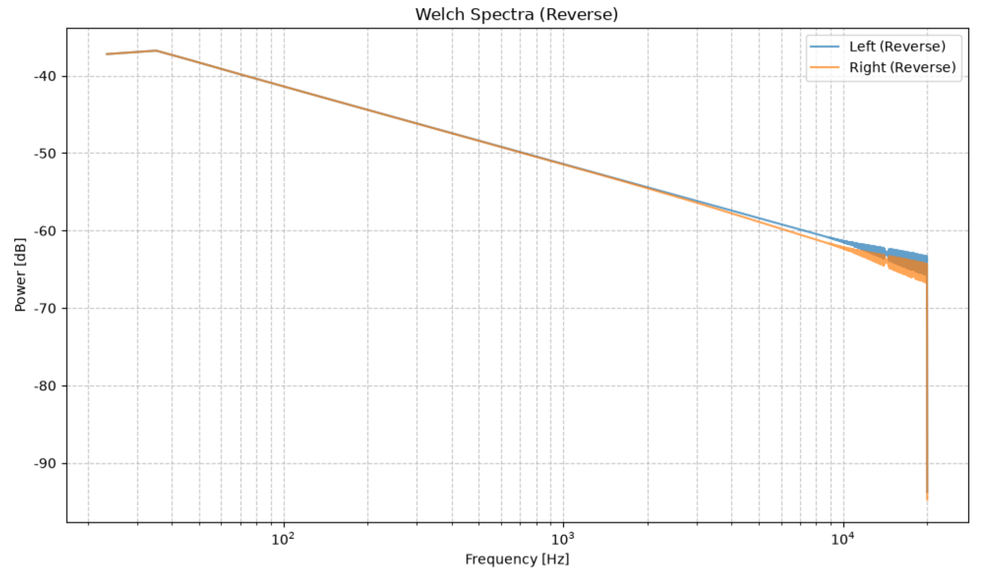
--- Channel Check ---
Energy Ratio (L/R) = 1.05
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, HSF 5 kHz -1 dB (Left)









Summary.txt

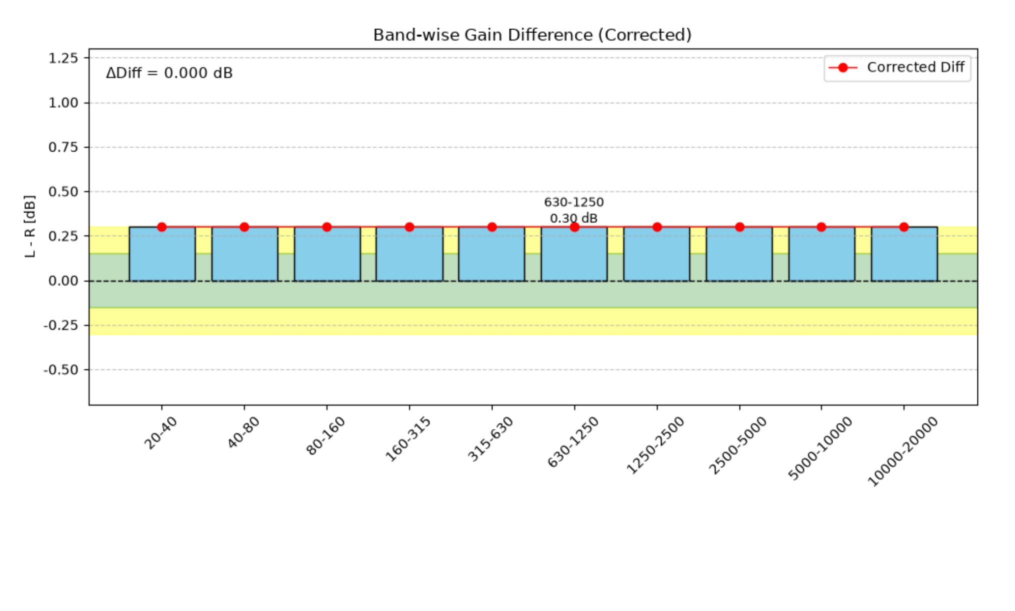
Corrected Gain Difference Analysis

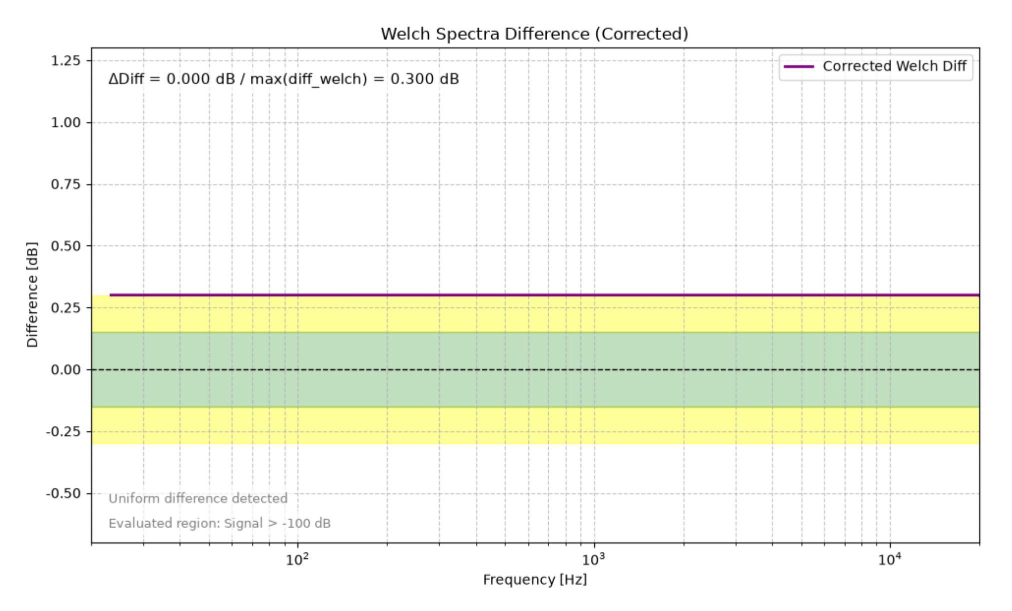
20-40 Hz: Diff=-0.000 dB
40-80 Hz: Diff=-0.000 dB
80-160 Hz: Diff=-0.000 dB
160-315 Hz: Diff=-0.002 dB
315-630 Hz: Diff=-0.007 dB
630-1250 Hz: Diff=-0.027 dB
1250-2500 Hz: Diff=-0.100 dB
2500-5000 Hz: Diff=-0.307 dB
5000-10000 Hz: Diff=-0.652 dB
10000-20000 Hz: Diff=-0.922 dB

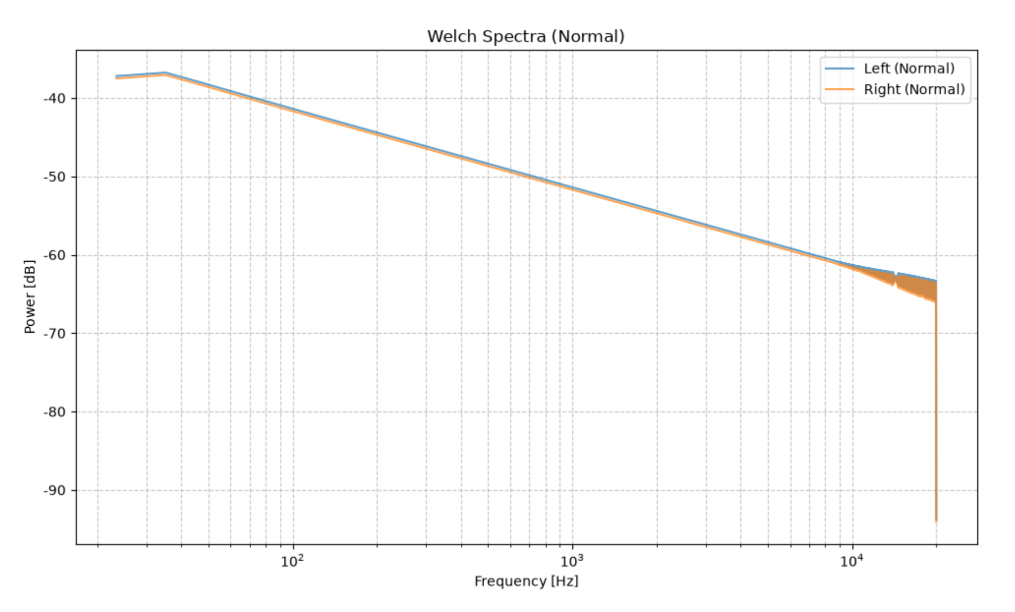
--- Summary ---
ΔDiff (Band) = 0.922 dB
ΔDiff (Welch max) = 0.991 dB
Max Diff Freq = 19933.6 Hz
Judgement: Check required: Possible equipment or measurement problem.

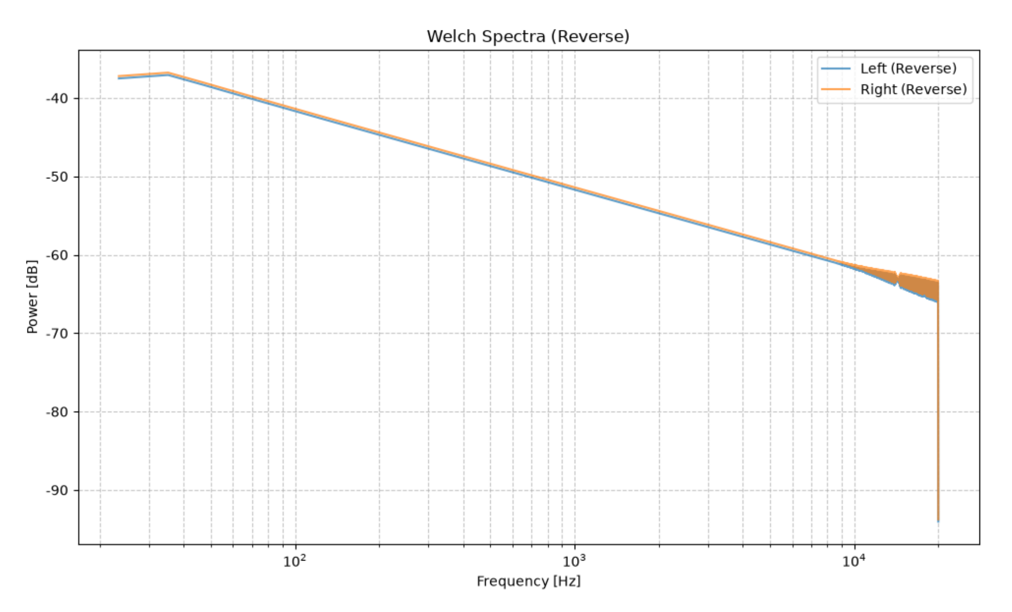
--- Channel Check ---
Energy Ratio (L/R) = 0.96
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Right)









Summary.txt

Corrected Gain Difference Analysis

20-40 Hz: Diff=0.300 dB
40-80 Hz: Diff=0.300 dB
80-160 Hz: Diff=0.300 dB
160-315 Hz: Diff=0.300 dB
315-630 Hz: Diff=0.300 dB
630-1250 Hz: Diff=0.300 dB
1250-2500 Hz: Diff=0.300 dB
2500-5000 Hz: Diff=0.300 dB
5000-10000 Hz: Diff=0.300 dB
10000-20000 Hz: Diff=0.300 dB

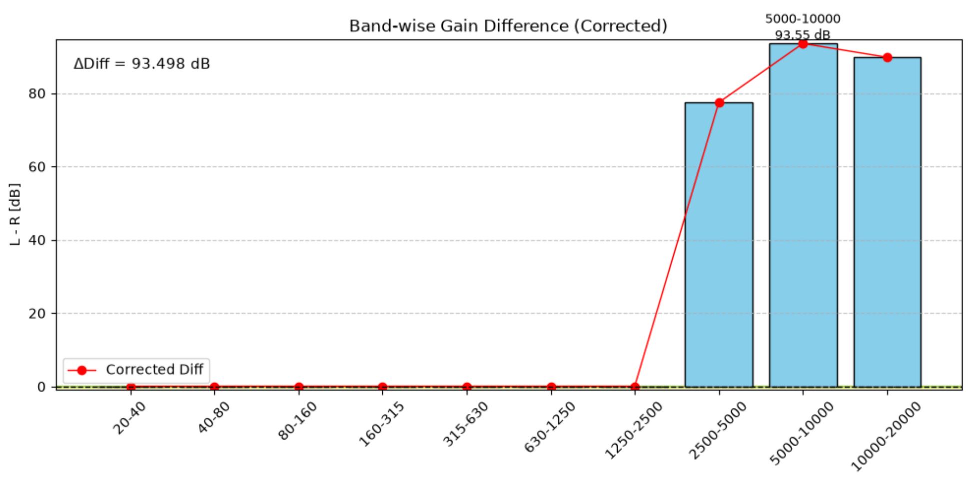
--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.300 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

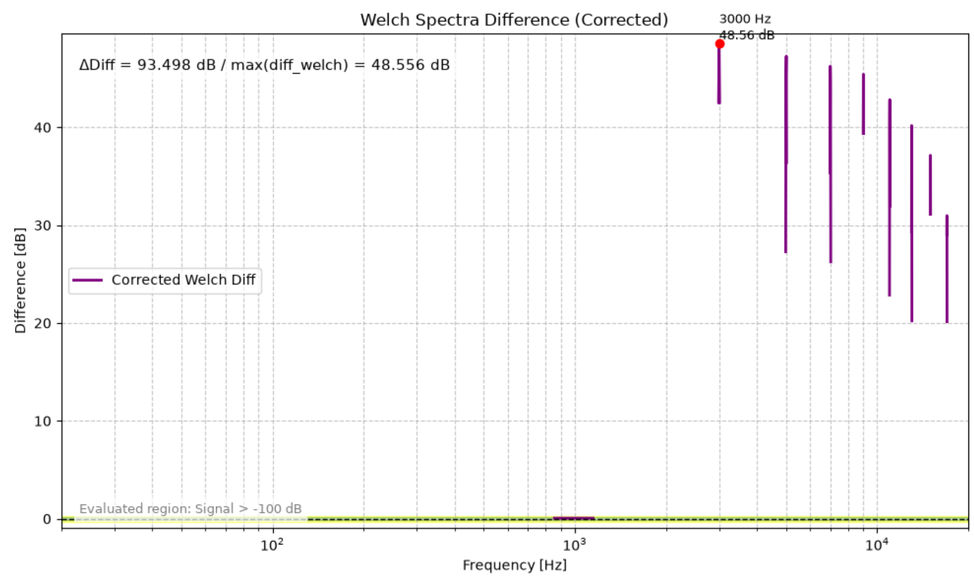
--- Channel Check ---
Energy Ratio (L/R) = 1.07
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

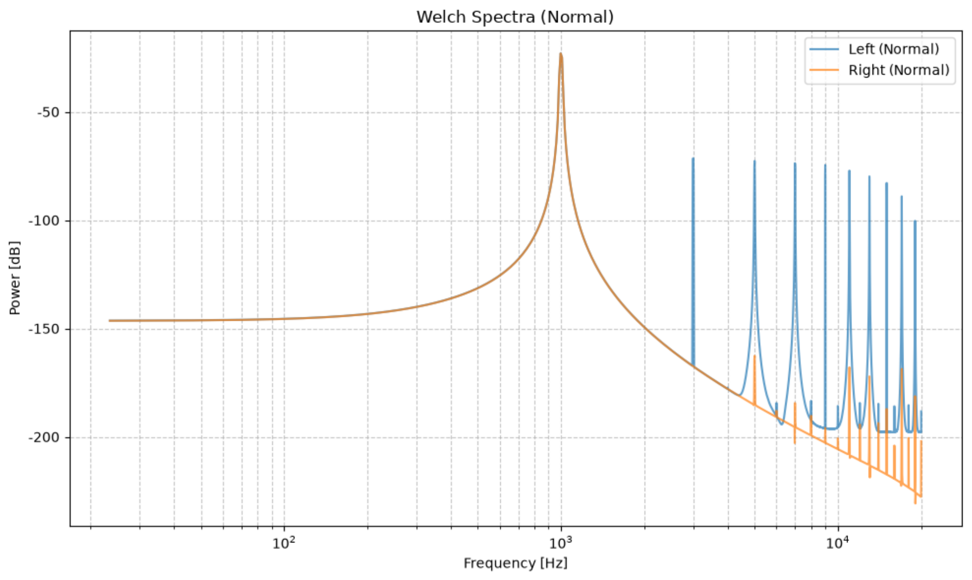
Test signals generated and processed using WaveGene and Cakewalk SONAR + GClip

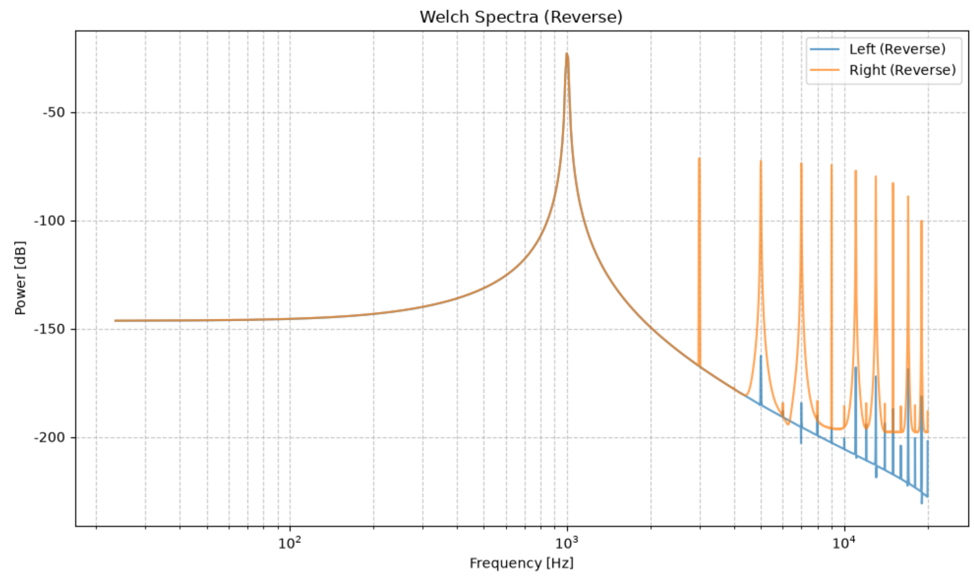
-10 dBFS 1 kHz Sine Wave, 1% THD (Left)

For these single-tone simulations, Band-analysis values outside the frequency regions containing meaningful signal energy should not be interpreted as broadband DUT characteristics. They are included here to show the script’s behavior with narrowband input signals.









Summary.txt

Corrected Gain Difference Analysis

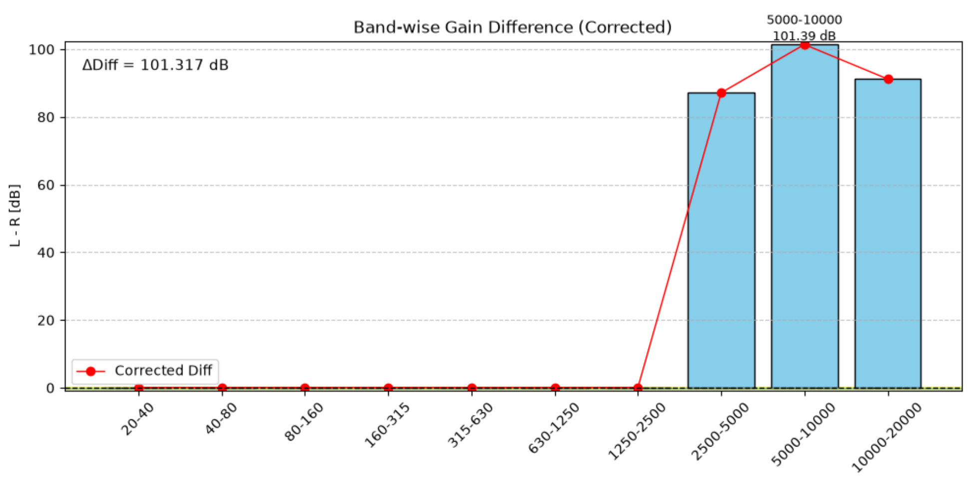
20-40 Hz: Diff=0.055 dB
40-80 Hz: Diff=0.055 dB
80-160 Hz: Diff=0.055 dB
160-315 Hz: Diff=0.055 dB
315-630 Hz: Diff=0.055 dB
630-1250 Hz: Diff=0.055 dB
1250-2500 Hz: Diff=0.055 dB
2500-5000 Hz: Diff=77.461 dB
5000-10000 Hz: Diff=93.553 dB
10000-20000 Hz: Diff=89.814 dB

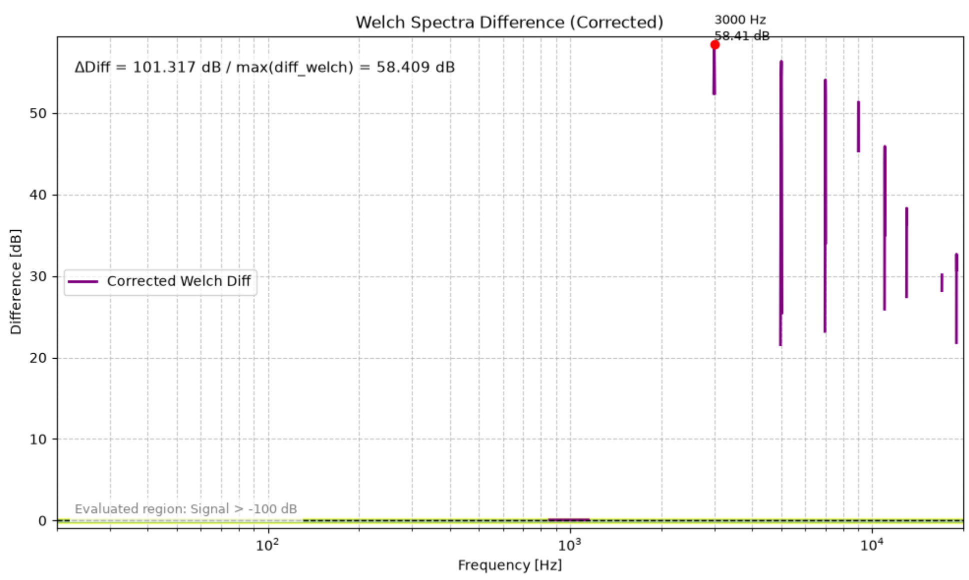
--- Summary ---
ΔDiff (Band) = 93.498 dB
ΔDiff (Welch max) = 48.556 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.

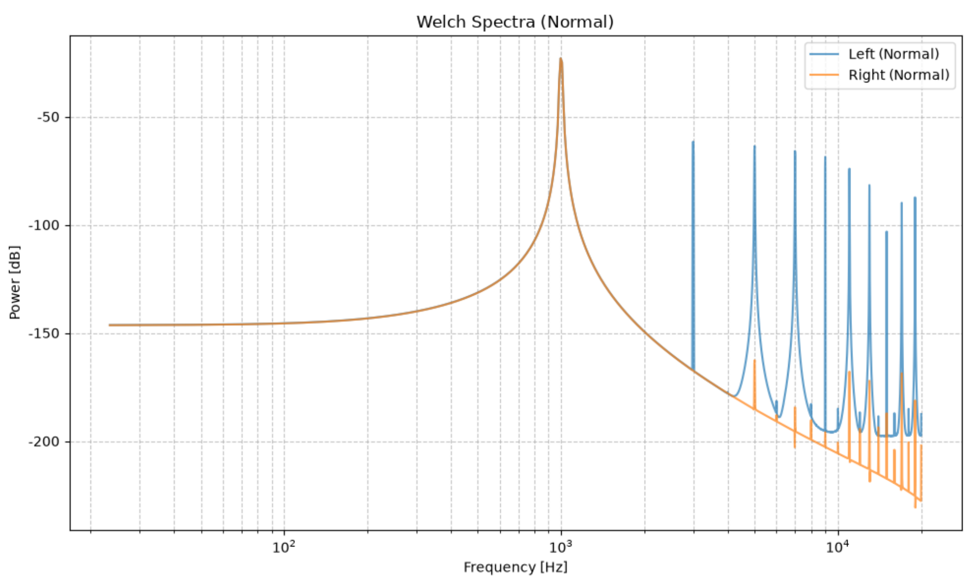
--- Channel Check ---
Energy Ratio (L/R) = 1.01
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

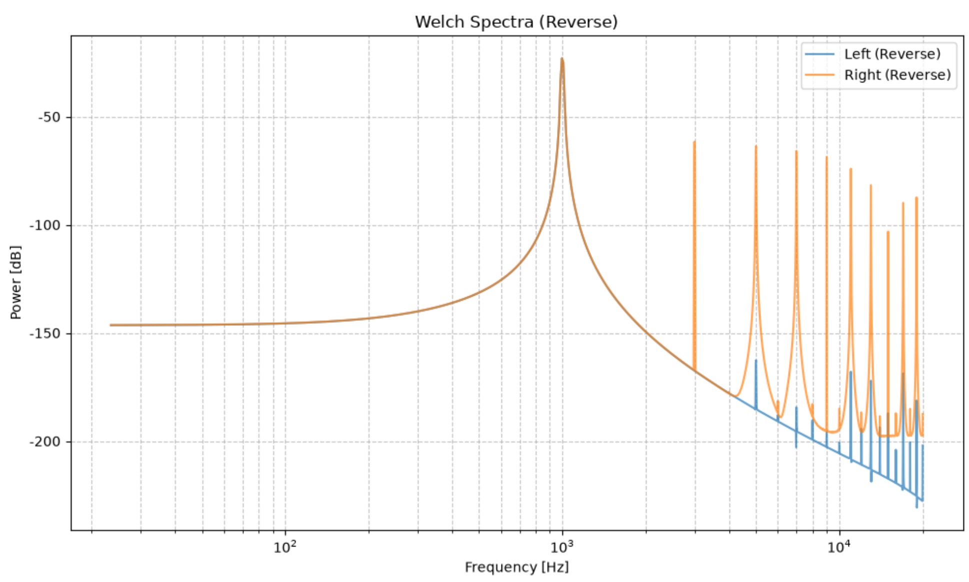
-10 dBFS 1 kHz Sine Wave, 3% THD (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

Corrected Gain Difference Analysis

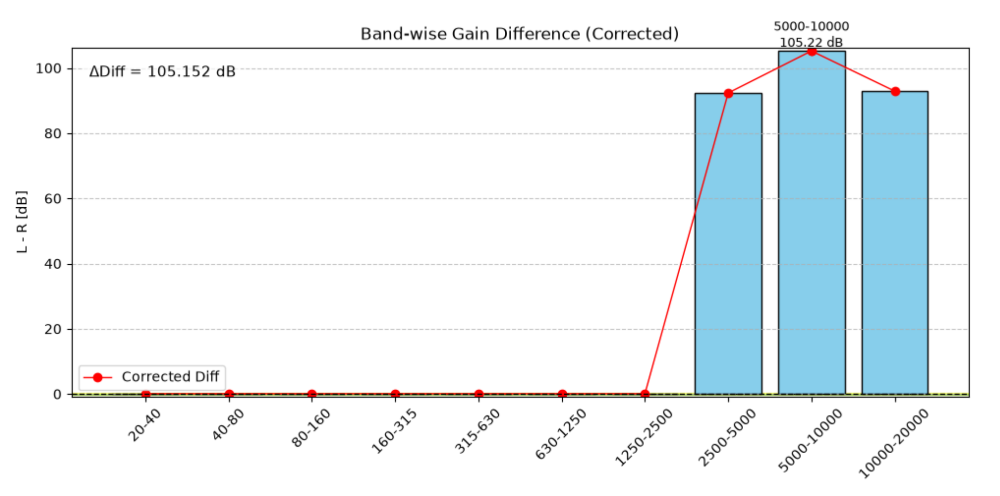
20-40 Hz: Diff=0.071 dB
40-80 Hz: Diff=0.071 dB
80-160 Hz: Diff=0.071 dB
160-315 Hz: Diff=0.071 dB
315-630 Hz: Diff=0.071 dB
630-1250 Hz: Diff=0.071 dB
1250-2500 Hz: Diff=0.071 dB
2500-5000 Hz: Diff=87.135 dB
5000-10000 Hz: Diff=101.388 dB
10000-20000 Hz: Diff=91.193 dB

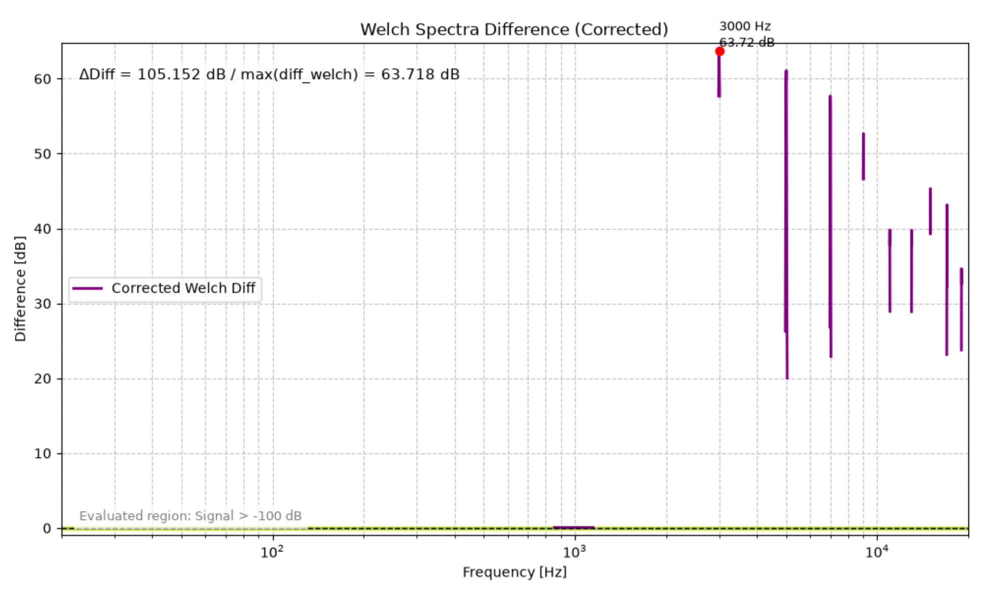
--- Summary ---
ΔDiff (Band) = 101.317 dB
ΔDiff (Welch max) = 58.409 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.

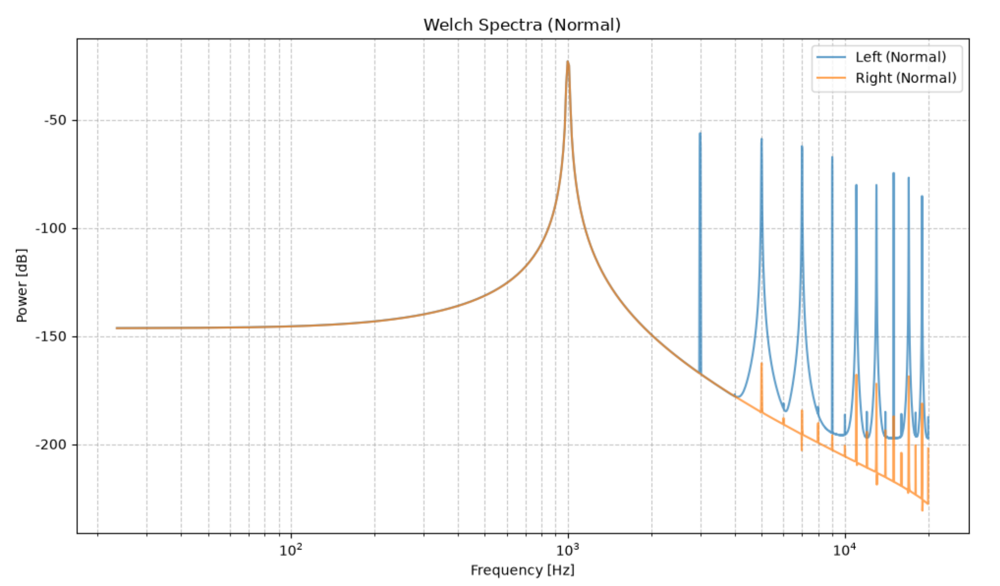
--- Channel Check ---
Energy Ratio (L/R) = 1.02
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

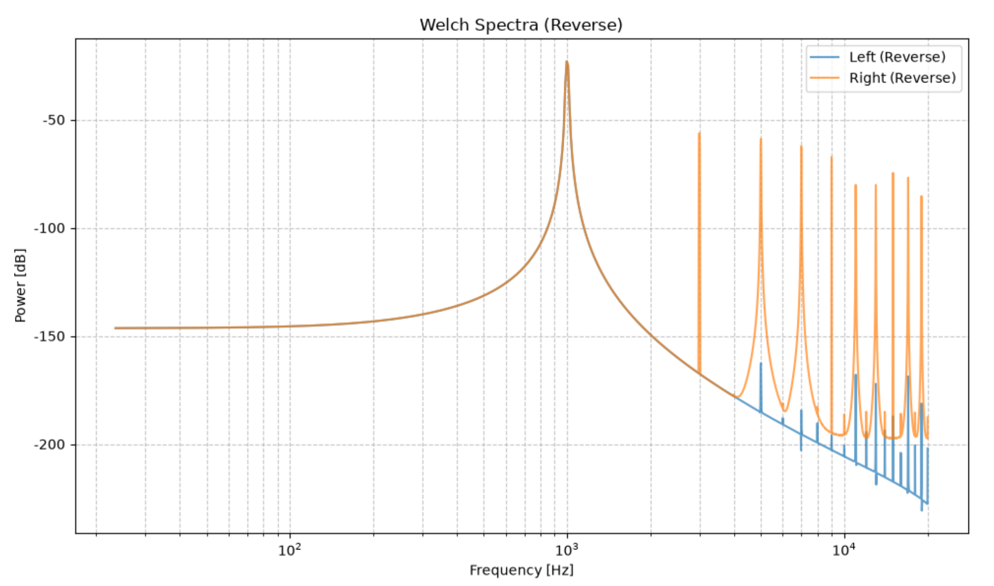
-10 dBFS 1 kHz Sine Wave, 5% THD (Left)

The simulation using a 1 kHz sine wave is not intended as a signal for measuring the device's bandwidth characteristics. It was used to evaluate the script's decision-making logic for narrowband signals.









Summary.txt

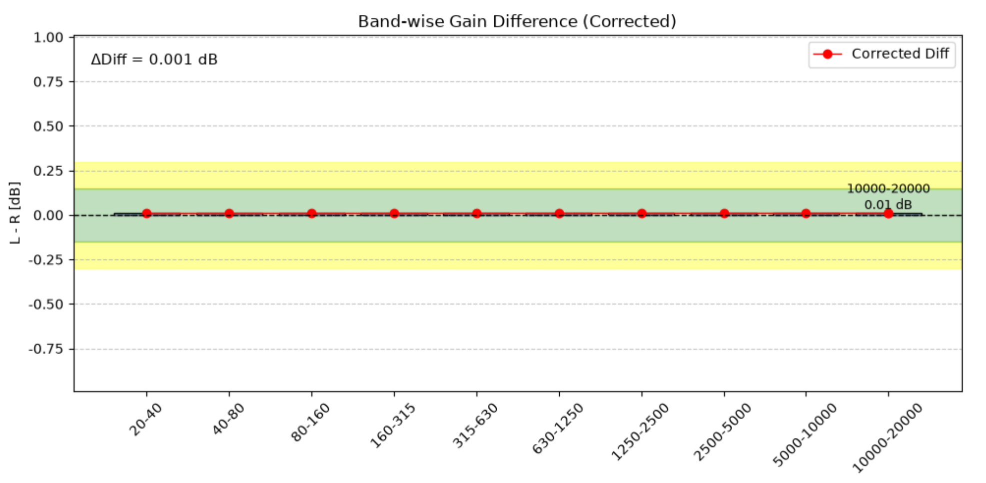
Corrected Gain Difference Analysis

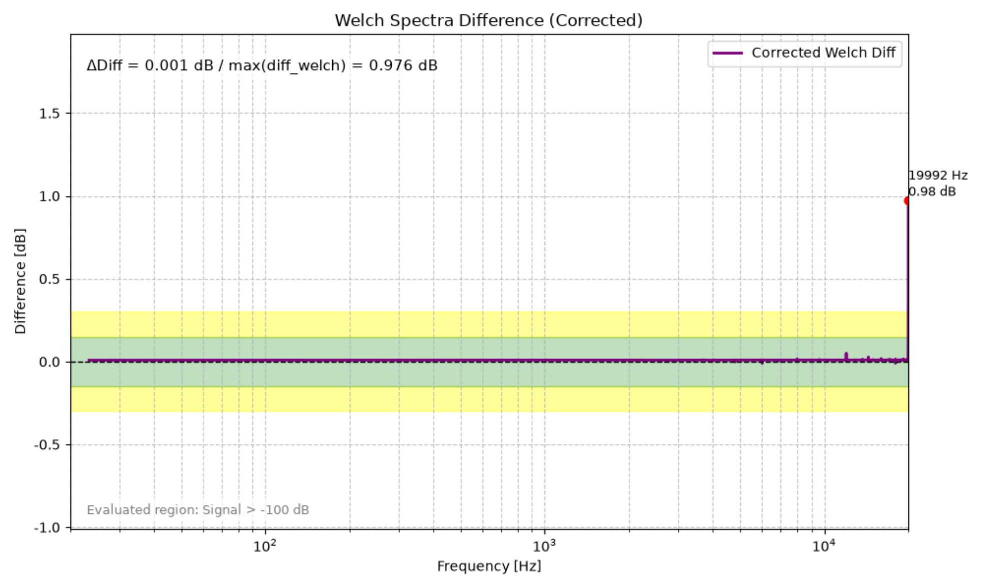
20-40 Hz: Diff=0.066 dB
40-80 Hz: Diff=0.066 dB
80-160 Hz: Diff=0.066 dB
160-315 Hz: Diff=0.067 dB
315-630 Hz: Diff=0.067 dB
630-1250 Hz: Diff=0.067 dB
1250-2500 Hz: Diff=0.067 dB
2500-5000 Hz: Diff=92.330 dB
5000-10000 Hz: Diff=105.218 dB
10000-20000 Hz: Diff=92.907 dB

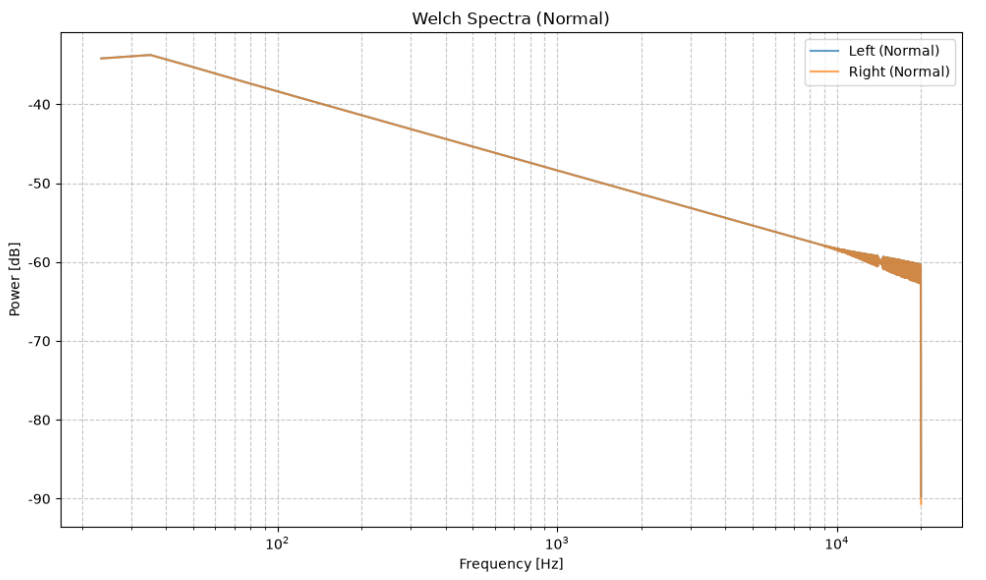
--- Summary ---
ΔDiff (Band) = 105.152 dB
ΔDiff (Welch max) = 63.718 dB
Max Diff Freq = 3000.0 Hz
Judgement: Check required: Possible equipment or measurement problem.

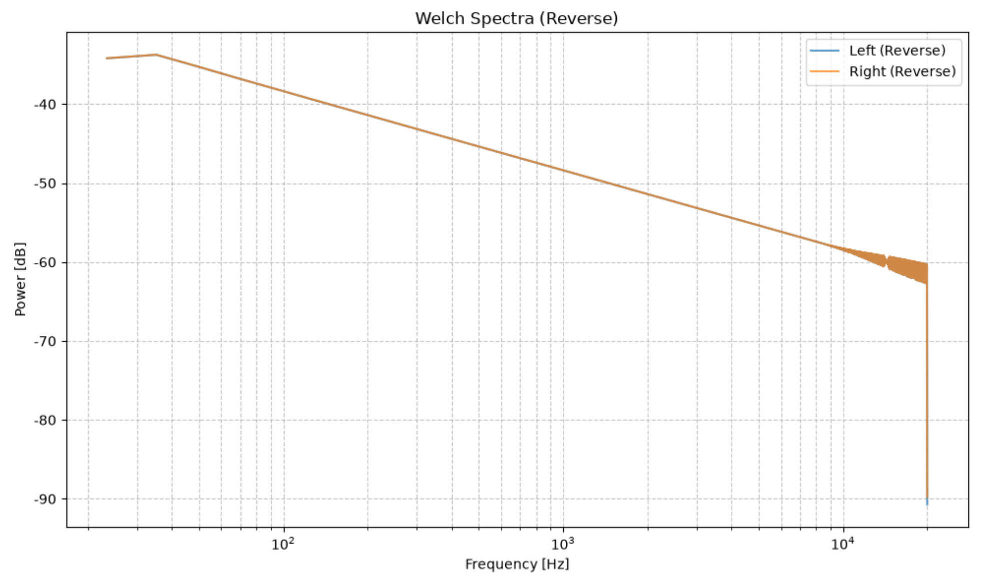
--- Channel Check ---
Energy Ratio (L/R) = 1.02
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 1% THD (Left)







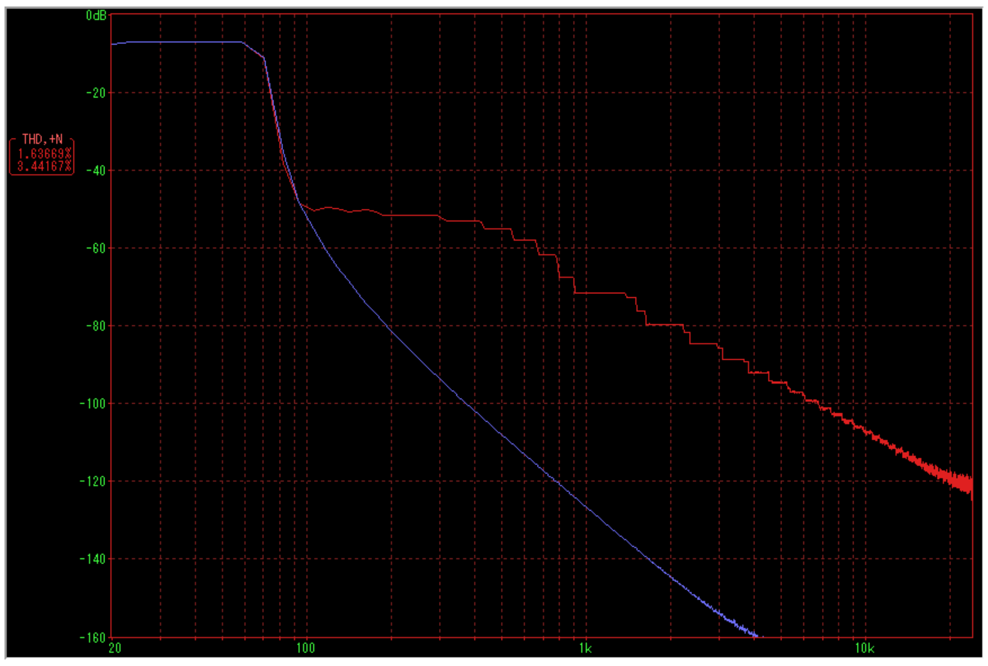


Summary.txt

Corrected Gain Difference Analysis

20-40 Hz: Diff=0.008 dB
40-80 Hz: Diff=0.008 dB
80-160 Hz: Diff=0.009 dB
160-315 Hz: Diff=0.009 dB
315-630 Hz: Diff=0.009 dB
630-1250 Hz: Diff=0.009 dB
1250-2500 Hz: Diff=0.009 dB
2500-5000 Hz: Diff=0.009 dB
5000-10000 Hz: Diff=0.009 dB
10000-20000 Hz: Diff=0.010 dB

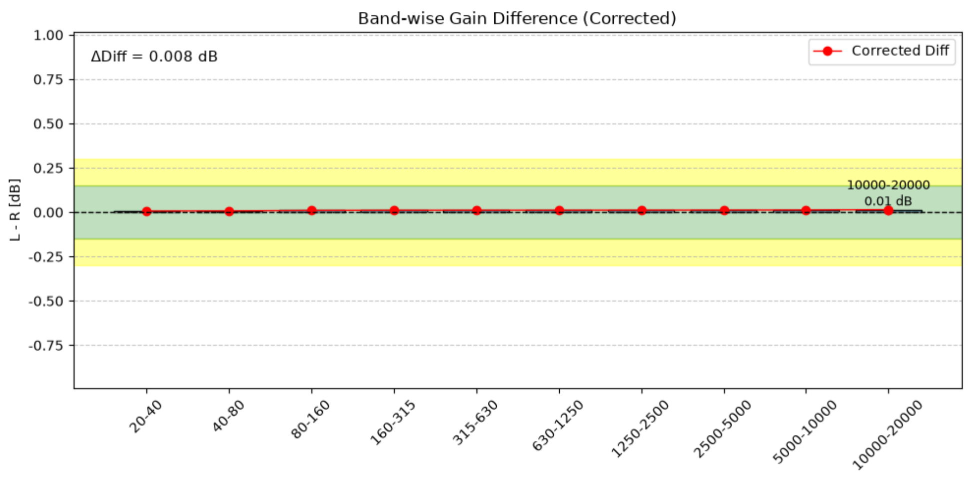
Show the level of harmonics superimposed on the Left-channel signal by the GClip plugin in an images captured during a sweep.

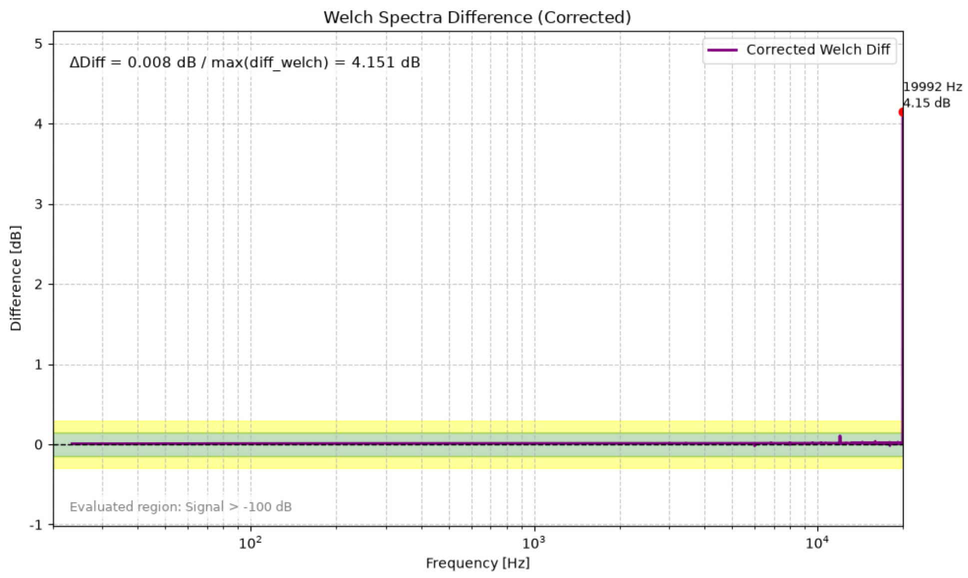


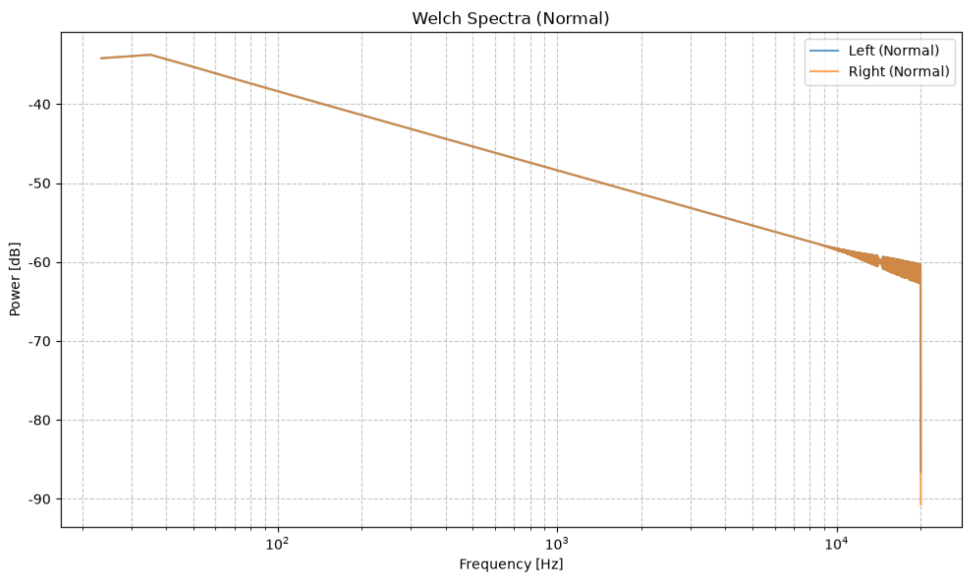
--- Summary ---
ΔDiff (Band) = 0.001 dB
ΔDiff (Welch max) = 0.976 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

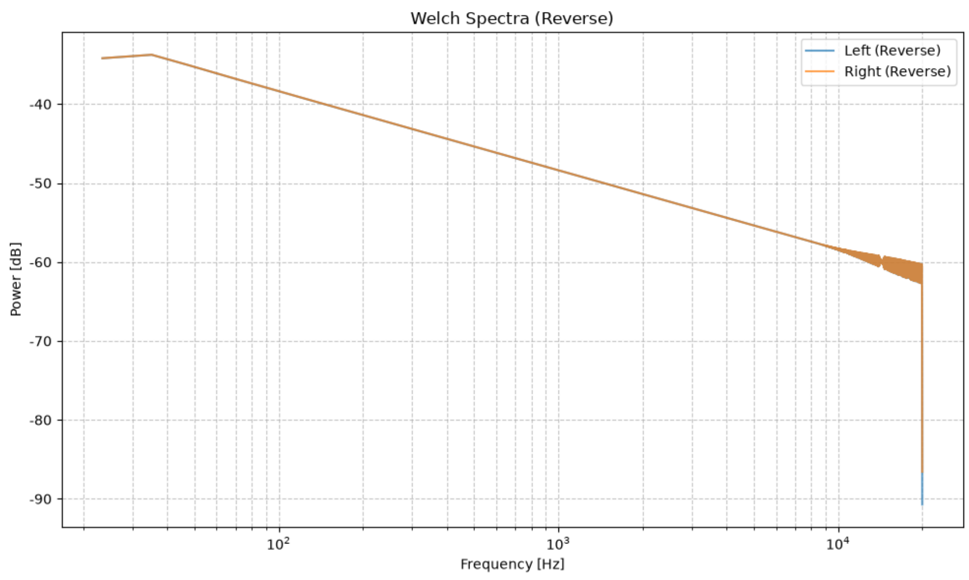
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 3% THD (Left)









Summary.txt

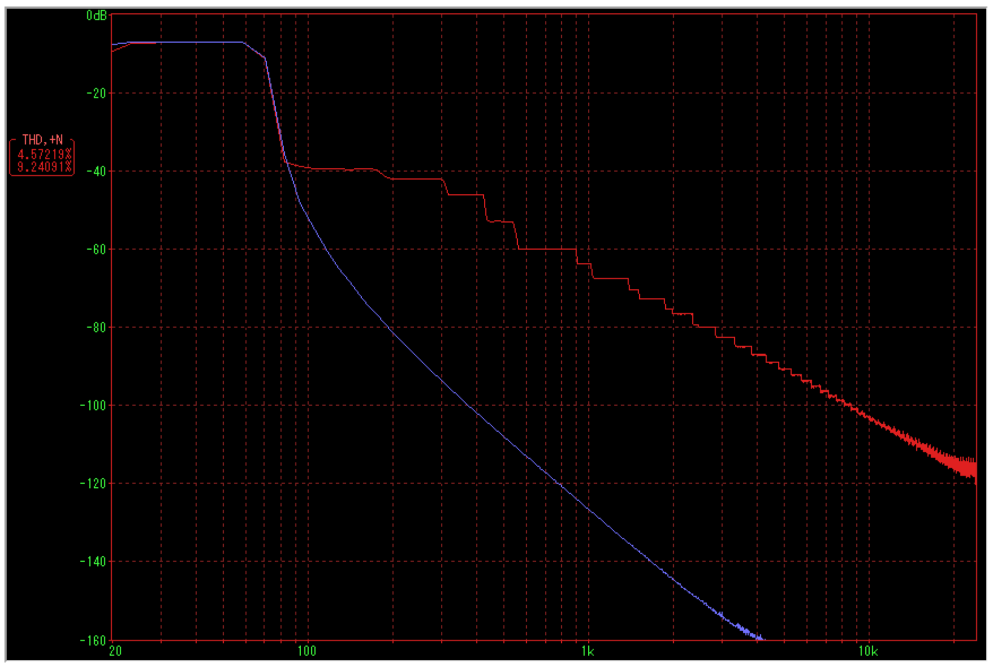
Corrected Gain Difference Analysis

20-40 Hz: Diff=0.007 dB
40-80 Hz: Diff=0.008 dB
80-160 Hz: Diff=0.010 dB
160-315 Hz: Diff=0.011 dB
315-630 Hz: Diff=0.011 dB
630-1250 Hz: Diff=0.011 dB
1250-2500 Hz: Diff=0.012 dB
2500-5000 Hz: Diff=0.012 dB
5000-10000 Hz: Diff=0.013 dB
10000-20000 Hz: Diff=0.014 dB

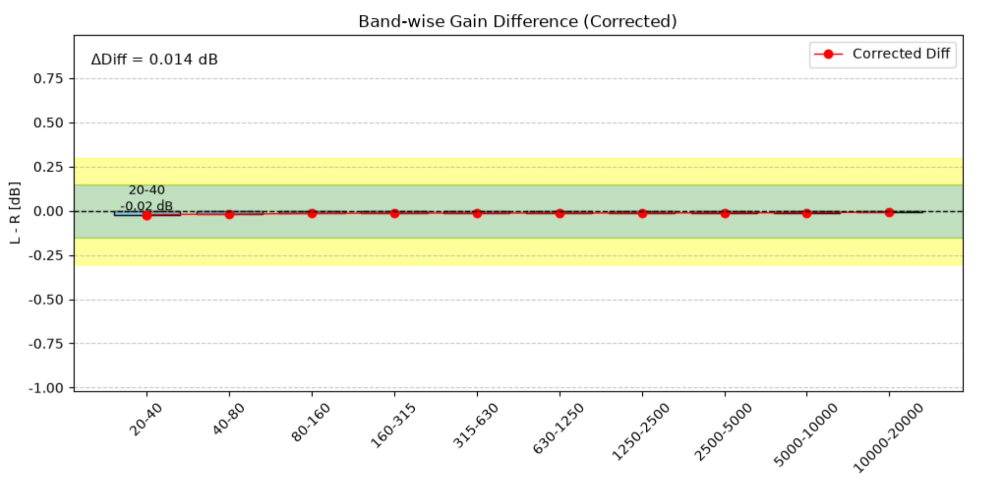
--- Summary ---
ΔDiff (Band) = 0.008 dB
ΔDiff (Welch max) = 4.151 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

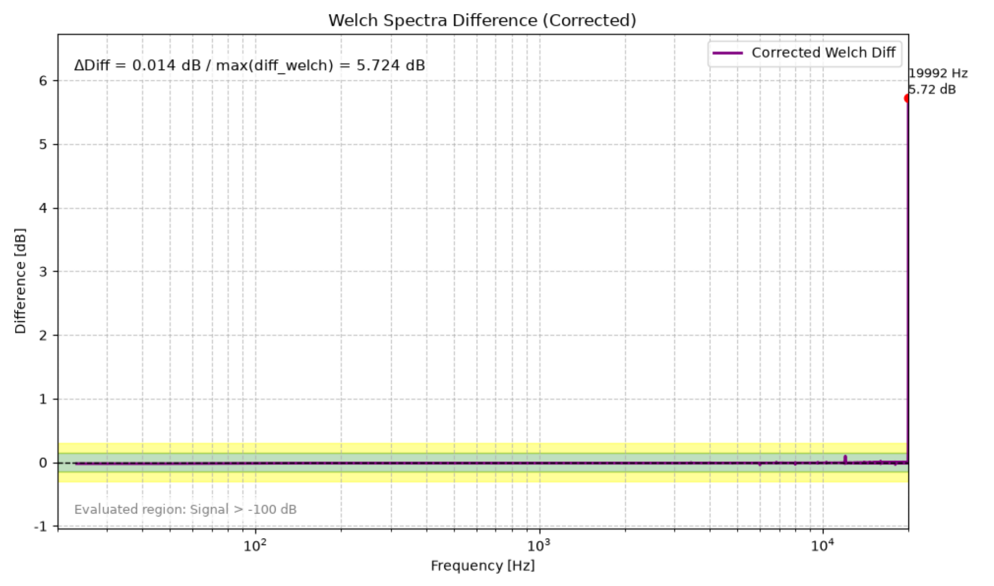
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

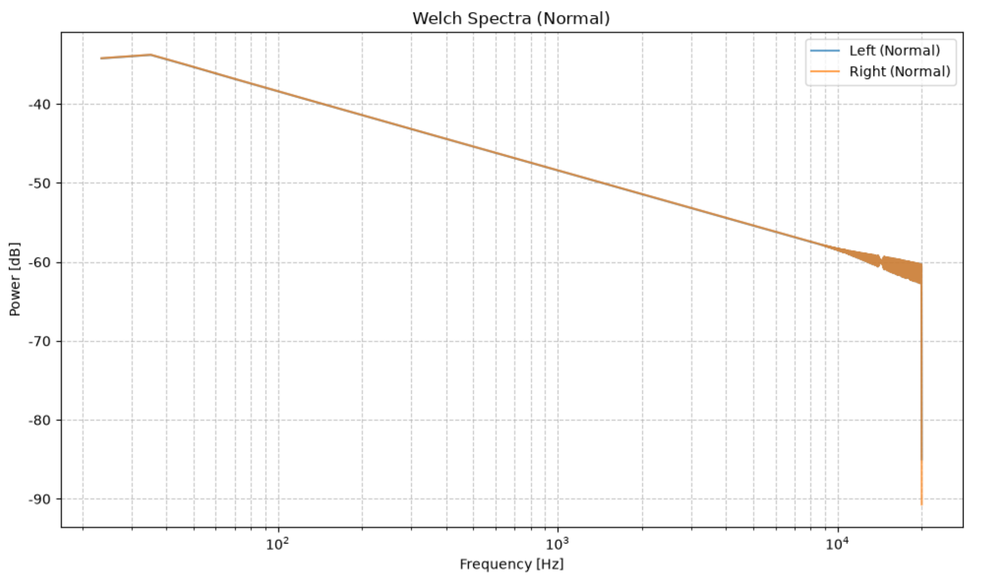
Show the level of harmonics superimposed on the Left-channel signal by the GClip plugin in an images captured during a sweep.

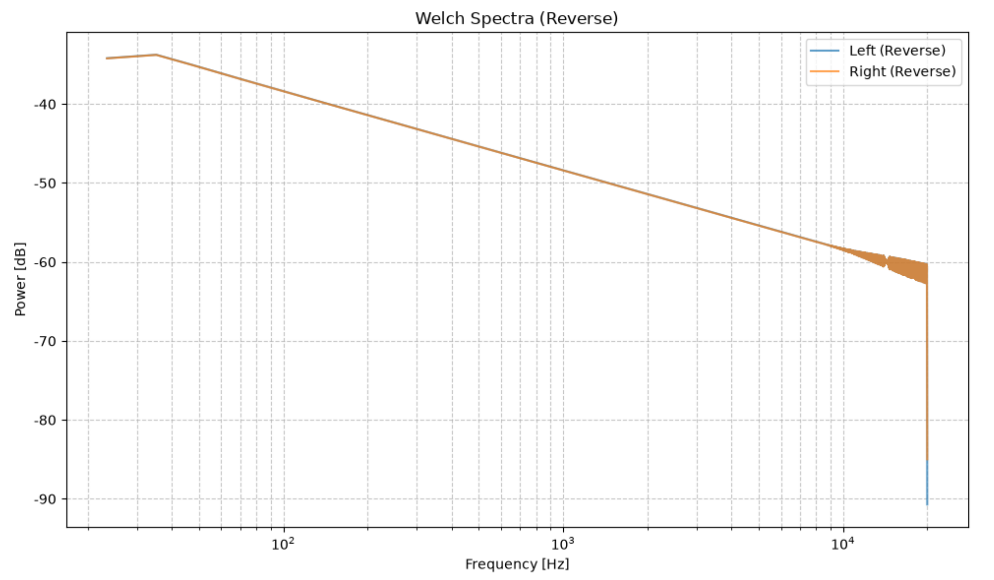


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, 5% THD (Left)









Summary.txt

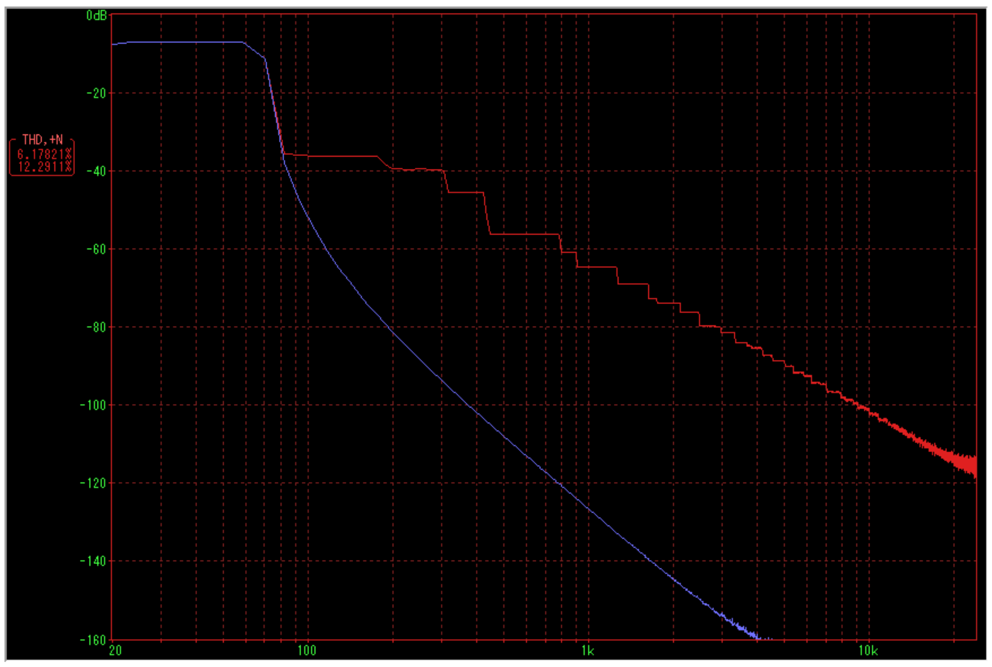
Corrected Gain Difference Analysis

20-40 Hz: Diff=-0.021 dB
40-80 Hz: Diff=-0.020 dB
80-160 Hz: Diff=-0.015 dB
160-315 Hz: Diff=-0.013 dB
315-630 Hz: Diff=-0.013 dB
630-1250 Hz: Diff=-0.013 dB
1250-2500 Hz: Diff=-0.013 dB
2500-5000 Hz: Diff=-0.012 dB
5000-10000 Hz: Diff=-0.011 dB
10000-20000 Hz: Diff=-0.008 dB

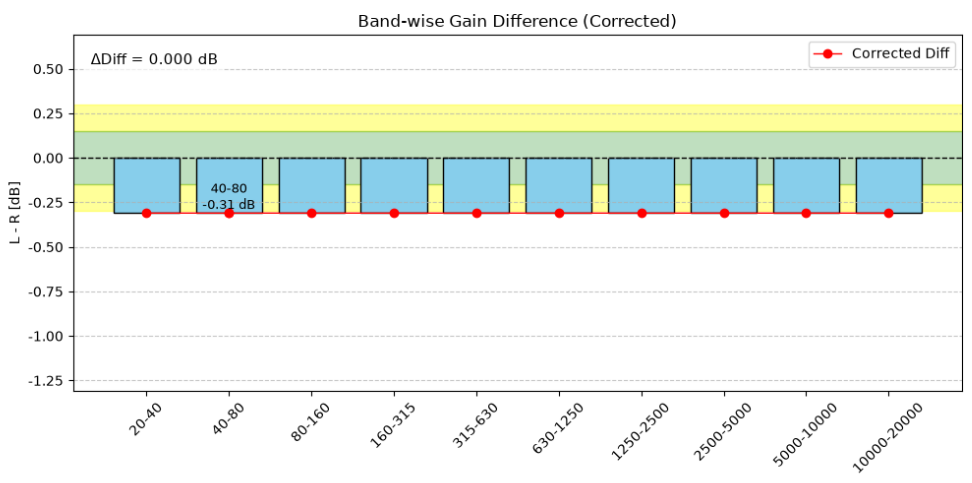
--- Summary ---
ΔDiff (Band) = 0.014 dB
ΔDiff (Welch max) = 5.724 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

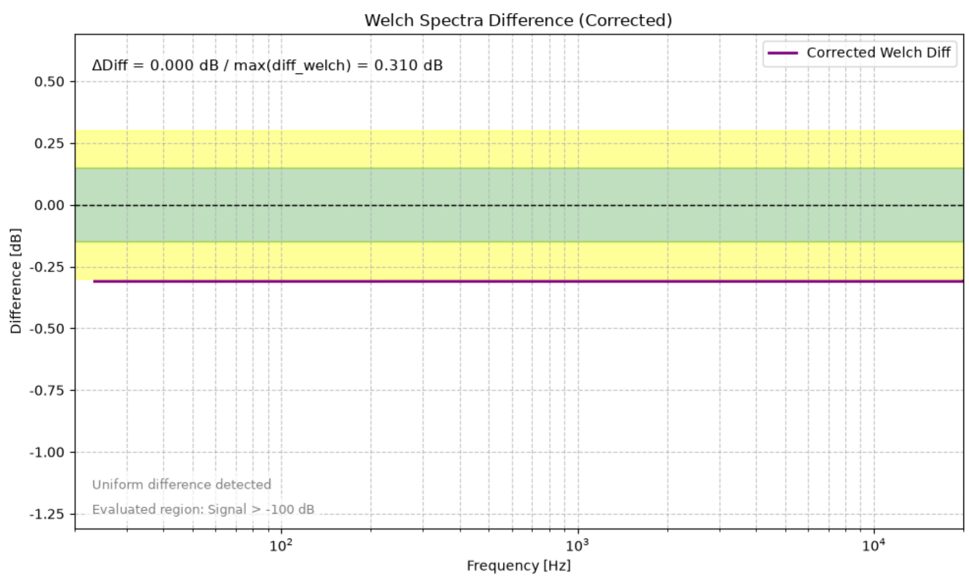
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

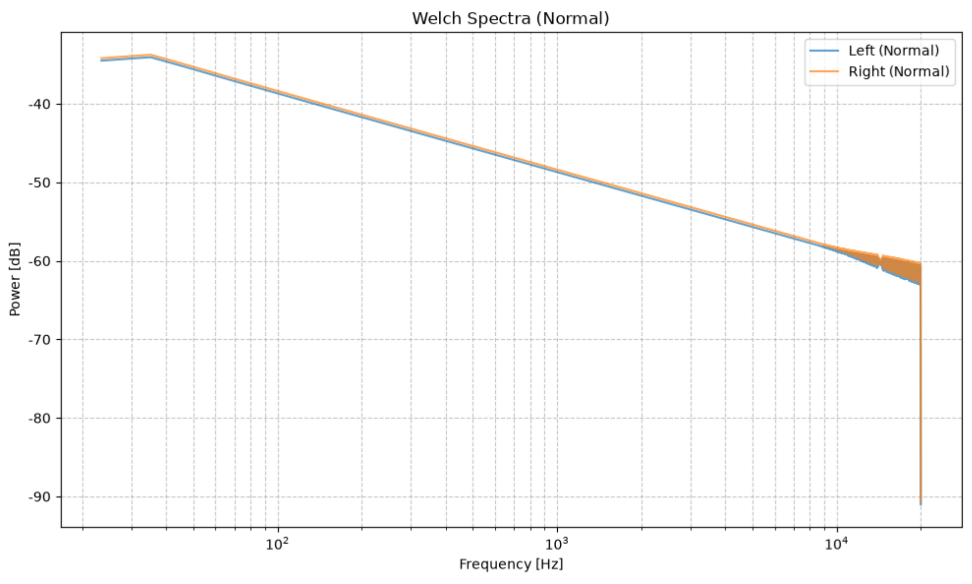
Show the level of harmonics superimposed on the Left-channel signal by the GClip plugin in an images captured during a sweep.

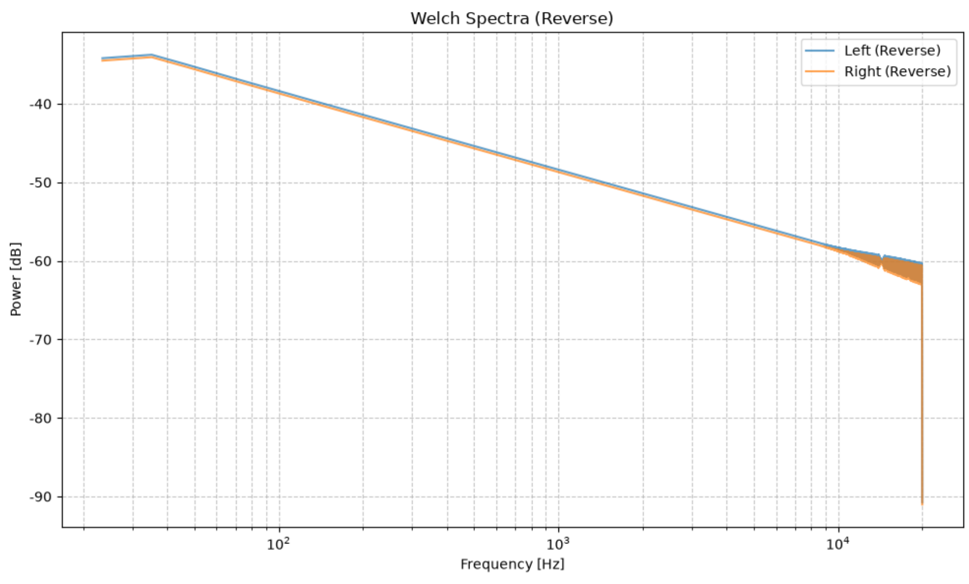


-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Gain -0.3 dB (Left)









Summary.txt

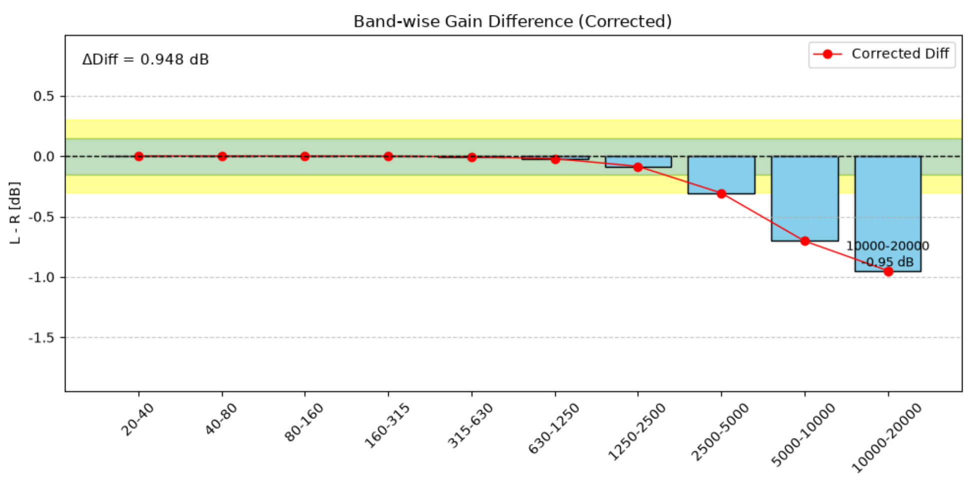
Corrected Gain Difference Analysis

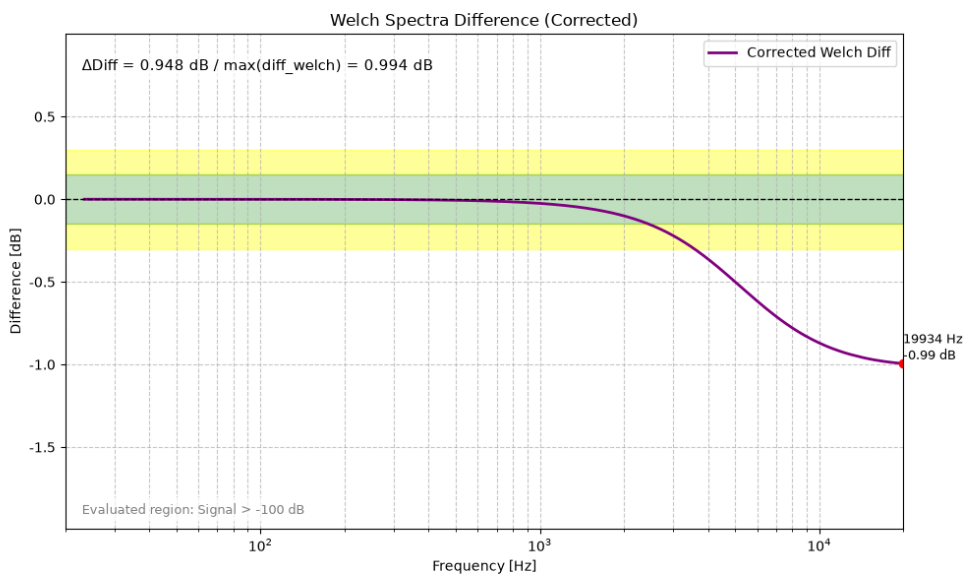
20-40 Hz: Diff=-0.310 dB
40-80 Hz: Diff=-0.310 dB
80-160 Hz: Diff=-0.310 dB
160-315 Hz: Diff=-0.310 dB
315-630 Hz: Diff=-0.310 dB
630-1250 Hz: Diff=-0.310 dB
1250-2500 Hz: Diff=-0.310 dB
2500-5000 Hz: Diff=-0.310 dB
5000-10000 Hz: Diff=-0.310 dB
10000-20000 Hz: Diff=-0.310 dB

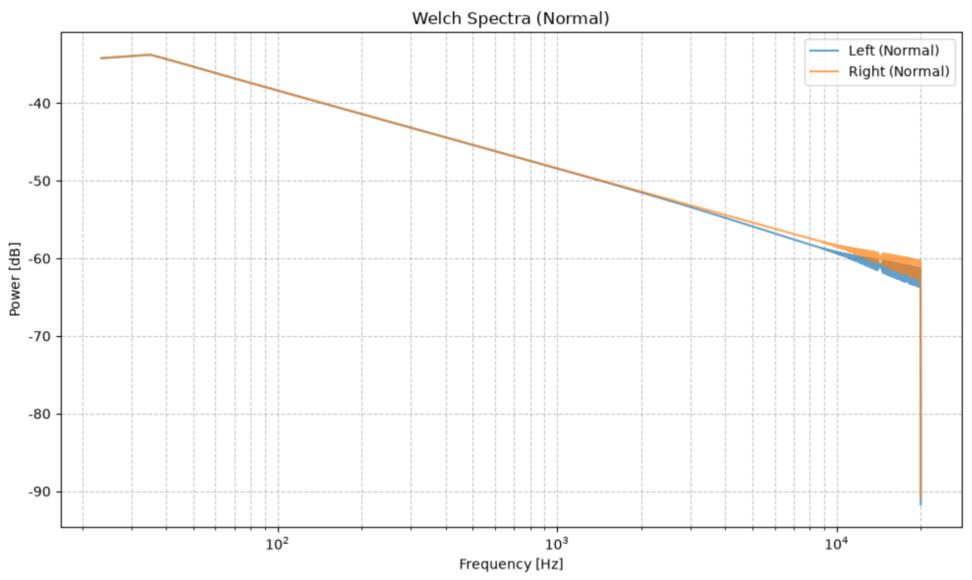
--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.310 dB
Judgement: Caution: Overall gain difference detected (possible DUT issue).

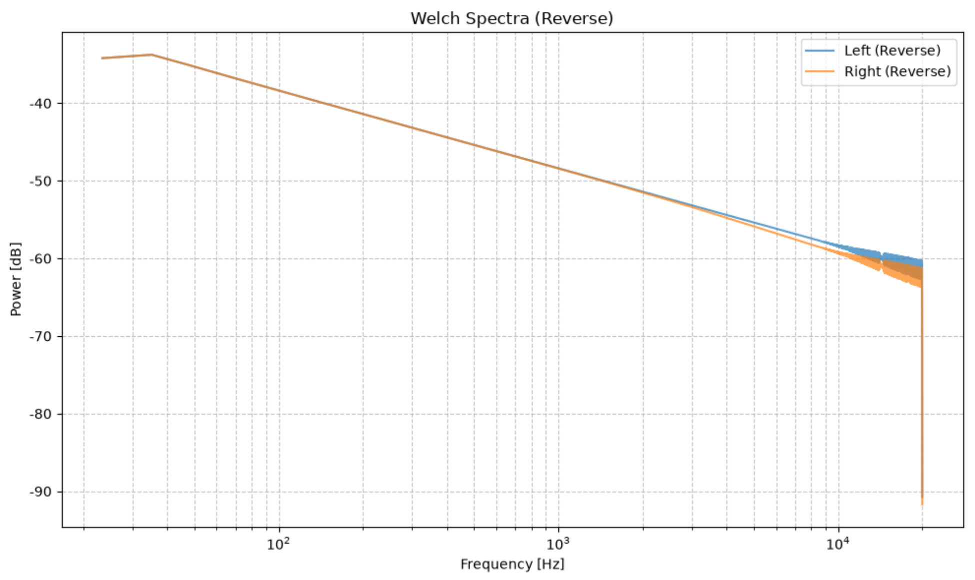
--- Channel Check ---
Energy Ratio (L/R) = 0.93
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, HSF 5 kHz -1 dB (Left)









Summary.txt

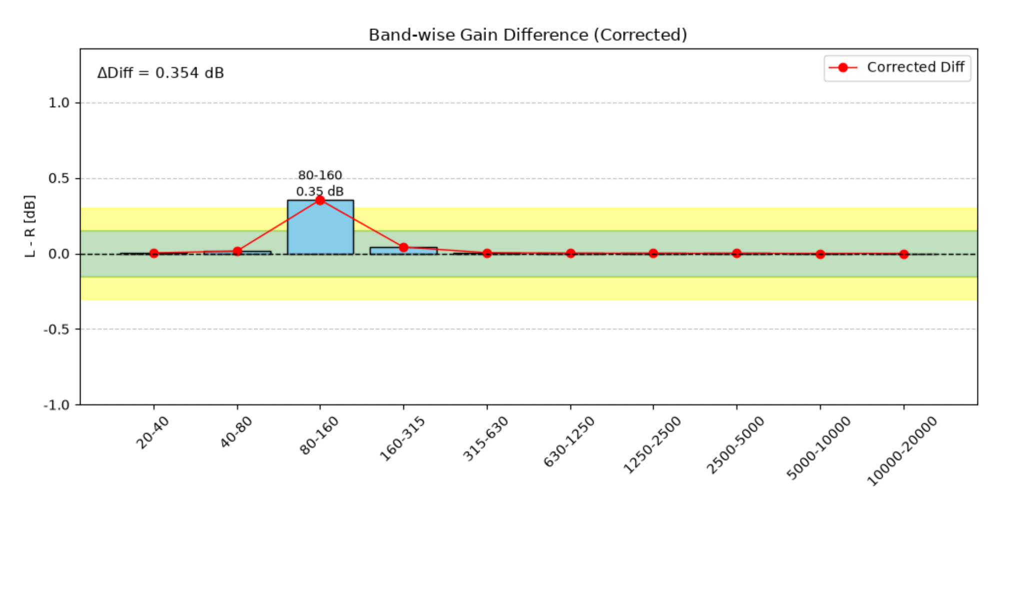
Corrected Gain Difference Analysis

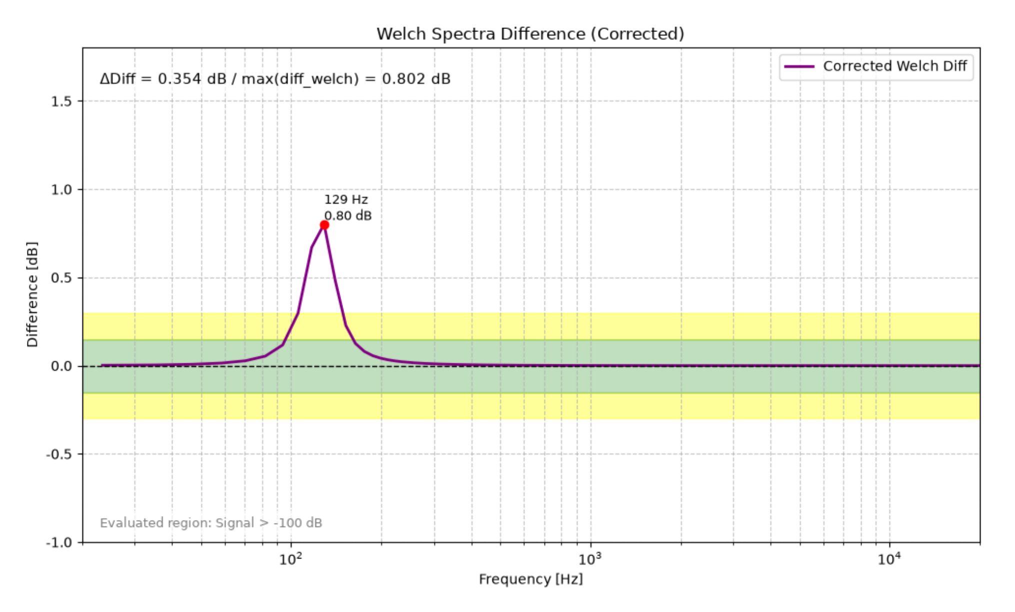
20-40 Hz: Diff=-0.000 dB
40-80 Hz: Diff=-0.000 dB
80-160 Hz: Diff=-0.000 dB
160-315 Hz: Diff=-0.001 dB
315-630 Hz: Diff=-0.005 dB
630-1250 Hz: Diff=-0.021 dB
1250-2500 Hz: Diff=-0.085 dB
2500-5000 Hz: Diff=-0.306 dB
5000-10000 Hz: Diff=-0.703 dB
10000-20000 Hz: Diff=-0.948 dB

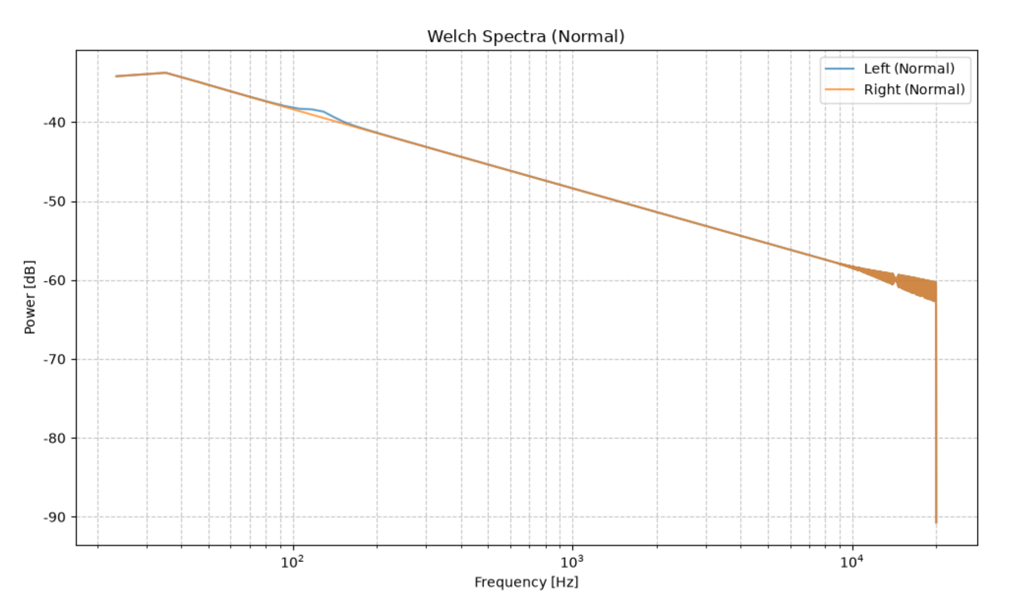
--- Summary ---
ΔDiff (Band) = 0.948 dB
ΔDiff (Welch max) = 0.994 dB
Max Diff Freq = 19933.6 Hz
Judgement: Check required: Possible equipment or measurement problem.

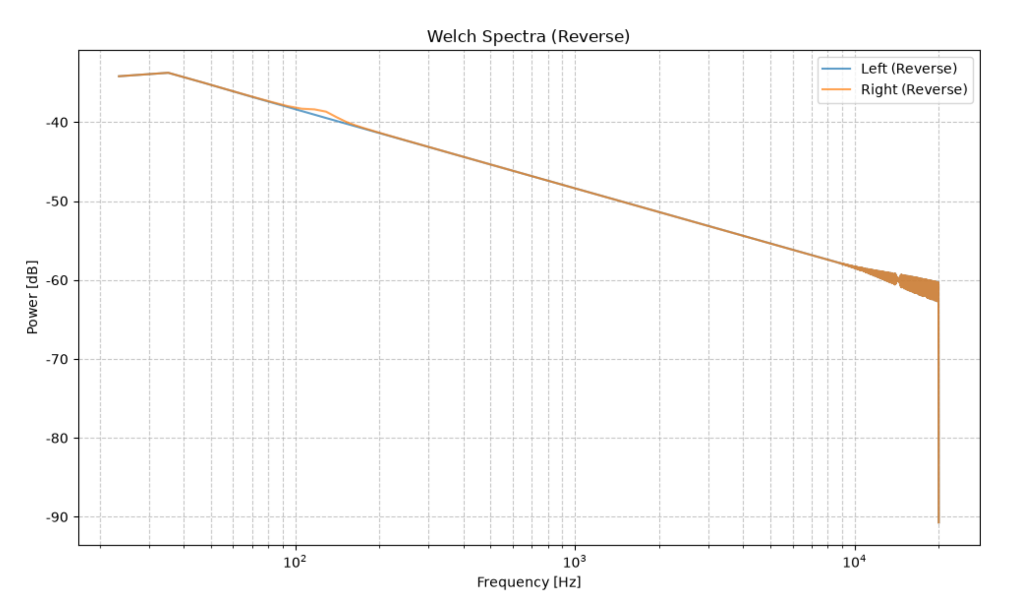
--- Channel Check ---
Energy Ratio (L/R) = 0.96
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, PEQ 125 Hz Q10 +1 dB (Left)









Summary.txt

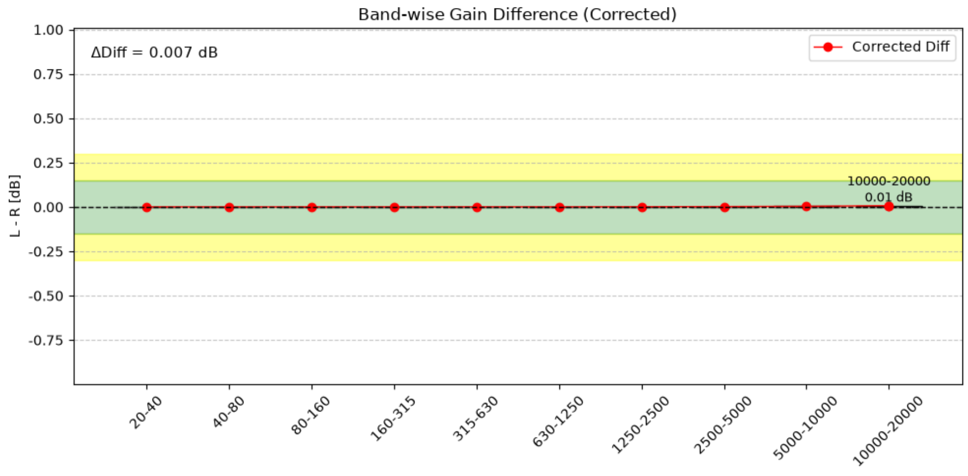
Corrected Gain Difference Analysis

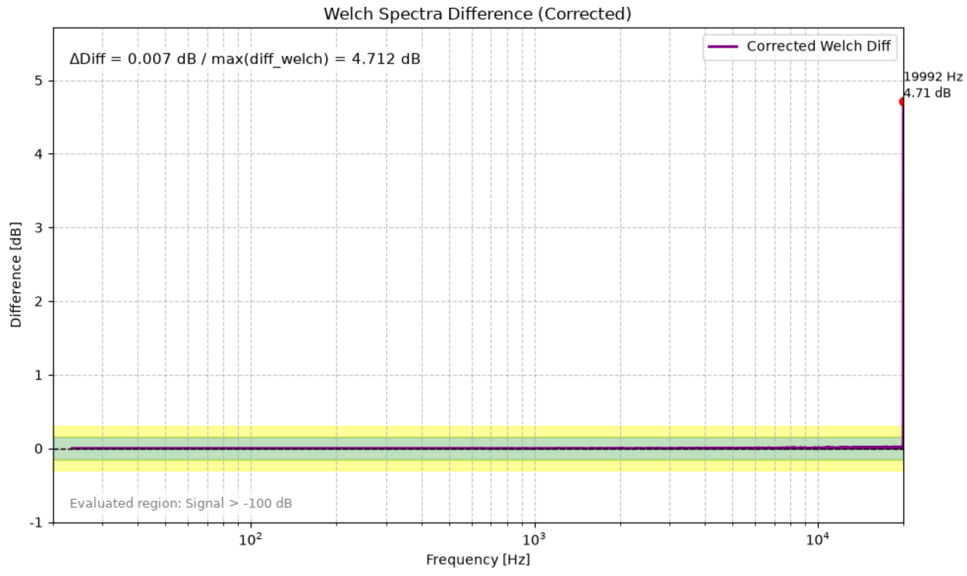
20-40 Hz: Diff=0.003 dB
40-80 Hz: Diff=0.015 dB
80-160 Hz: Diff=0.354 dB
160-315 Hz: Diff=0.042 dB
315-630 Hz: Diff=0.004 dB
630-1250 Hz: Diff=0.001 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.000 dB
5000-10000 Hz: Diff=0.000 dB
10000-20000 Hz: Diff=0.000 dB

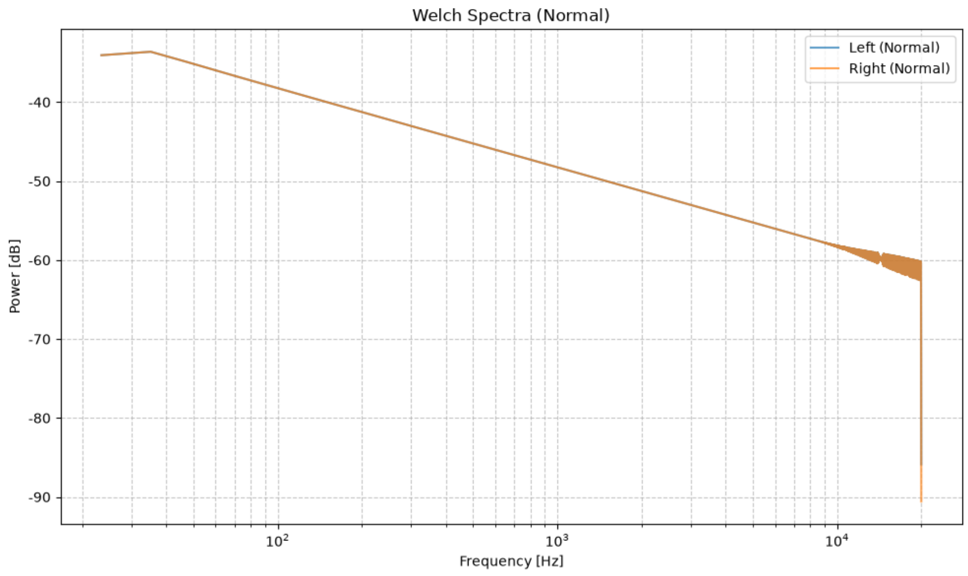
--- Summary ---
ΔDiff (Band) = 0.354 dB
ΔDiff (Welch max) = 0.802 dB
Max Diff Freq = 128.9 Hz
Judgement: Check required: Possible equipment or measurement problem.

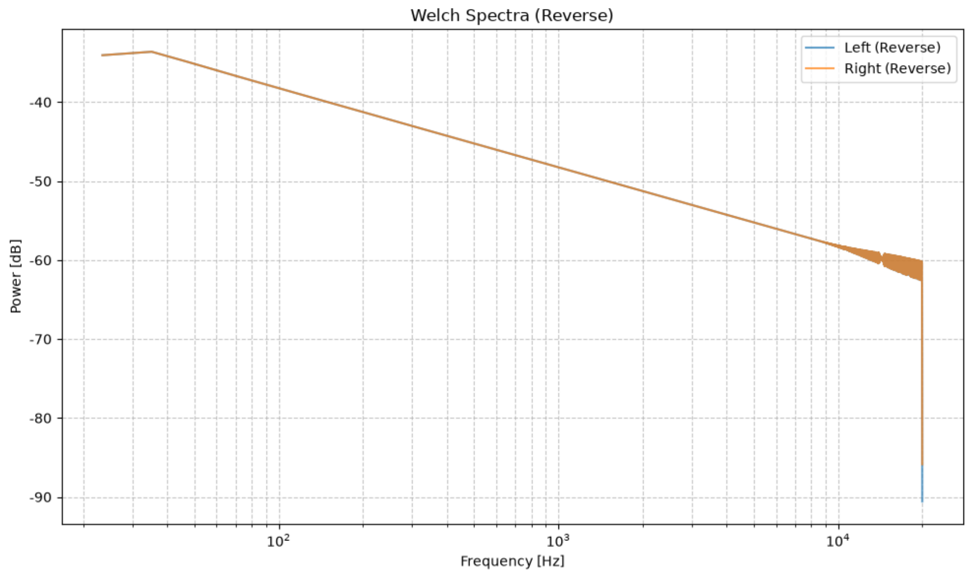
--- Channel Check ---
Energy Ratio (L/R) = 1.01
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, WhiteNoise Mix -26 dB (Left)







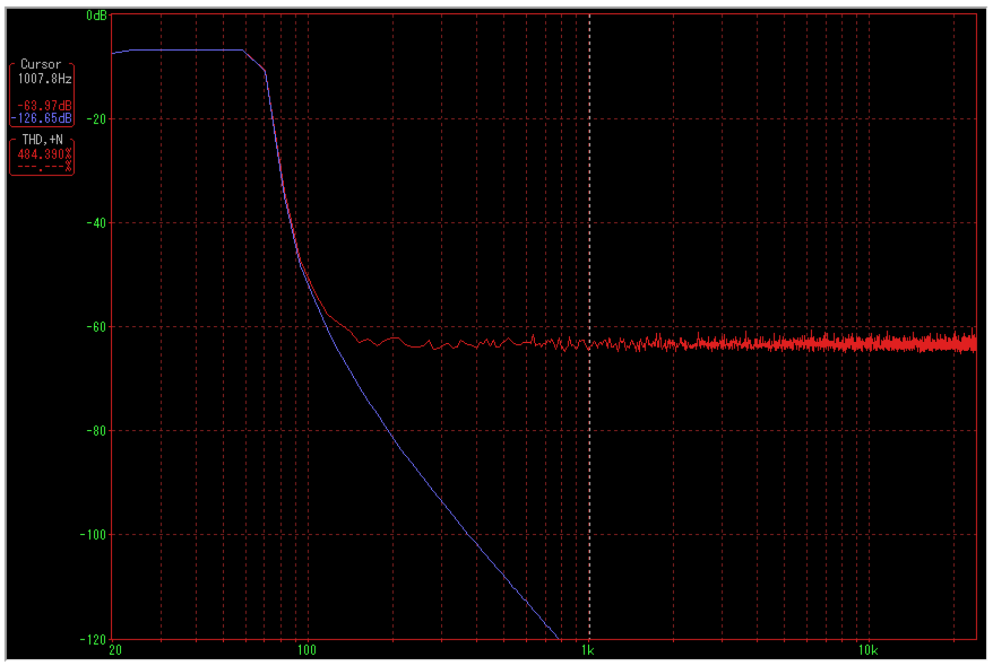


Summary.txt

Corrected Gain Difference Analysis

20-40 Hz: Diff=0.000 dB
40-80 Hz: Diff=0.000 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.001 dB
2500-5000 Hz: Diff=0.002 dB
5000-10000 Hz: Diff=0.003 dB
10000-20000 Hz: Diff=0.007 dB

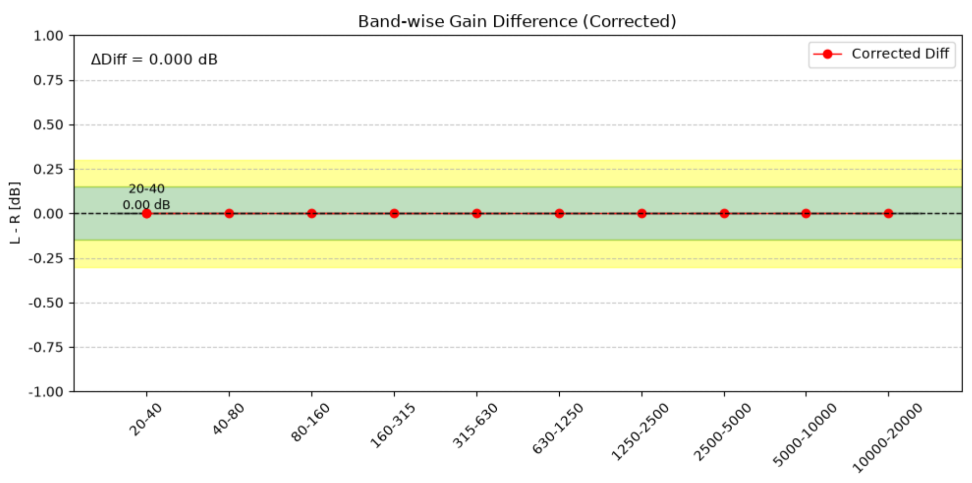
Shows the level of noise superimposed on the Left-channel signal in an image taken during a sweep

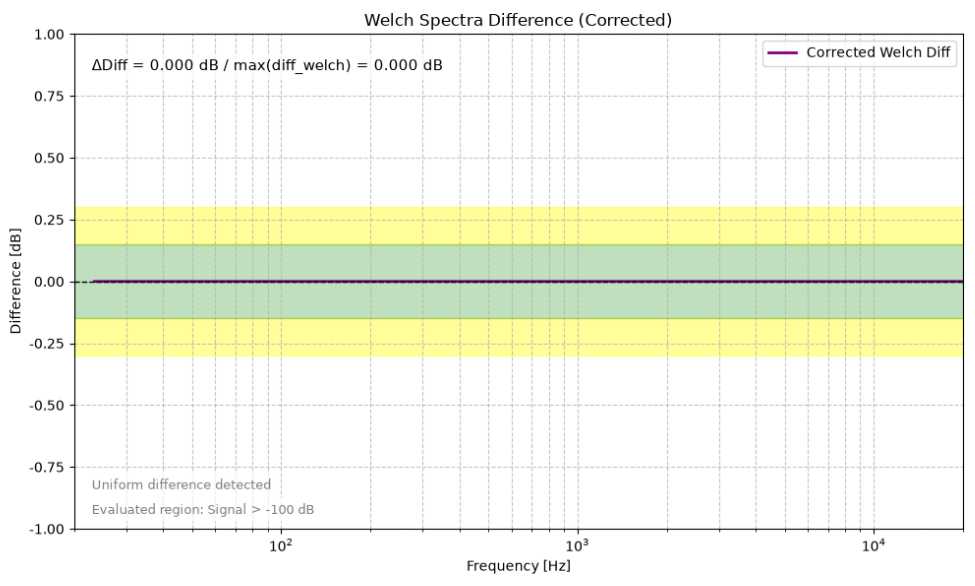


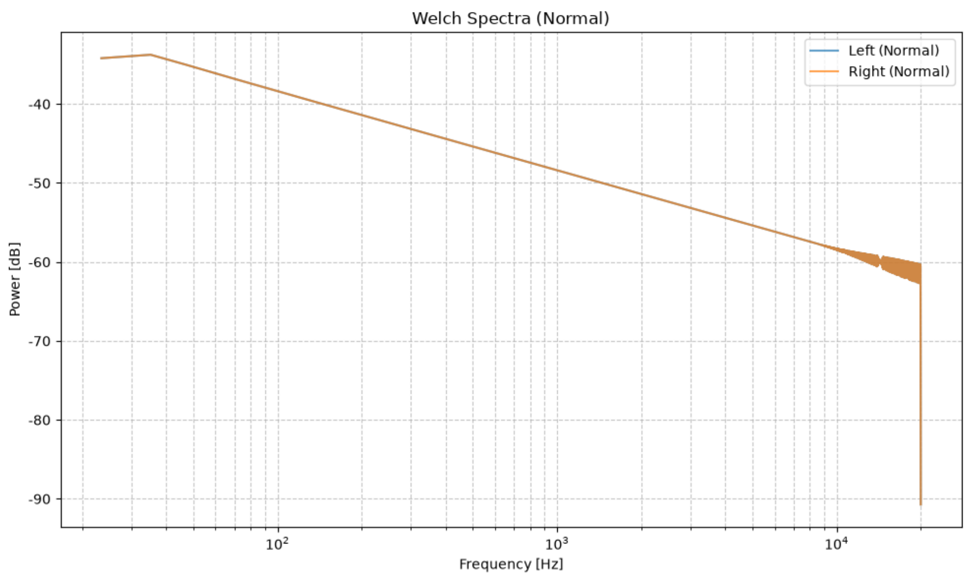
--- Summary ---
ΔDiff (Band) = 0.007 dB
ΔDiff (Welch max) = 4.712 dB
Max Diff Freq = 19992.2 Hz
Judgement: Caution: Local peak difference detected (possible contact issue, resonance, or cable difference).

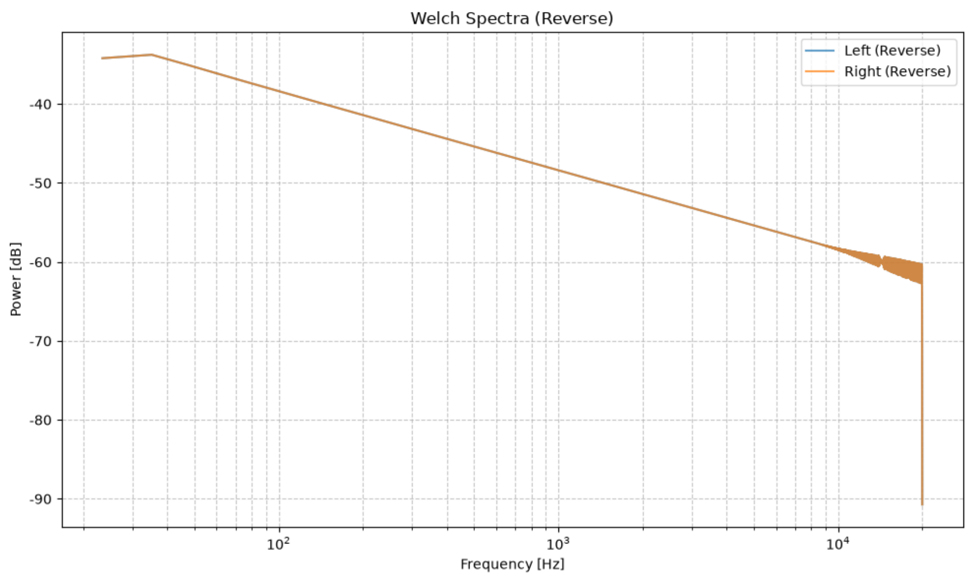
--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.

-10 dBFS 20 Hz to 20 kHz Sine Wave Log-Sweep, Symmetrical









Summary.txt
Corrected Gain Difference Analysis

20-40 Hz: Diff=0.000 dB
40-80 Hz: Diff=0.000 dB
80-160 Hz: Diff=0.000 dB
160-315 Hz: Diff=0.000 dB
315-630 Hz: Diff=0.000 dB
630-1250 Hz: Diff=0.000 dB
1250-2500 Hz: Diff=0.000 dB
2500-5000 Hz: Diff=0.000 dB
5000-10000 Hz: Diff=0.000 dB
10000-20000 Hz: Diff=0.000 dB

--- Summary ---
ΔDiff (Band) = 0.000 dB
ΔDiff (Welch max) = 0.000 dB
Judgement: OK: Channel difference is within measurement error.

--- Channel Check ---
Energy Ratio (L/R) = 1.00
Channel balance is within the normal range.
This Welch analysis evaluates mainly the frequency regions containing signal energy.
For narrow-band signals (such as single tones), frequency regions without signal are excluded from the evaluation.
Please also refer to the band analysis results for differences outside the detected signal region.